

# Generative AI for Software Development





Facebook interface for the page **somkiat.cc**. The top navigation bar includes the Facebook logo, the page name **somkiat.cc**, a search icon, and user profile icons for **Somkiat** and **Home**. The main navigation bar features **Page**, **Messages**, **Notifications** (with a red badge showing 3), **Insights**, **Publishing Tools**, **Settings**, and **Help**.

The page cover image shows a man in a white Superman t-shirt with "SOMKIAT.CC" printed on it, posing against a white wall. The profile picture is a smaller version of the same image. The page name **somkiat.cc** and handle **@somkiat.cc** are displayed below the profile picture.

The left sidebar contains a menu with **Home**, **Posts**, **Videos**, and **Photos**.

Below the cover image, there are buttons for **Liked**, **Following**, **Share**, and a three-dot menu icon. A blue call-to-action button **+ Add a Button** is also present. A blue tooltip message reads: "Help people take action on this Page."



**[https://github.com/up1/  
workshop-ai-with-technical-team](https://github.com/up1/workshop-ai-with-technical-team)**





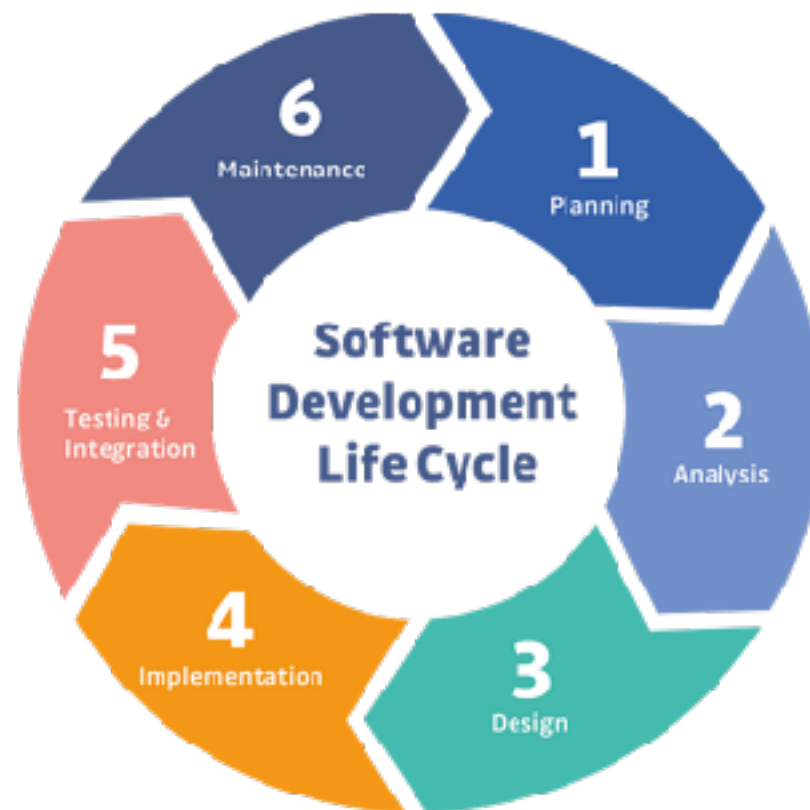
# Goals

Integrate Generative AI in Software Development

Optimize code quality

Team up with AI on coding tasks

Develop innovative solutions



# Learning Path

AI/ML

AI Model

Prompt Engineer

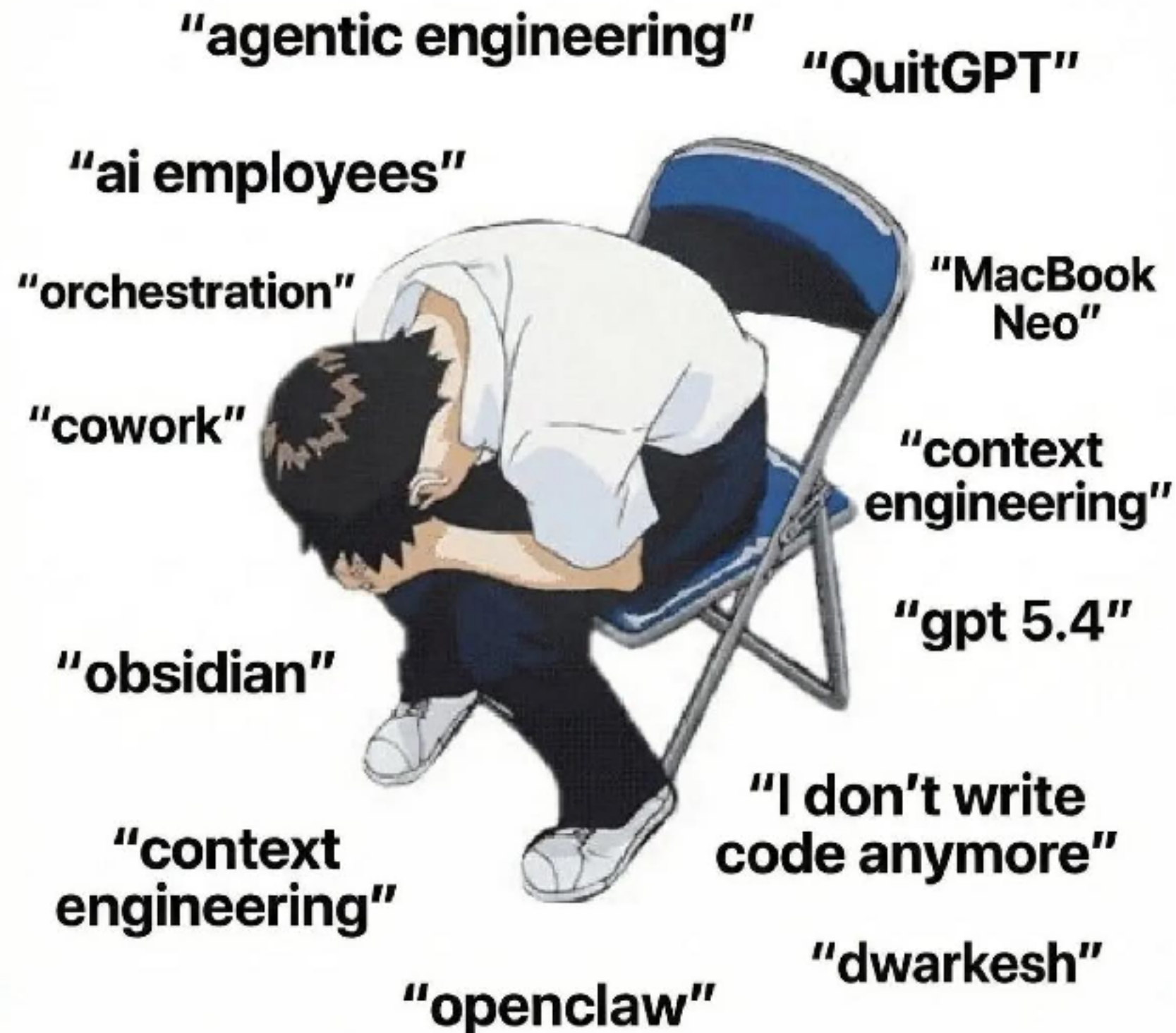
AI in Software  
development

Develop AI/LLM  
app

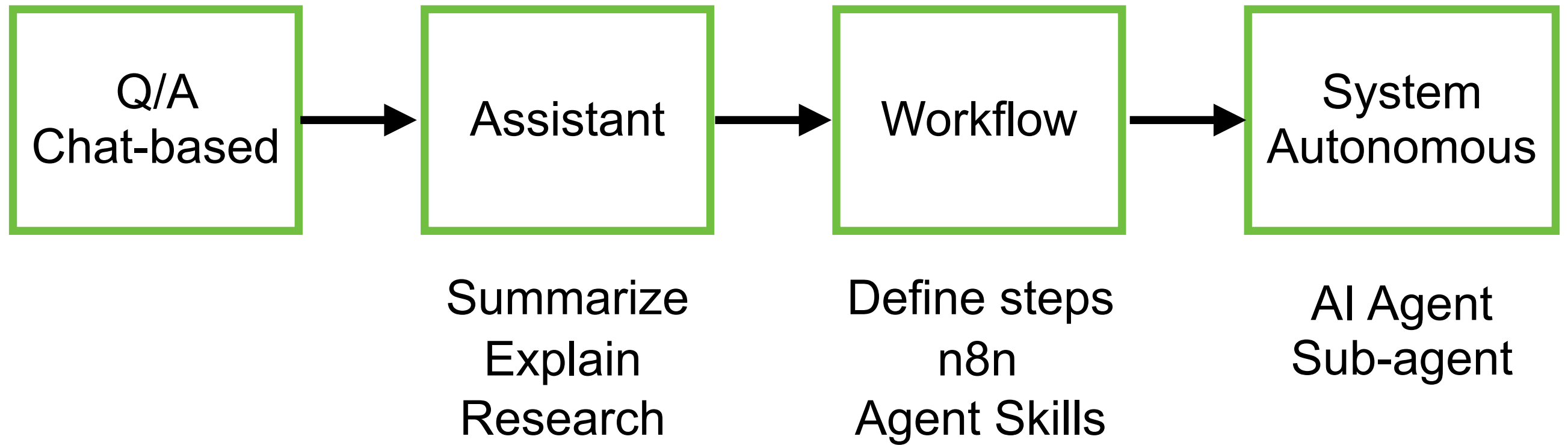
RAG

Use cases and workshops

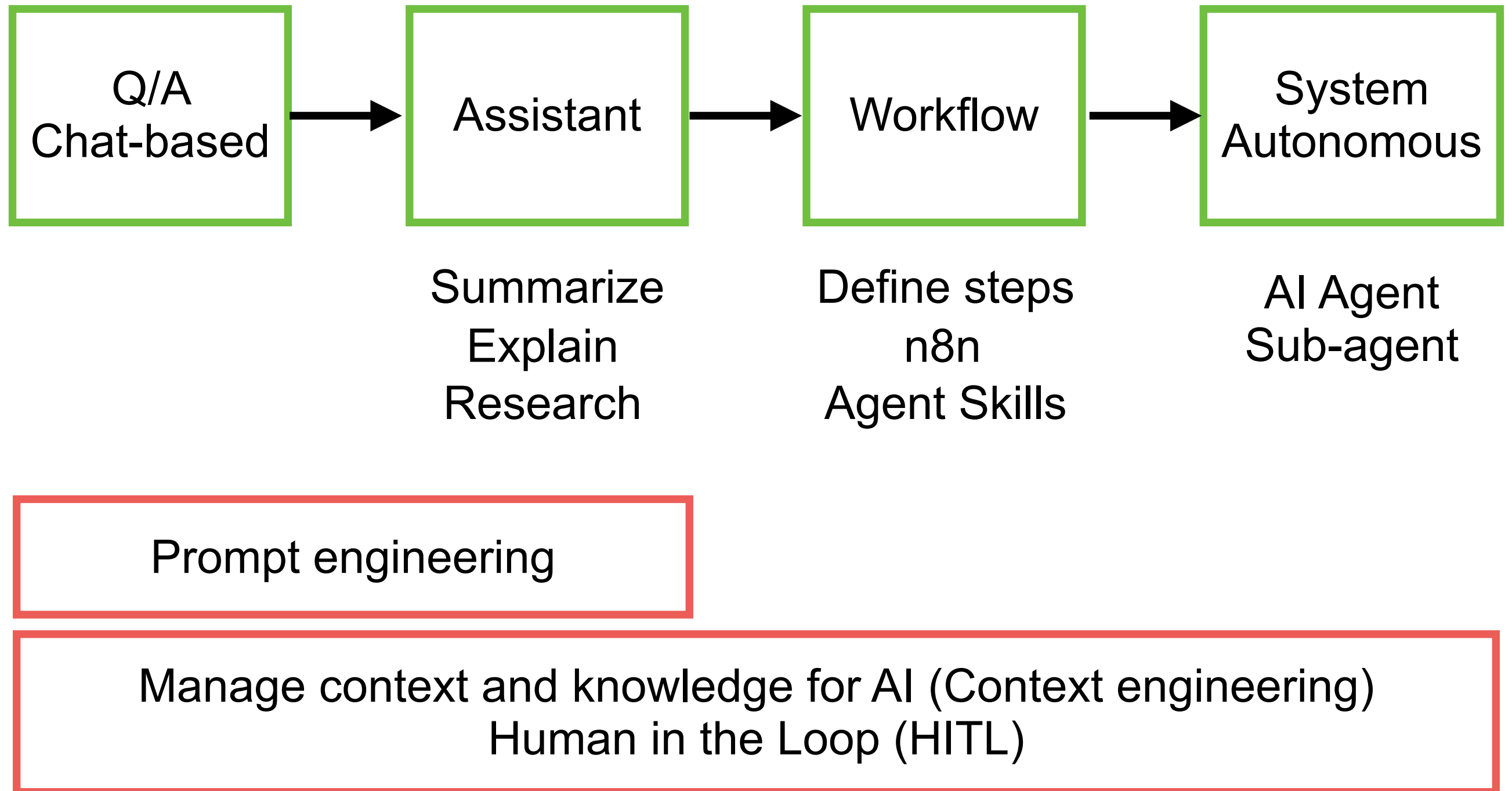




# Steps to use AI ?



# Steps to use AI ?



# Software Development

Requirement

Design

Develop

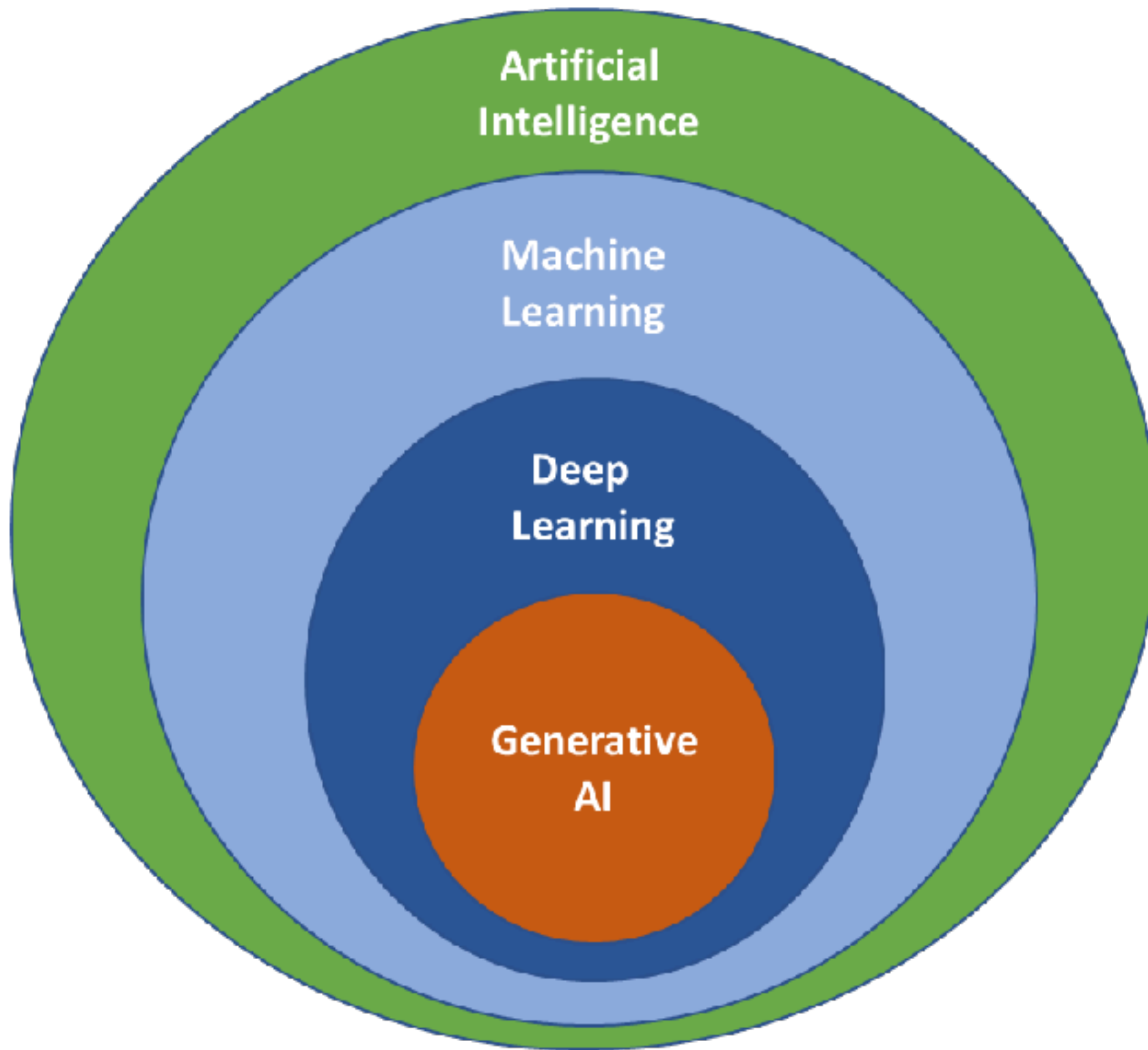
Testing

Deploy

Generative AI

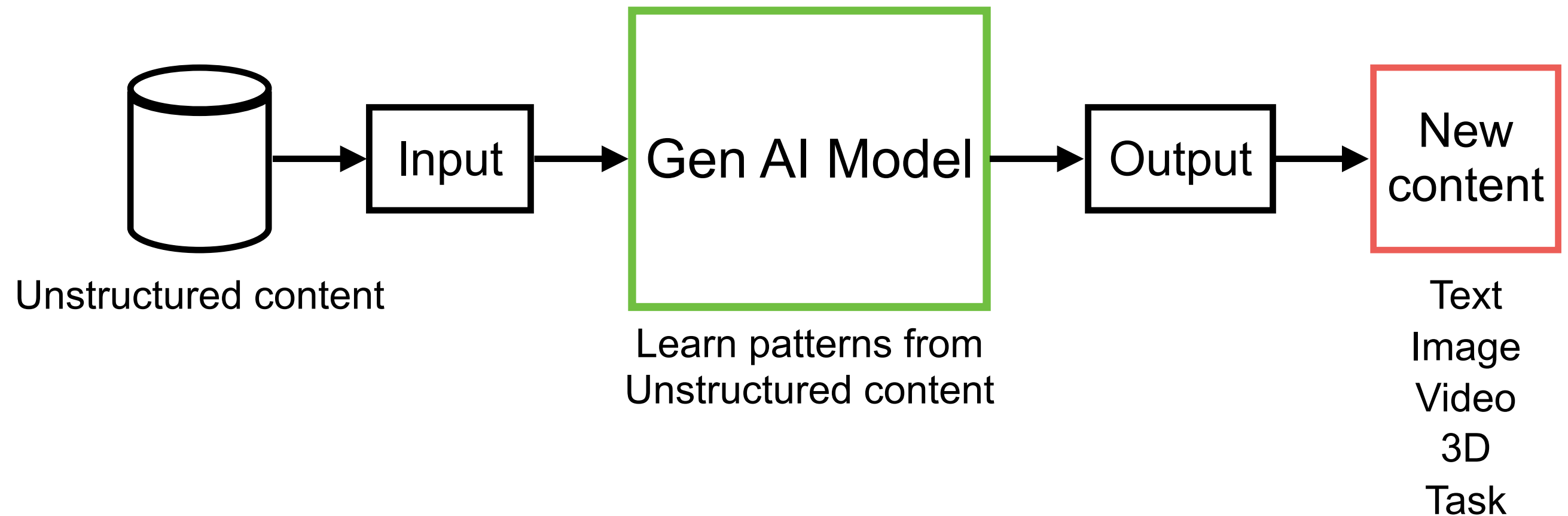
Improve Productivity ... (Replace human !!)







# Generative AI



<https://grow.google/ai-essentials/>





# Generative AI

**LLMs**

**Large Language Models**

Text generation  
Code generation  
Chatbot  
Conversation AI

**GANs**

**Generative Adversarial Network**

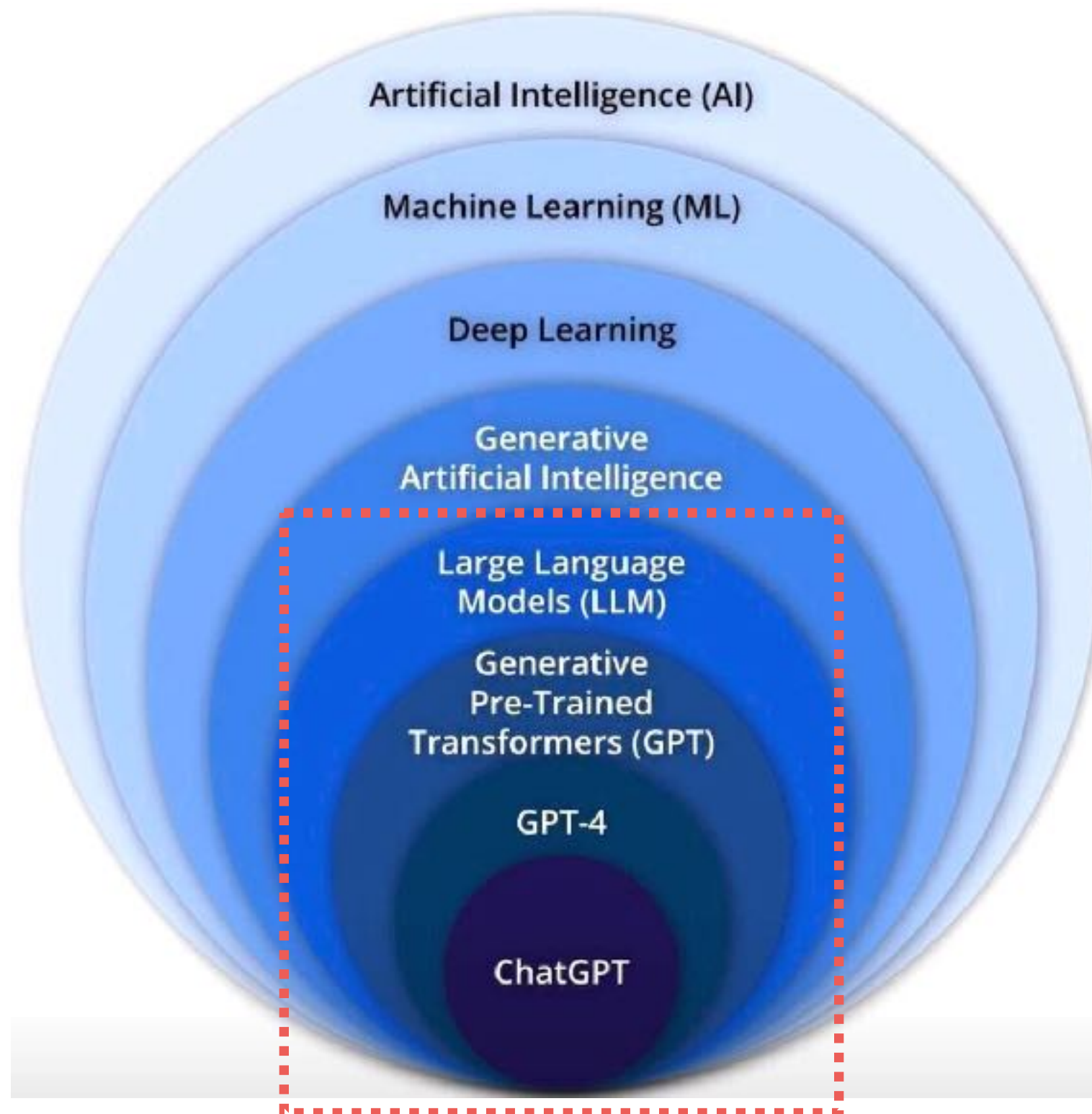
Image generation  
Deep fake  
Art creation  
Simulate financial market

**VAEs**

**Variational Autoencoders**

Data compression  
Synthetic data generation  
Image reconstruction





# Large Language Model (LLM)

Type of AI model  
Process, understand  
Generate human readable data

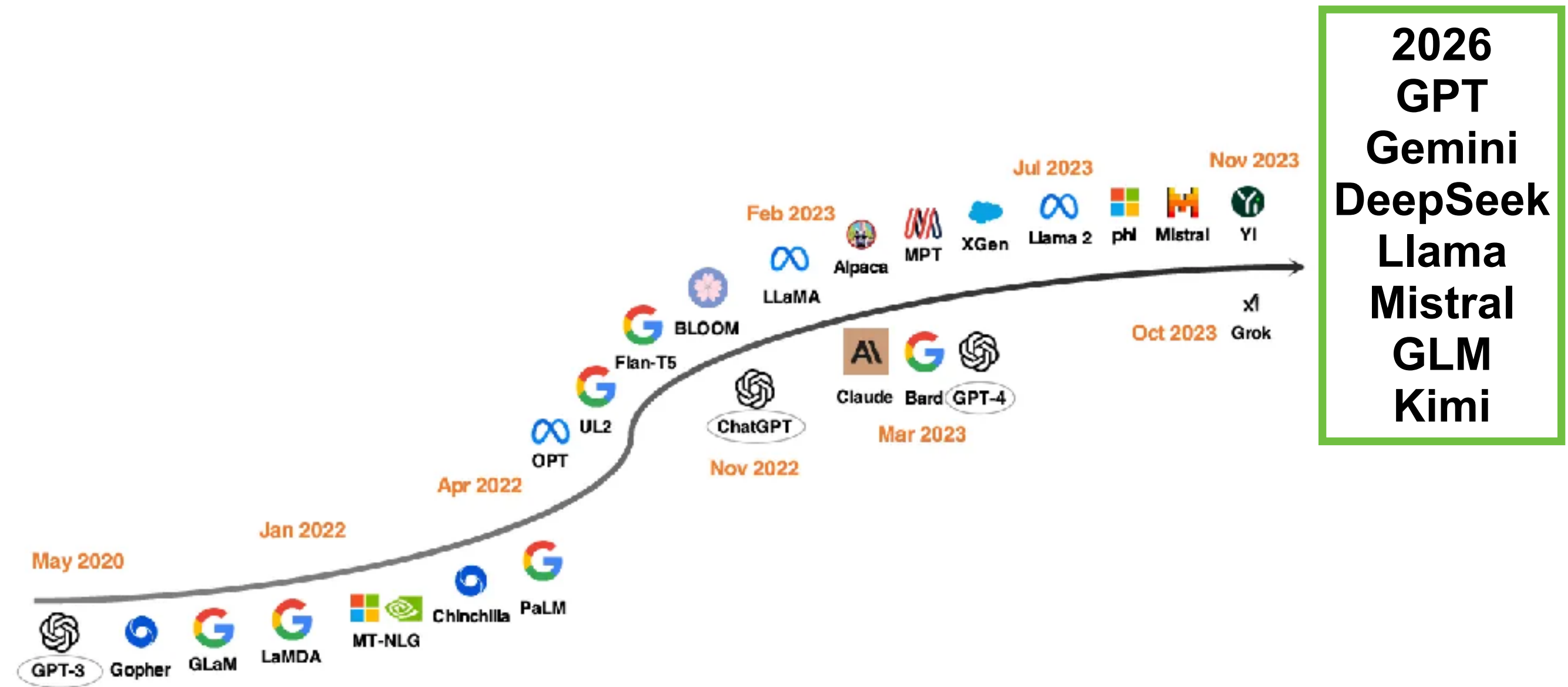


# LLMs in industry

| Model name                                                    | Company   |
|---------------------------------------------------------------|-----------|
| Bidirectional Encoder Representation from Transformers (BERT) | Google AI |
| Generative Pre-trained transformer-5 (GPT-5)                  | OpenAI    |
| Pathways Language Model-E (PaLM-E)                            | Google AI |
| BLOOM                                                         | NVIDIA AI |
| Llama 4                                                       | Facebook  |
| Claude 4.5 Sonnet/Opus                                        | Anthropic |



# LLM Development timeline



2026  
GPT  
Gemini  
DeepSeek  
Llama  
Mistral  
GLM  
Kimi

Figure 3: LLM development timeline. The models below the arrow are closed-source while those above the arrow are open-source.

<https://arxiv.org/abs/2311.16989>



# Let's go !!



# สี่เหลืง

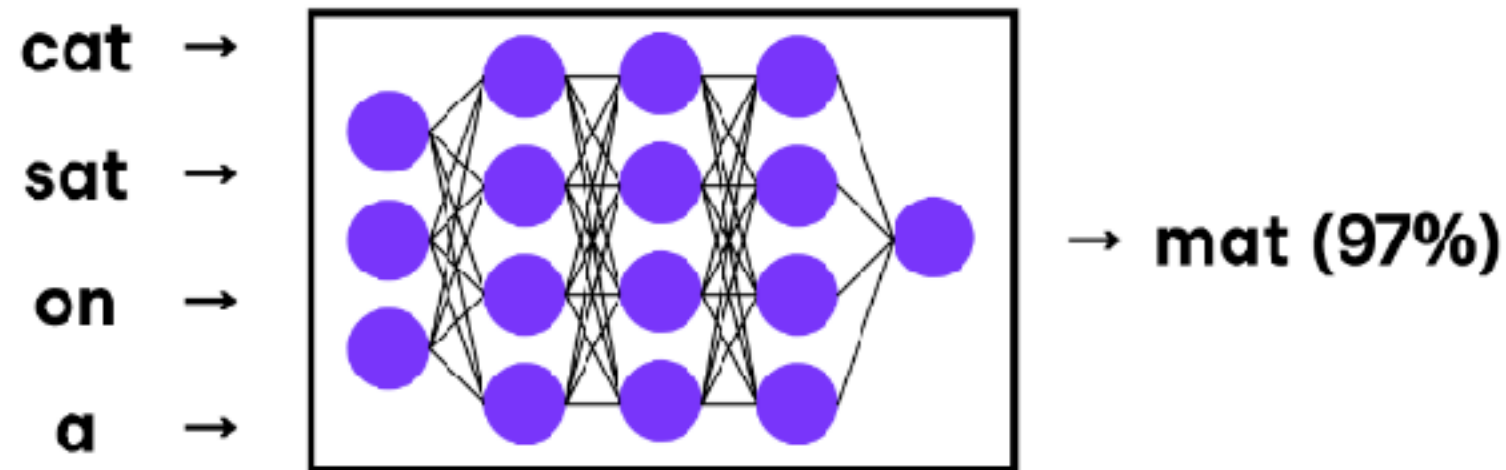




# Large Language Model (LLM)

Neural network

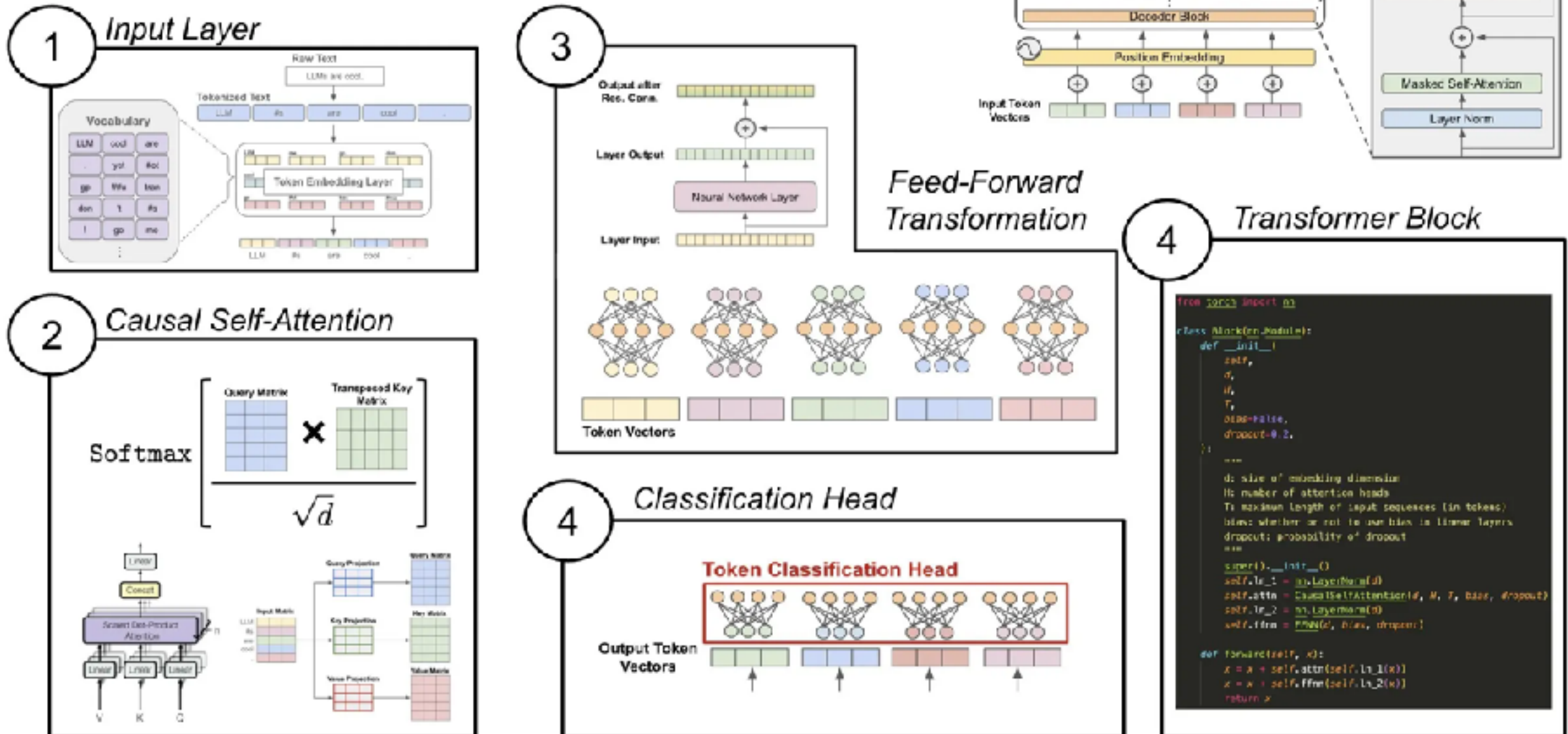
Predicts the next word in a sequence





# LLM components !!

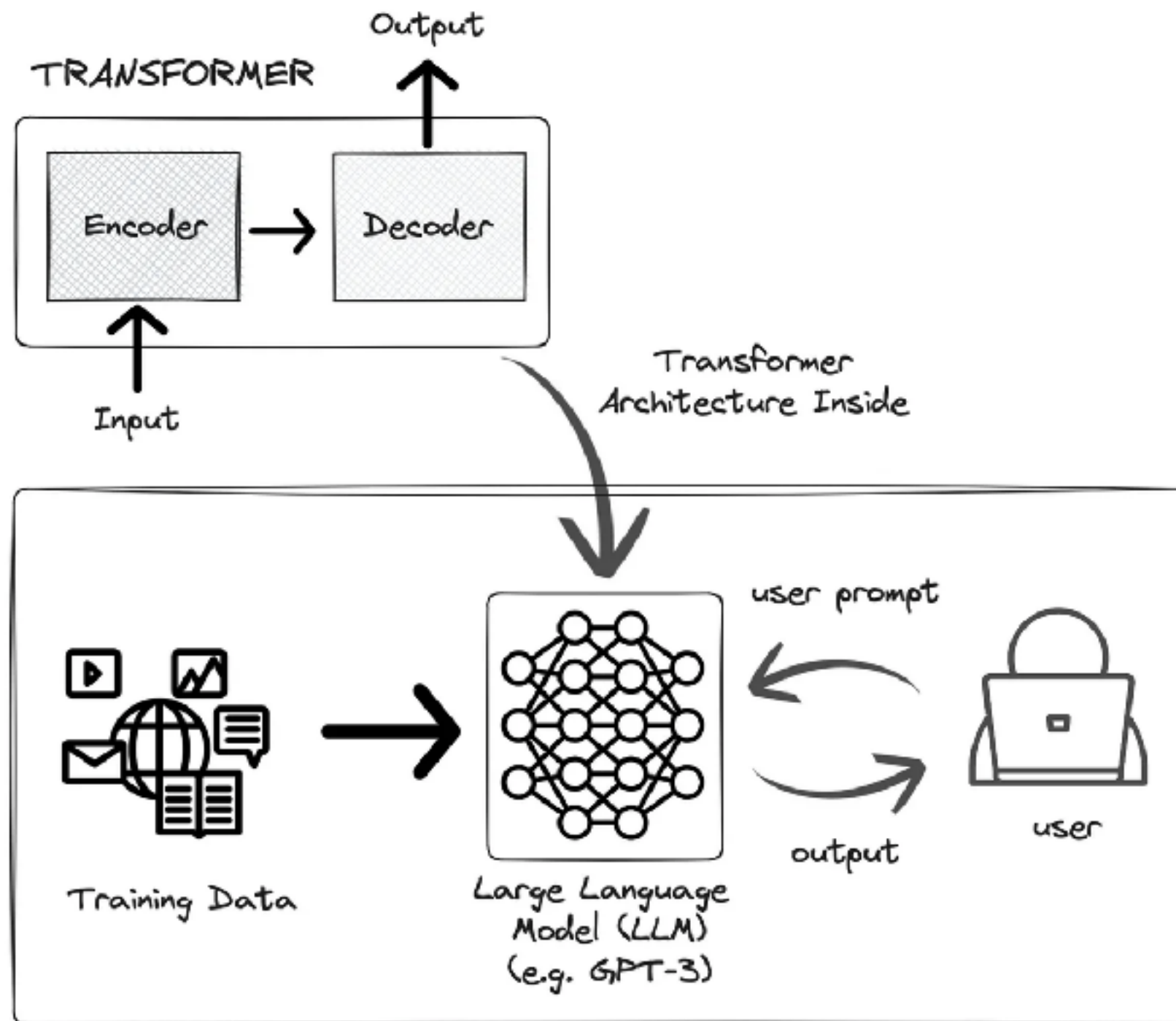
## Components of the Decoder-only Transformer



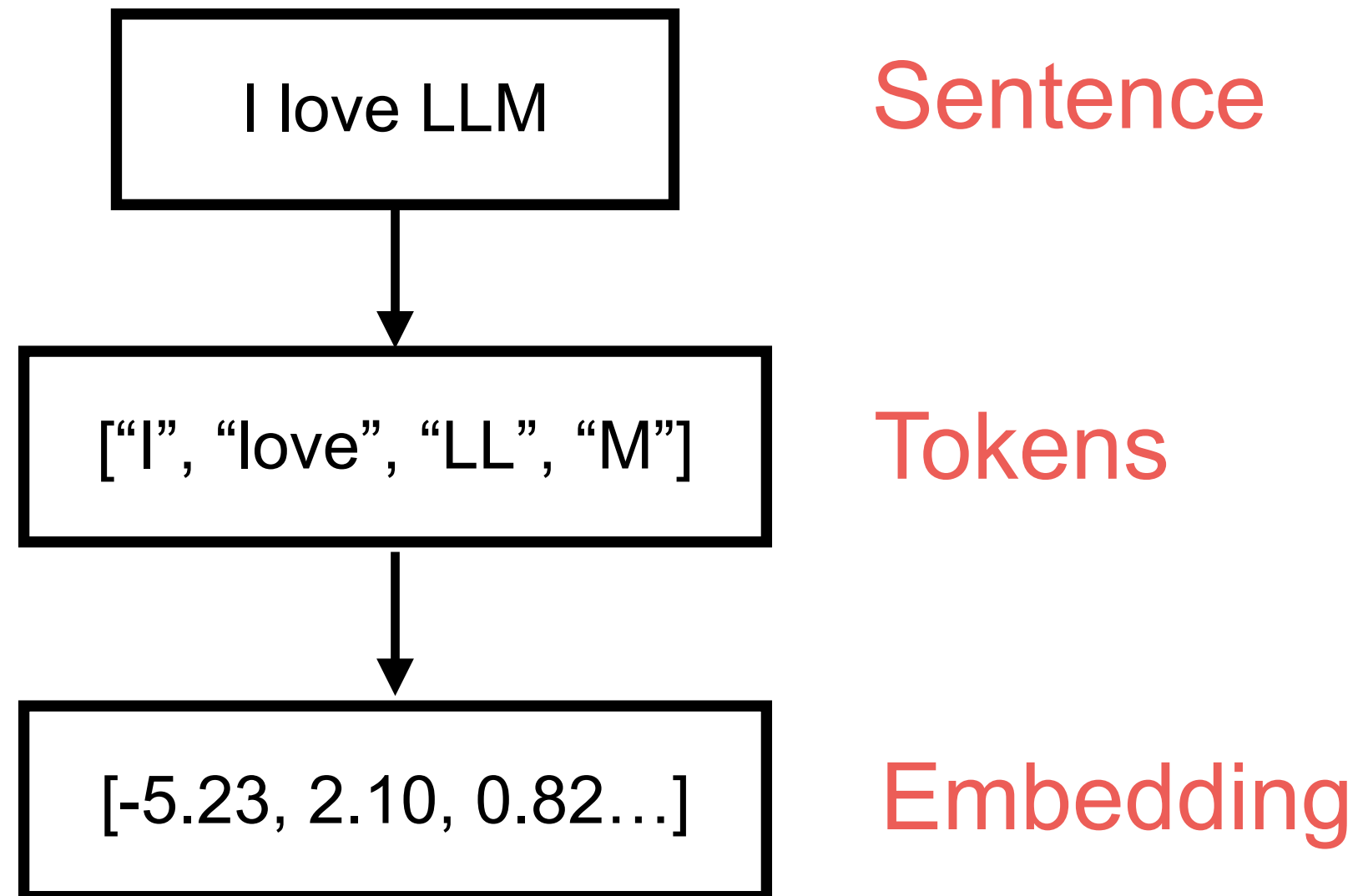
<https://stackoverflow.blog/2024/08/22/llms-evolve-quickly-their-underlying-architecture-not-so-much>



# Transformer inside



# Transformer process



# OpenAI Tokenizer

GPT-4o & GPT-4o mini (coming soon) **GPT-3.5 & GPT-4** GPT-3 (Legacy)

ประเทศไทย

Clear Show example

Tokens Characters  
10 9

ประเทศไทย

Text Token IDs

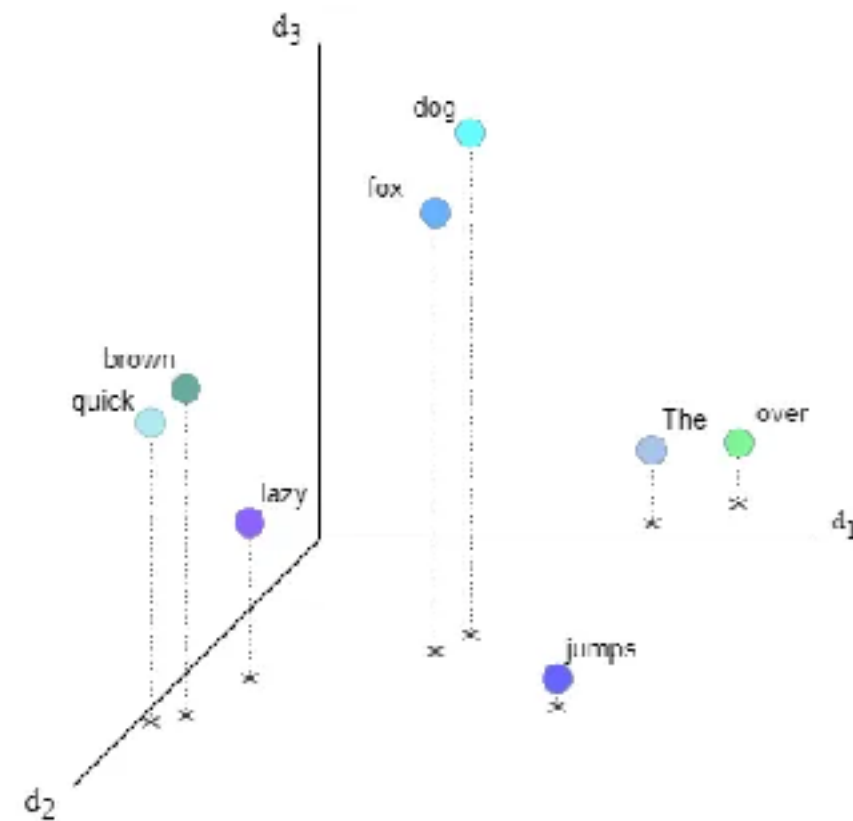
<https://platform.openai.com/tokenizer>



# Embedding ?

Map items of **unstructured data** to high-dimensional real vectors

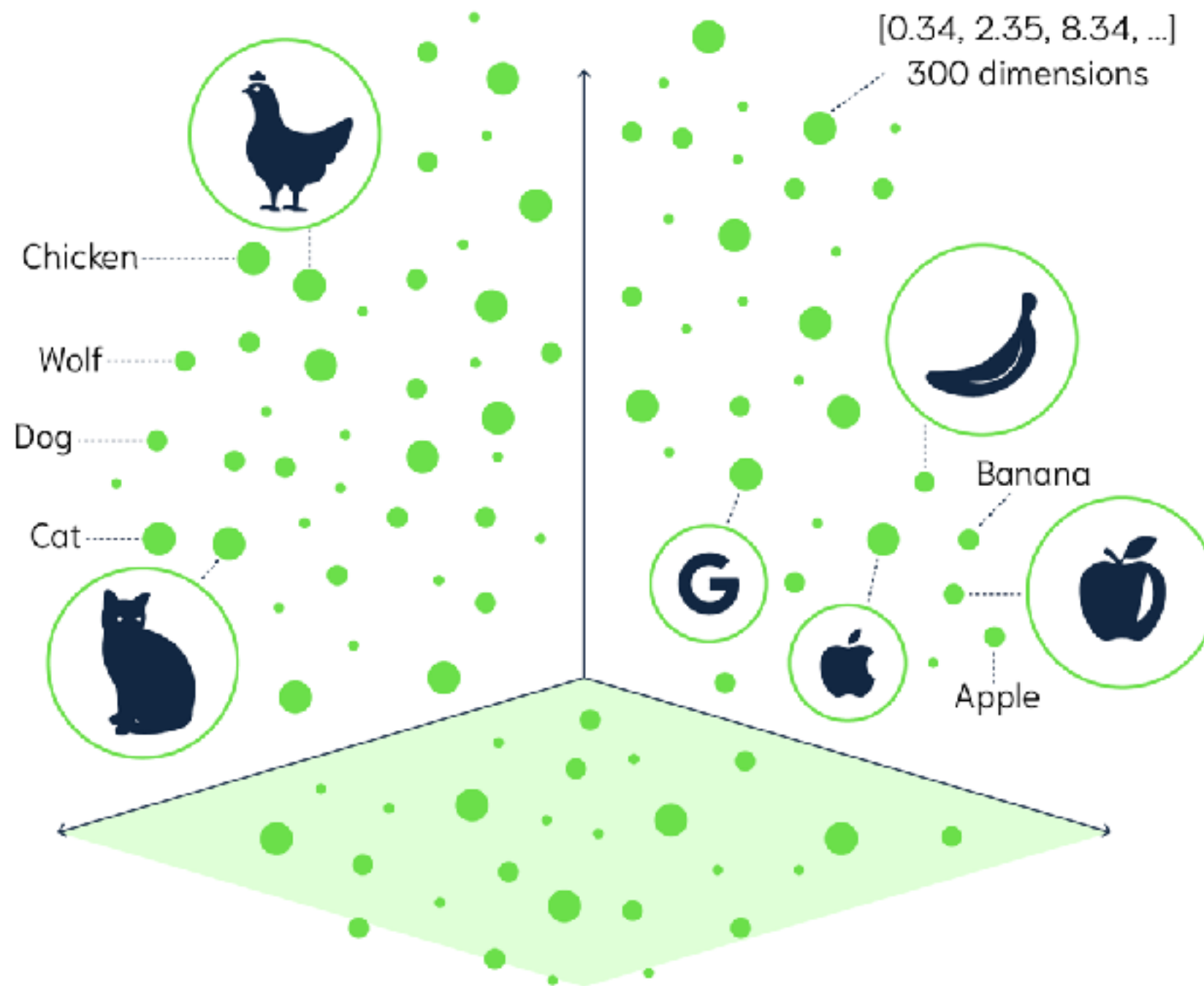
|       |   | 0    | 1     | 2    |       | $d_{\text{model}}$ |
|-------|---|------|-------|------|-------|--------------------|
| The   | → | 0.64 | -0.09 | 0.23 | ..... | 0.005              |
| quick | → | 0.05 | 0.79  | 0.47 | ..... | -0.54              |
| brown | → | 0.12 | 0.74  | 0.51 | ..... | -0.3               |
| fox   | → | 0.42 | 0.52  | 0.83 | ..... | 0.01               |
| jumps | → | 0.88 | 0.69  | 0.02 | ..... | 0.27               |
| over  | → | 0.84 | 0.15  | 0.13 | ..... | 0.05               |
| the   | → | 0.64 | 0.09  | 0.23 | ..... | 0.005              |
| lazy  | → | 0.1  | 0.65  | 0.28 | ..... | -0.19              |
| dog   | → | 0.54 | 0.49  | 0.90 | ..... | 0.000              |



<https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca>



# Visual of Vector space





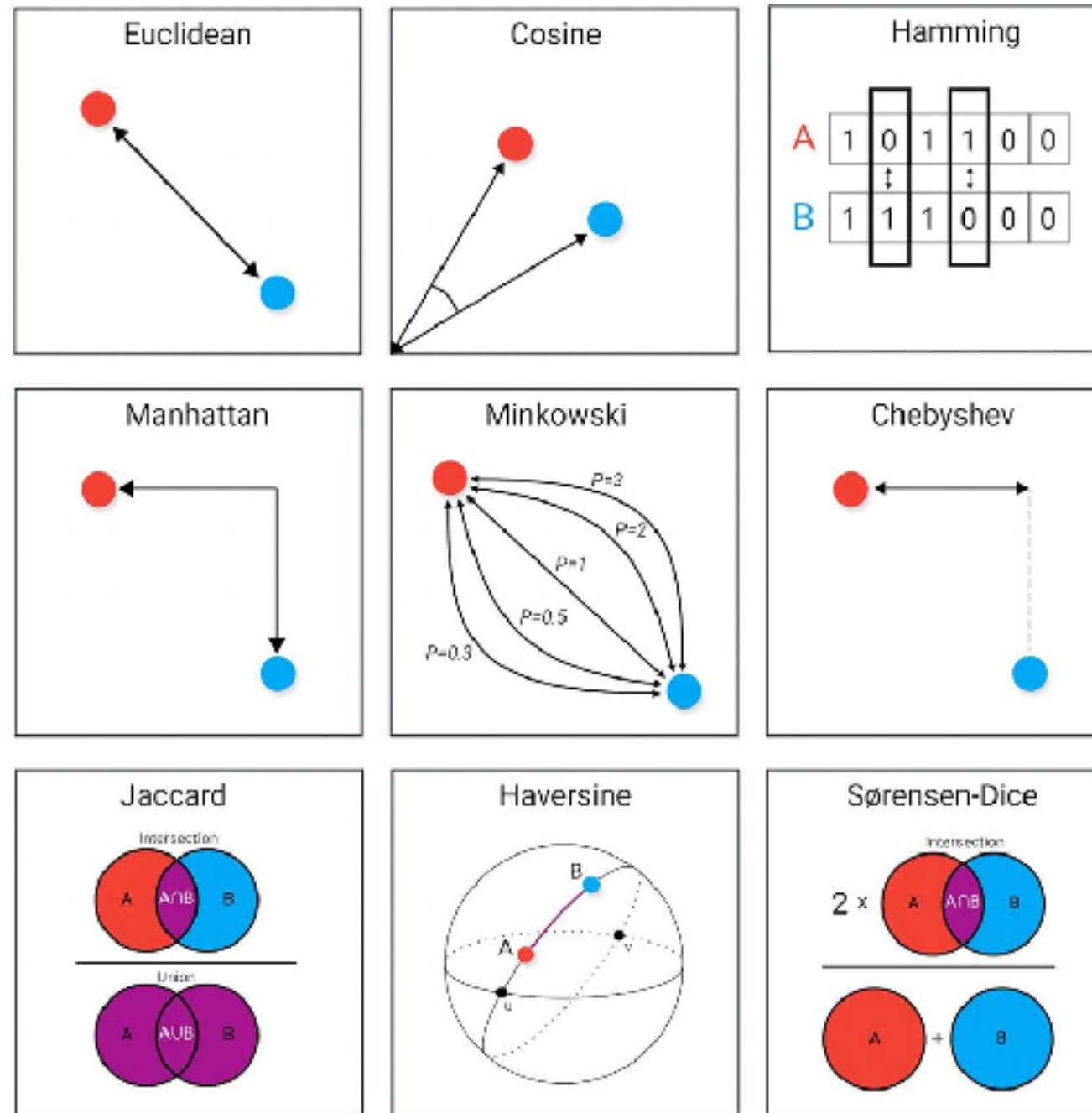
# Embedding Leaderboard

| Rank (Boxed) | Model                                           | Zero-shot | Memory Usage | Number of Parameters | Embedding Dimension | Max Tokens | Mean (Text) | Mean (TaskText) | Bitext | Classification |
|--------------|-------------------------------------------------|-----------|--------------|----------------------|---------------------|------------|-------------|-----------------|--------|----------------|
| 1            | <a href="#">llama-embed-nemotron-8b</a>         | 99%       | 28629        | 7B                   | 4096                | 32768      | 69.46       | 61.09           | 81.72  | 73.21          |
| 2            | <a href="#">Qwen3-Embedding-8B</a>              | 99%       | 28866        | 7B                   | 4096                | 32768      | 70.58       | 61.69           | 80.89  | 74.00          |
| 3            | <a href="#">gemini-embedding-001</a>            | 99%       | Unknown      | Unknown              | 3072                | 2048       | 68.37       | 59.59           | 79.28  | 71.82          |
| 4            | <a href="#">Qwen3-Embedding-4B</a>              | 99%       | 15341        | 4B                   | 2560                | 32768      | 69.45       | 60.86           | 79.36  | 72.33          |
| 5            | <a href="#">Qwen3-Embedding-0.6B</a>            | 99%       | 2272         | 595M                 | 1024                | 32768      | 64.34       | 56.01           | 72.23  | 66.83          |
| 6            | <a href="#">gte-Qwen2-7B-instruct</a>           | ⚠️ NA     | 29040        | 7B                   | 3584                | 32768      | 62.51       | 55.93           | 73.92  | 61.55          |
| 7            | <a href="#">Linq-Embed-Mistral</a>              | 99%       | 13563        | 7B                   | 4096                | 32768      | 61.47       | 54.14           | 70.34  | 62.24          |
| 8            | <a href="#">multilingual-e5-large-instruct</a>  | 99%       | 1068         | 560M                 | 1024                | 514        | 63.22       | 55.08           | 80.13  | 64.94          |
| 9            | <a href="#">embeddinggemma-300m</a>             | 99%       | 578          | 367M                 | 768                 | 2048       | 61.15       | 54.31           | 64.40  | 60.90          |
| 10           | <a href="#">SFR-Embedding-Mistral</a>           | 96%       | 13563        | 7B                   | 4096                | 32768      | 60.90       | 53.92           | 70.00  | 60.02          |
| 11           | <a href="#">text-multilingual-embedding-002</a> | 99%       | Unknown      | Unknown              | 768                 | 2048       | 62.16       | 54.25           | 70.73  | 64.64          |

<https://huggingface.co/spaces/mteb/leaderboard>

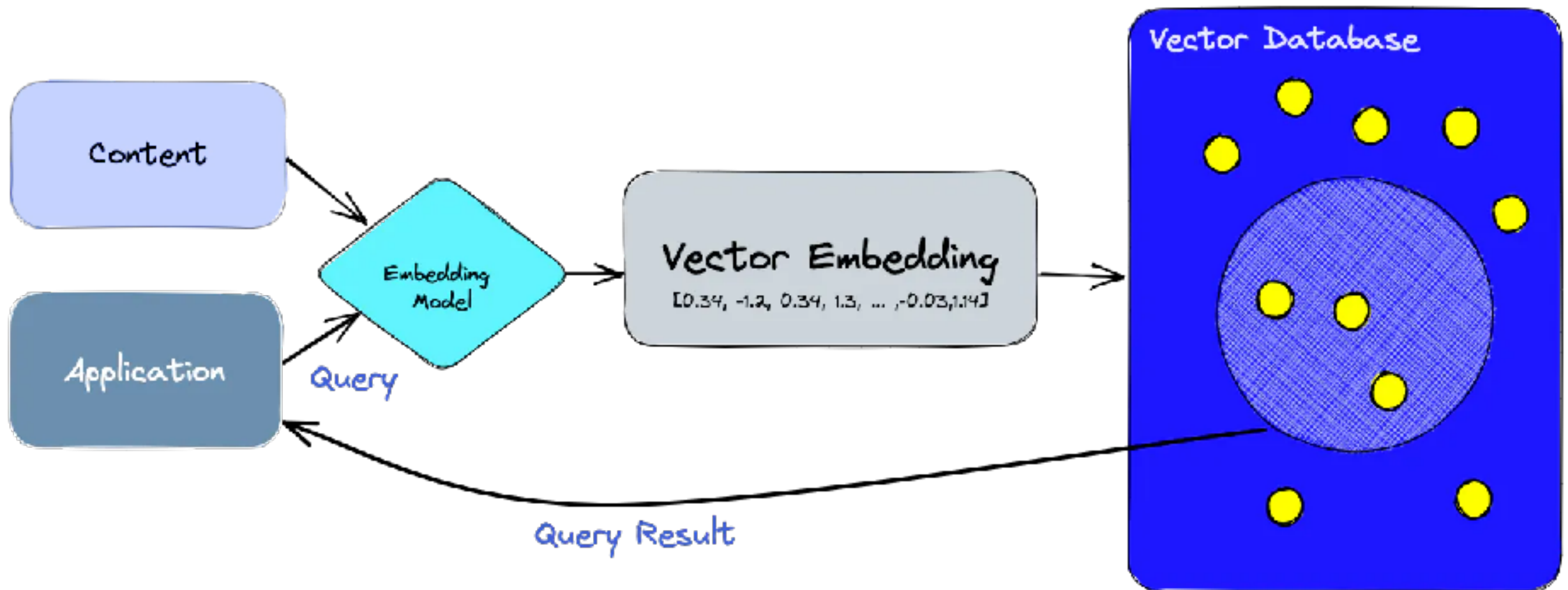


# Distance measure in Data Science



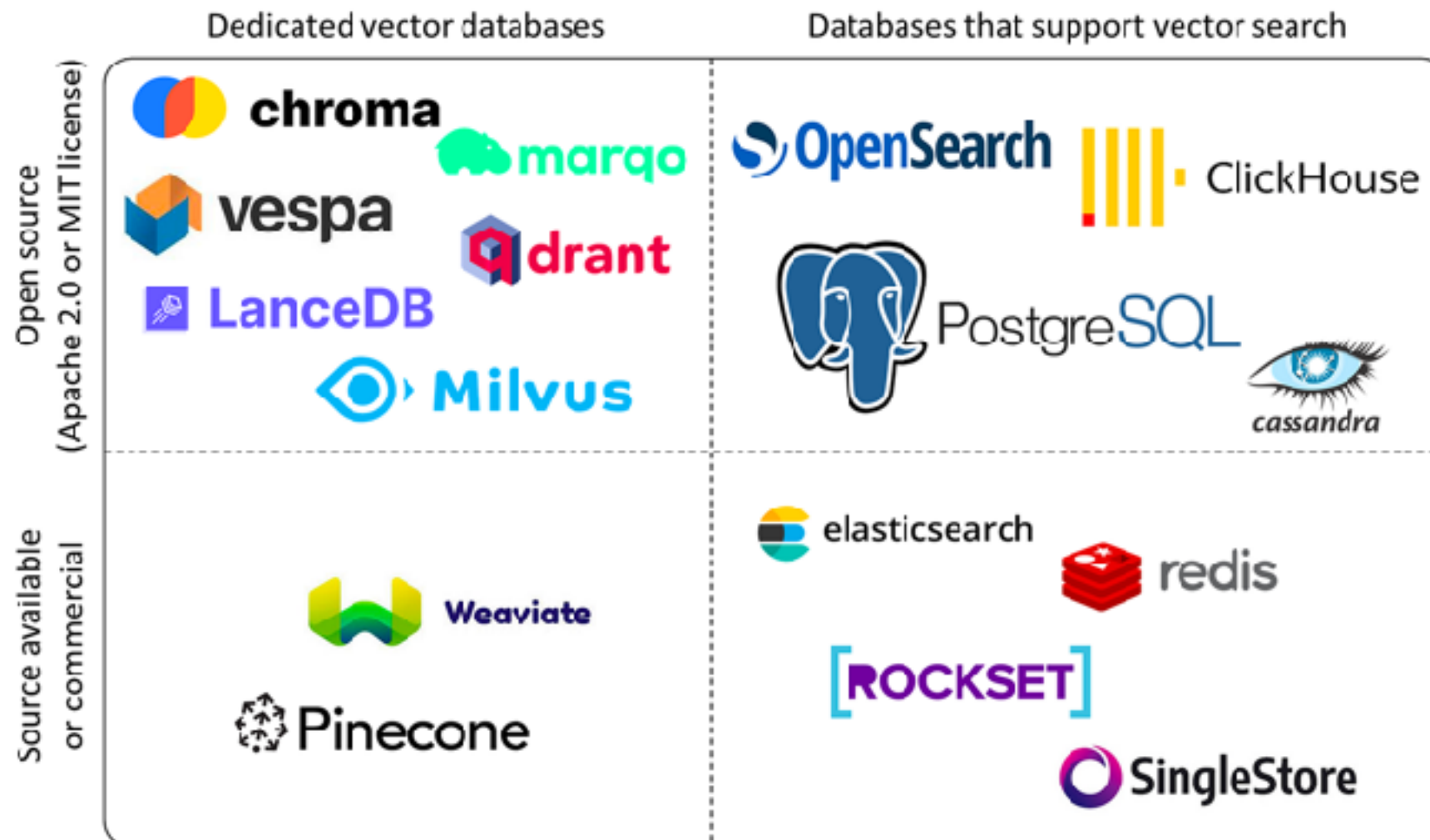


# Store data in Vector Database



# Vector Database ?

Map items of unstructured data to high-dimensional  
real **vectors**



<https://towardsdatascience.com/transformers-in-depth-part-1-introduction-to-transformer-models-in-5-minutes-ad25da6d3cca>



Application

Infrastructure

Model



Application

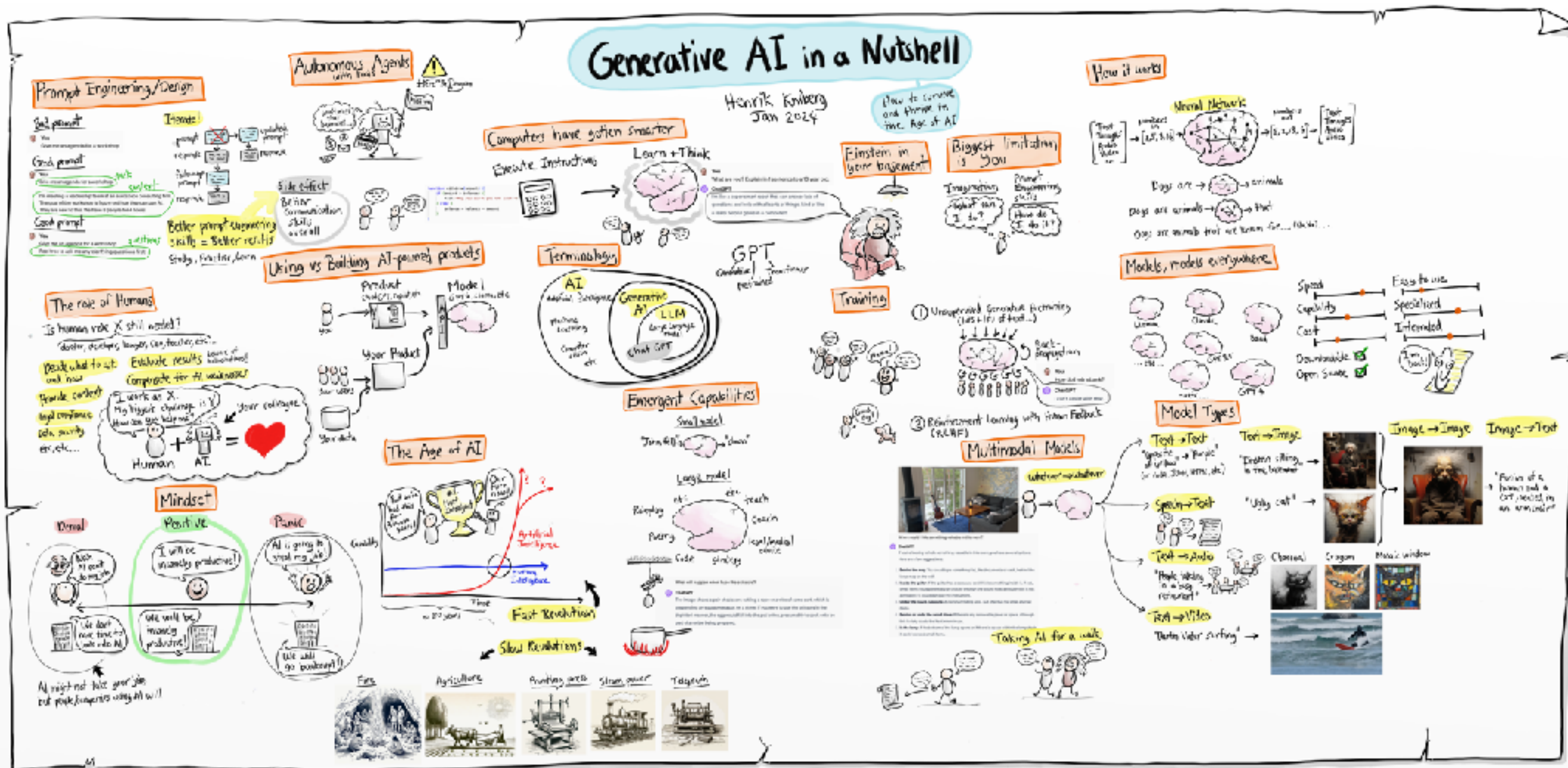
REST API

Infrastructure

Model



# Generative AI in Nutshell



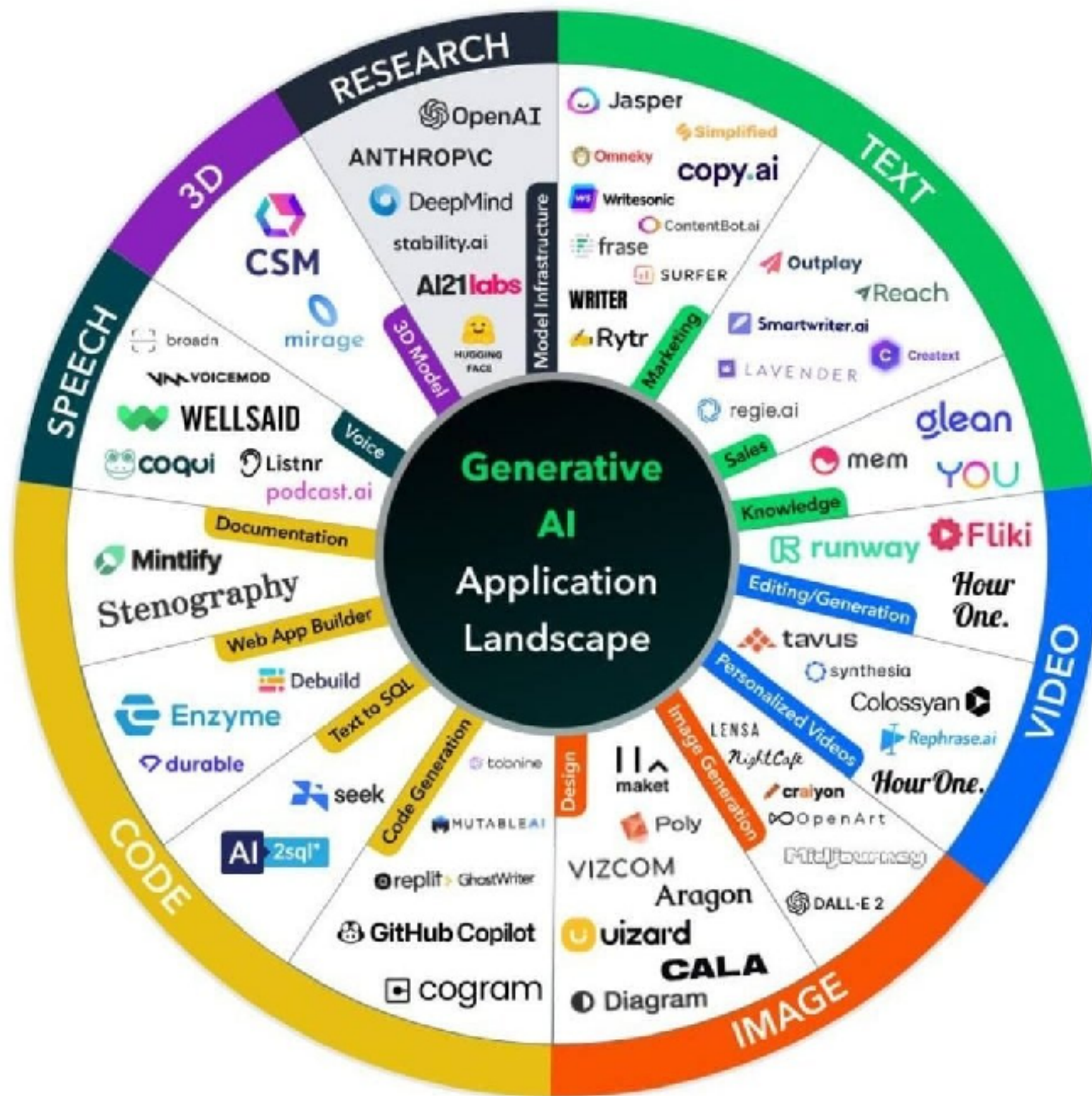
<https://www.youtube.com/watch?v=2IK3DFHRFfw>



# Application Tool chains







# Tool chains category

Assist  
tasks

Interaction  
modes

Prompt  
composition

Properties of  
model

LLM models





# Assist tasks

Finding information faster in context

Generating code

Reasoning about code

Transforming code into something ..

Requirement

Design

Develop

Testing

Deploy

Software Delivery Lifecycle



# Interaction modes

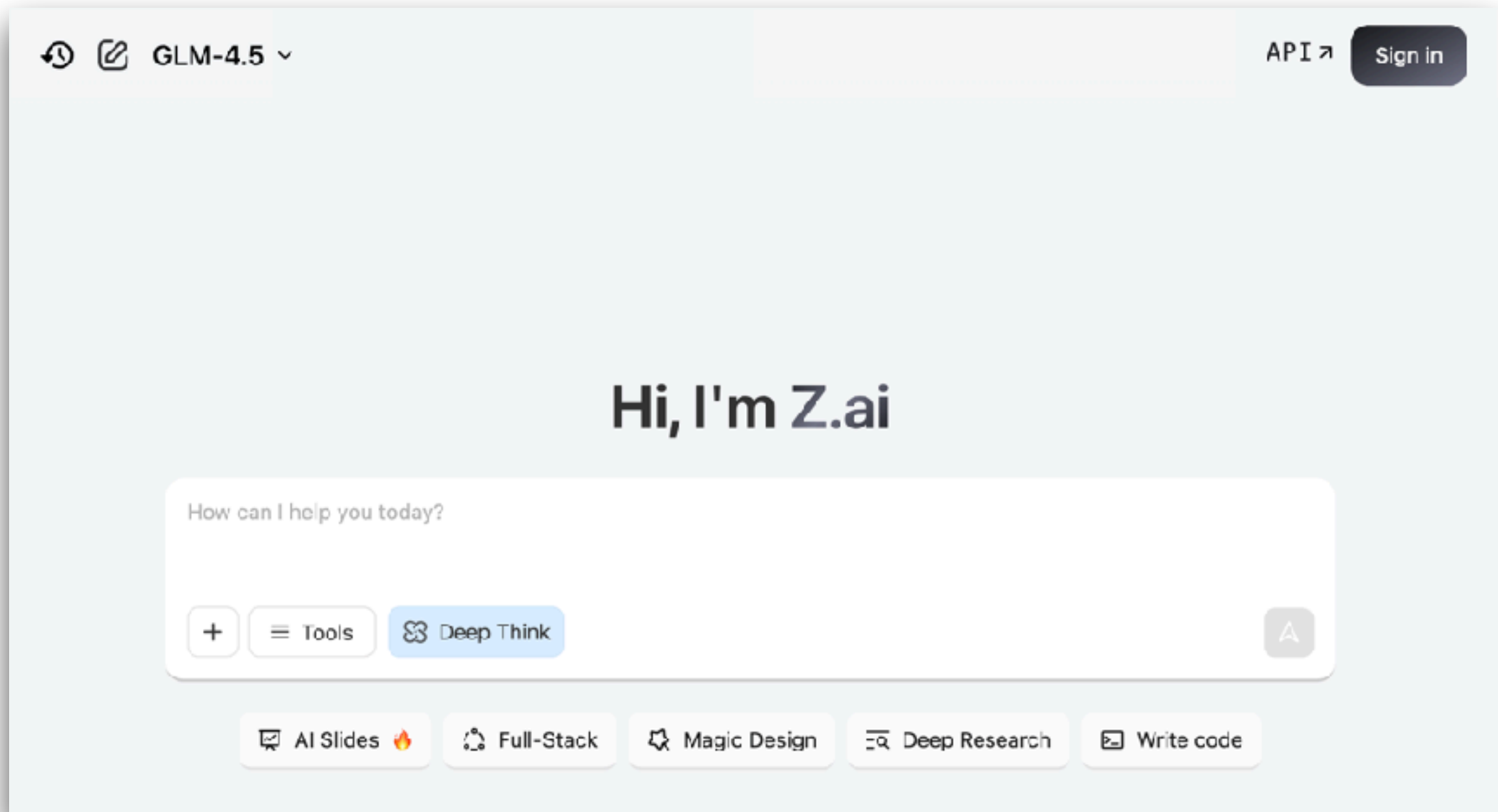
## Chat interfaces

In-line assistance (typing in code editor)

CLI (command-line interface)



# Z.ai



<https://z.ai/>



# Models

The screenshot displays the 'Models' section of the Z.ai website. It features a dark theme with a header containing three filter buttons: 'Language Models', 'Image Models', and 'Video Generation Models'. Below the filters, there are five model cards, each marked with a 'New' badge in the top right corner. Each card includes a category icon (purple dot for Reasoning, yellow square for Visual Reasoning, and teal dot for Language), the model name, a brief description, and a 'Learn More' link with a right-pointing arrow.

- GLM-4.6** (Reasoning Model): Z.ai's latest model achieves SOTA among open-source models! Context window expanded to 200K. Brings you superior performance in real-world coding, reasoning, tool using and role-playing.
- GLM-4.5-Air** (Reasoning Model): Z.ai's new lightweight flagship model delivers SOTA performance with exceptional cost-effectiveness!
- GLM-4.5-Flash** (Reasoning Model): The free version of GLM-4.5 now stands as Z.ai's most powerful offering, delivering unparalleled performance at no cost.
- GLM-4.5V** (Visual Reasoning Model): Achieve the state-of-the-art (SOTA) performance among open-source VLMs of the same level in various benchmark tests.
- GLM-4-32B-0414-128K** (Language Model): A general-purpose, cost-efficient LLM for advanced Q&A, coding, search, and structured task automation across business and technical domains.

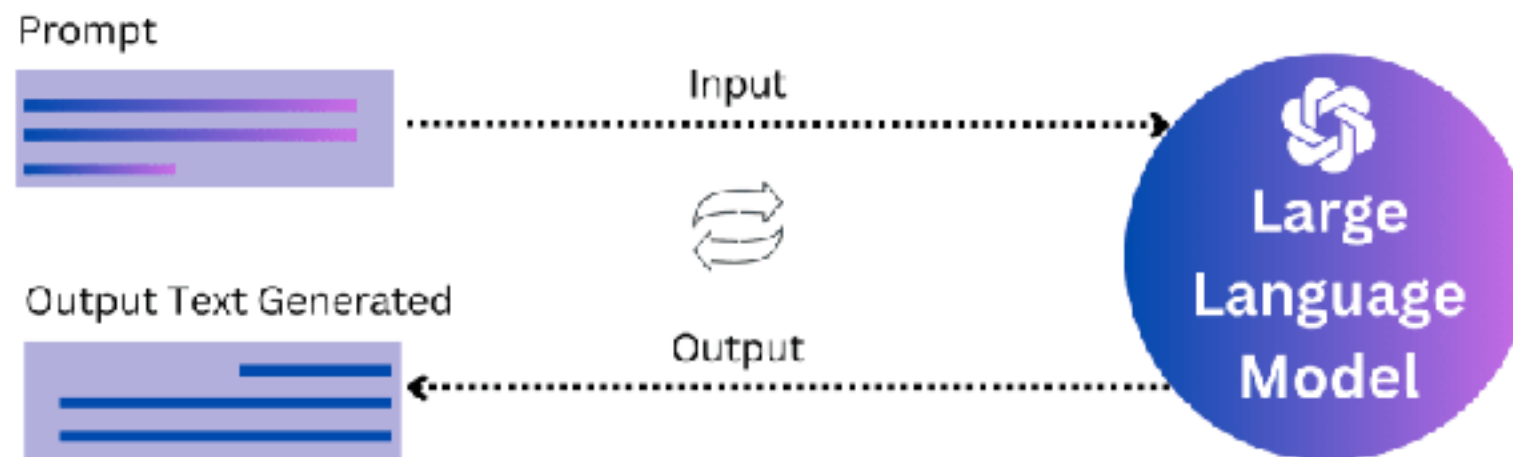
<https://z.ai/>



# Prompt composition

## Prompt engineering

Compose prompts from user inputs and context



<https://platform.openai.com/docs/guides/prompt-engineering>



# Better Prompt

Write clear instructions

Provide reference text

Split complex tasks into simpler subtasks

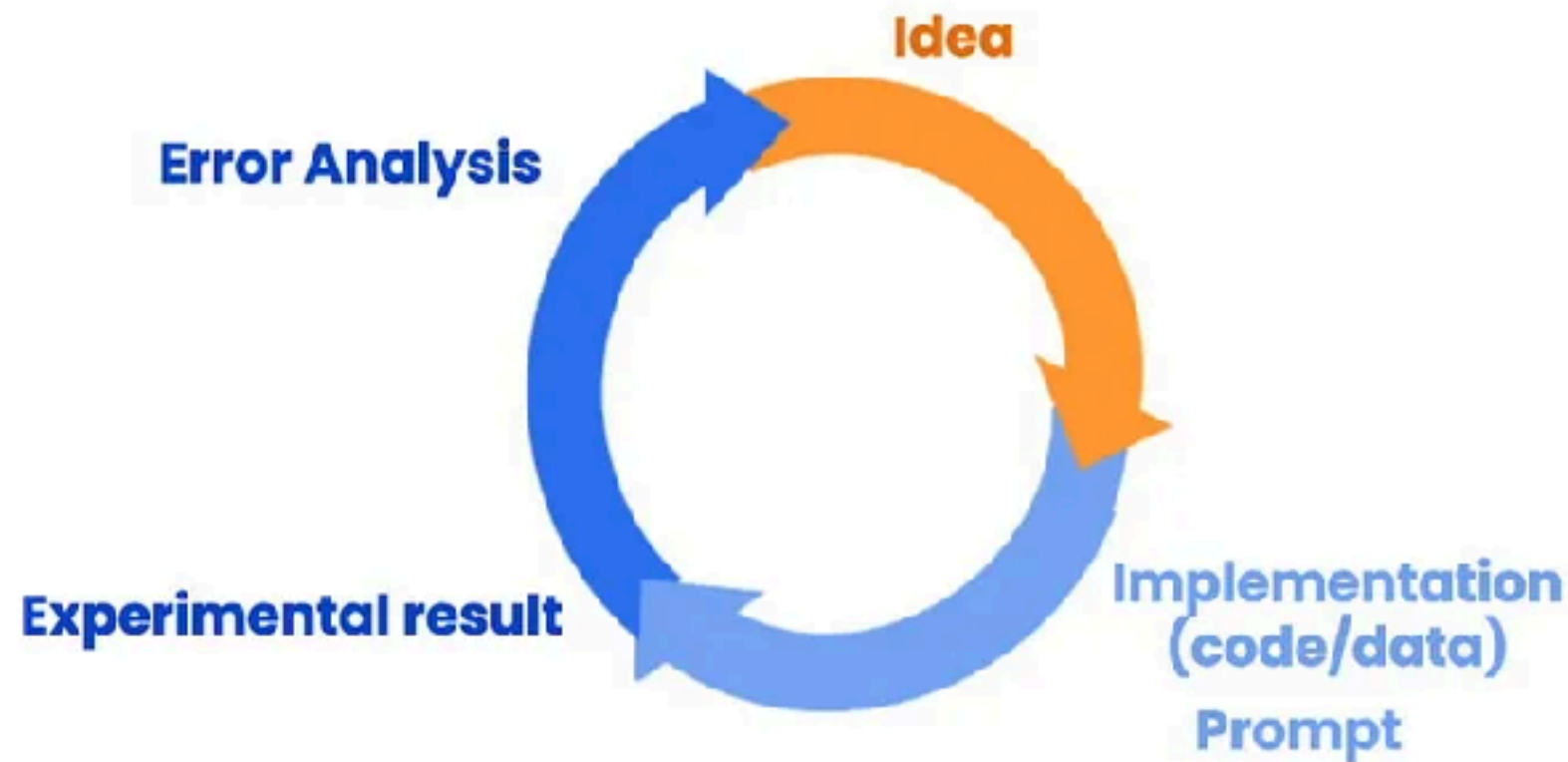
Give the model time to think

Testing and improve ...

<https://platform.openai.com/docs/guides/prompt-engineering>



# Iterative Prompt Development



## Iterative Process

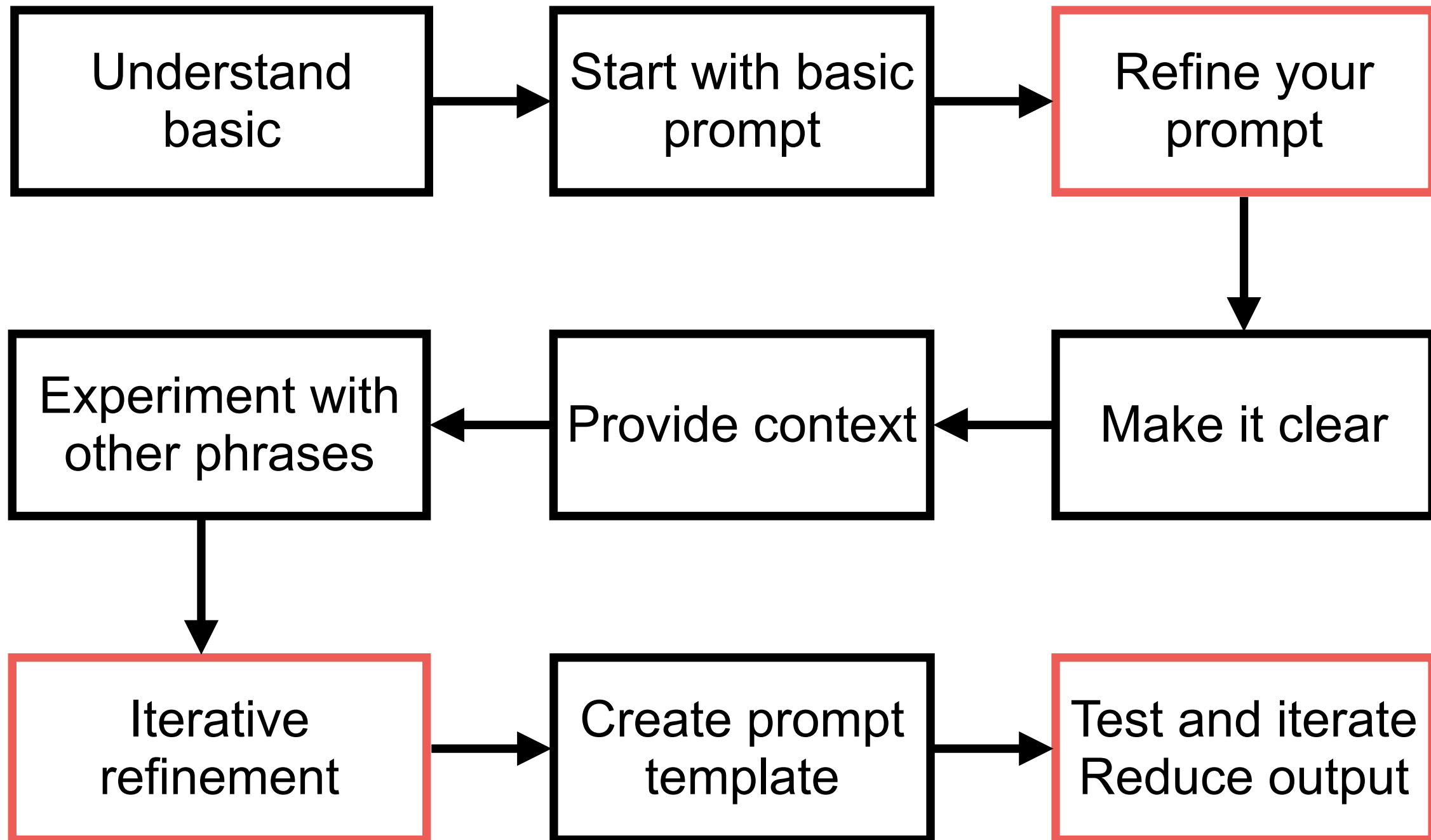
- Try something
- Analyze where the result does not give what you want
- Clarify instructions, give more time to think
- Refine prompts with a batch of examples

<https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/>





# Basic of Prompt Engineer



# Better Prompt

Write clear instructions

Provide reference text

Split complex tasks into simpler subtasks

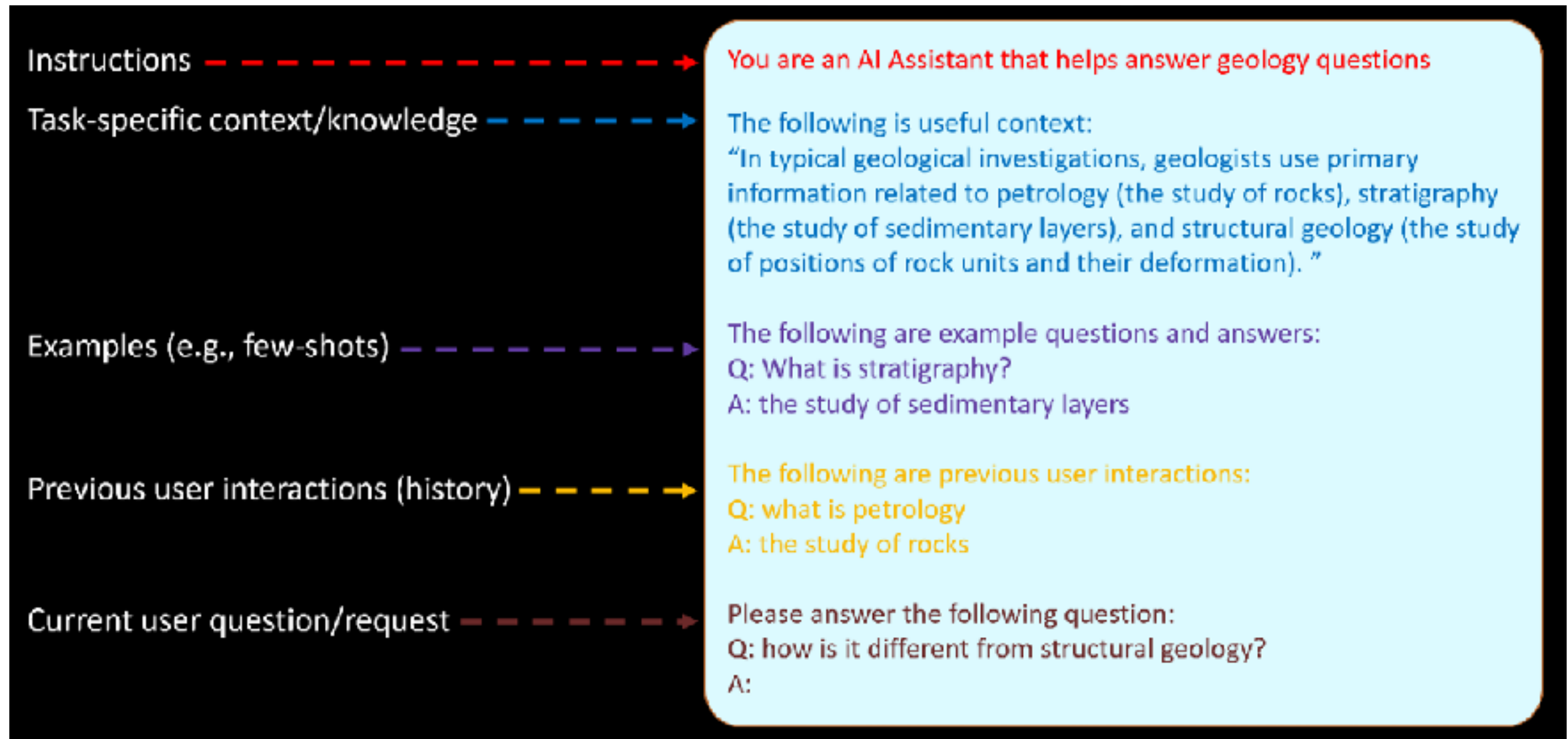
Give the model time to think

Testing and improve ...

<https://platform.openai.com/docs/guides/prompt-engineering/six-strategies-for-getting-better-results>



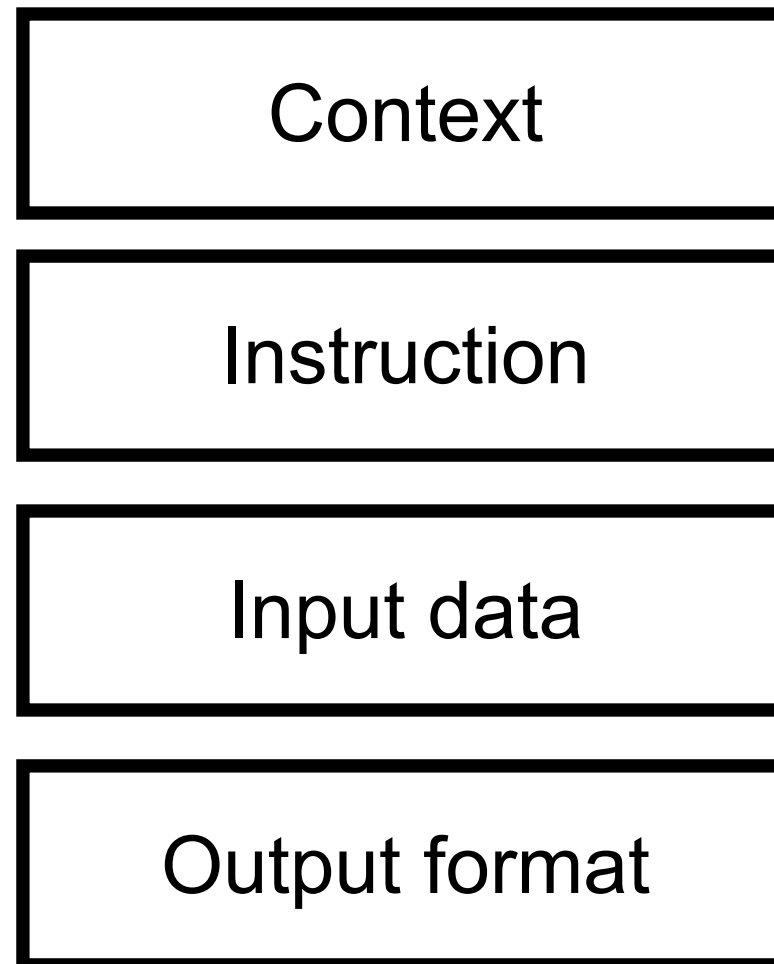
# Prompt Structure



<https://learn.microsoft.com/en-us/ai/playbook/technology-guidance/generative-ai/working-with-llms/prompt-engineering>



# Structure of Prompt



<https://platform.openai.com/docs/guides/prompt-engineering>



# Prompting Guide

Prompt Engineering

## Prompt Engineering Guide

Prompt engineering is a relatively new discipline for developing and optimizing prompts to efficiently use language models (LMs) for a wide variety of applications and research topics. Prompt engineering skills help to better understand the capabilities and limitations of large language models (LLMs).

Researchers use prompt engineering to improve the capacity of LLMs on a wide range of common and complex tasks such as question answering and arithmetic reasoning. Developers use prompt engineering to design robust and effective prompting techniques that interface with LLMs and other tools.

Prompt engineering is not just about designing and developing prompts. It encompasses a wide range of skills and techniques that are useful for interacting and developing with LLMs. It's an important skill to interface, build with, and understand capabilities of LLMs. You can use prompt engineering to improve safety of LLMs and build new capabilities like augmenting LLMs with domain knowledge and external tools.

<https://www.promptingguide.ai/>



# Structure of Prompt !!

## **APE**

Action, Purpose,  
Expectation

## **RACE**

Role, Action,  
Context,  
Expectation

## **TAG**

Task, Action, Goal

## **COAST**

Context, Objective,  
Action, Scenario,  
Task

## **RISE**

Role, Input, Step,  
Expectation

<https://twitter.com/pradeepeth/status/1673271866696544257>



# Prompt Techniques

Zero-shot

Chain-of  
Thought (CoT)

Few-shot

Meta or structure

<https://www.promptingguide.ai/techniques>





# Chain of Thought Prompting (CoT)

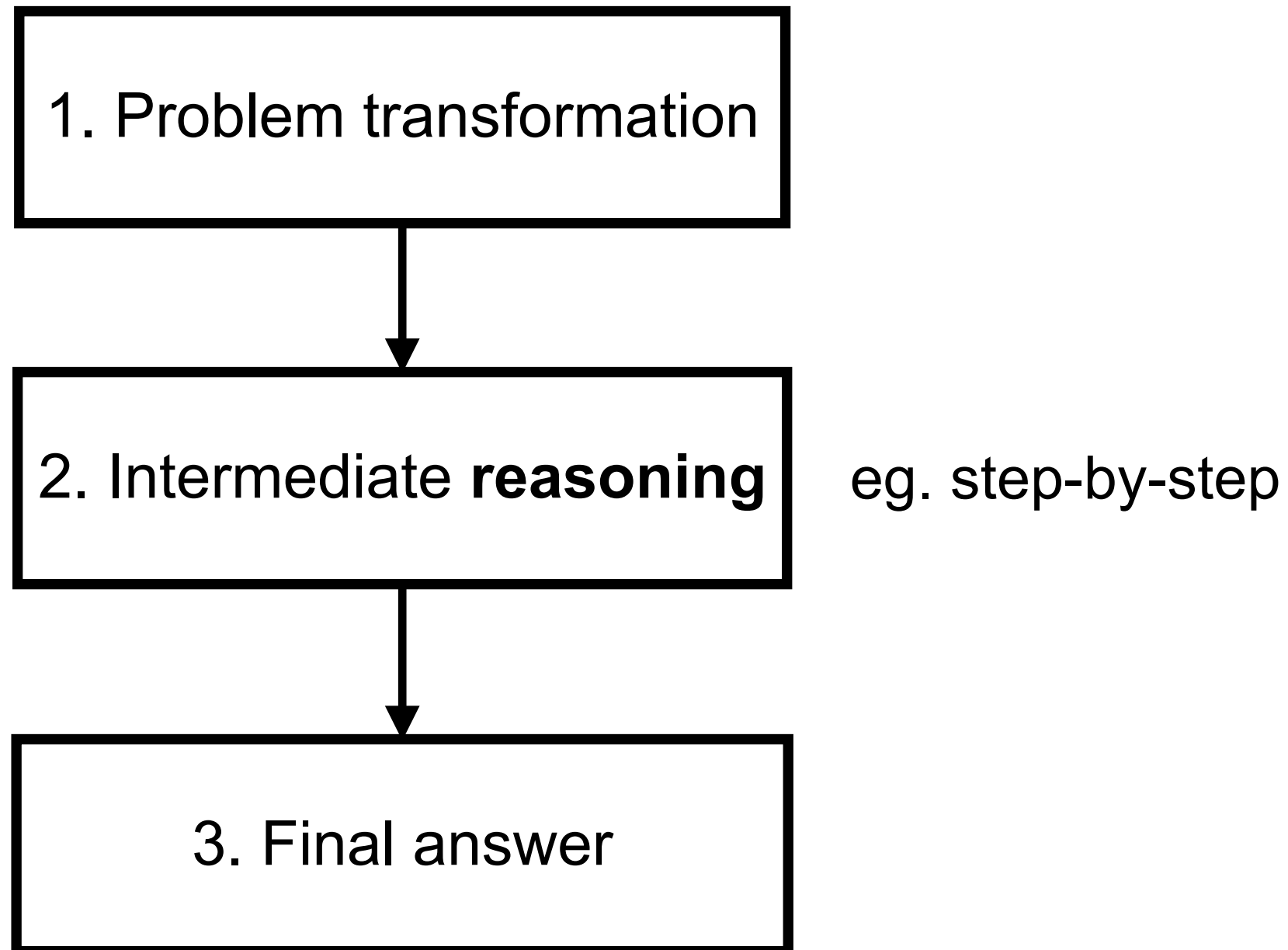
Technique used to improve the reasoning ability of LLM

Try to break down a complex problem into smaller, More manageable steps, lead to final answer

Reasoning model !!



# Chain of Thought Prompting (CoT)



# Chain of Thought Prompting (CoT)

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls.  $5 + 6 = 11$ . The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are  $16 / 2 = 8$  golf balls. Half of the golf balls are blue. So there are  $8 / 2 = 4$  blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

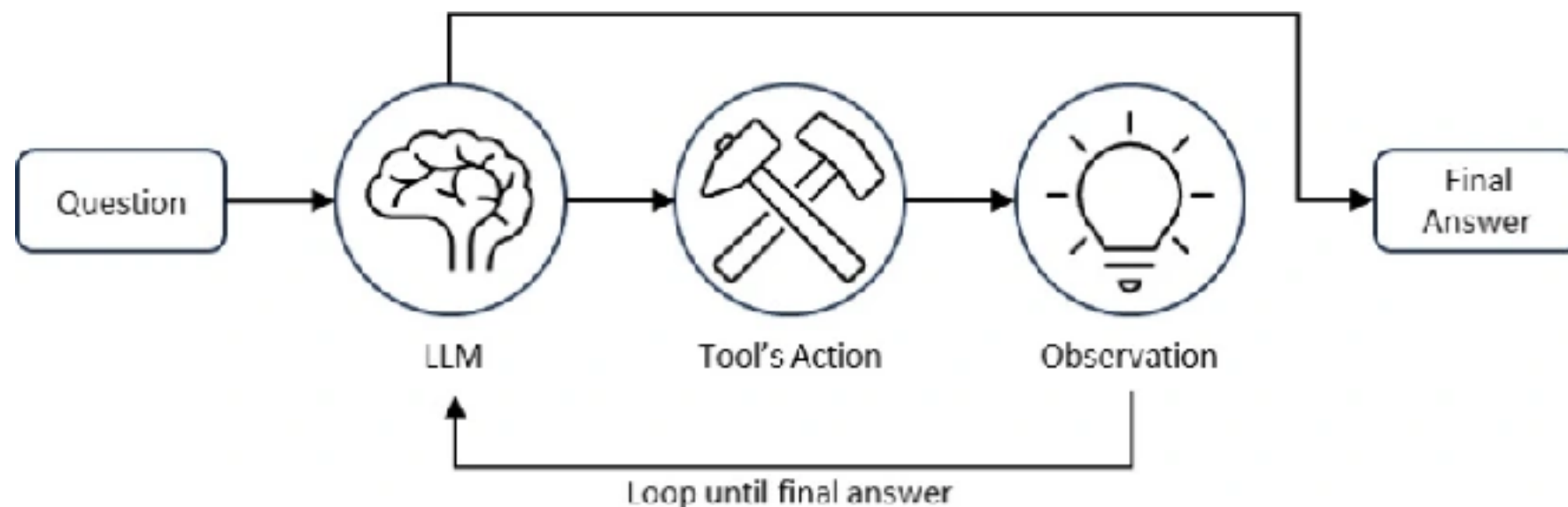
(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

<https://www.promptingguide.ai/techniques/cot>



# ReAct Prompt

LLM reasoning and additional tools (expert)  
Improve better answer



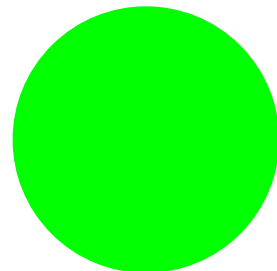
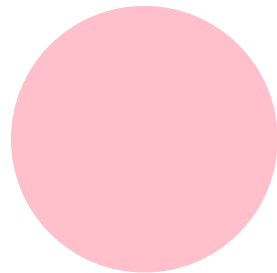
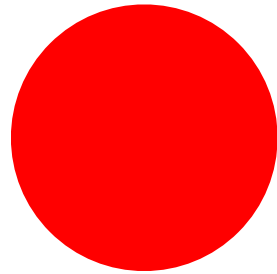
<https://www.promptingguide.ai/techniques/react>



# Workshop



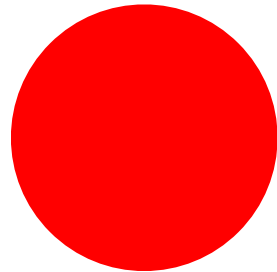
# RGB ?



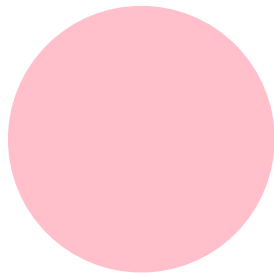
<https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb>



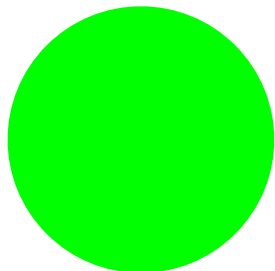
# RGB ?



[255, 0, 0]



[255, 192, 203]



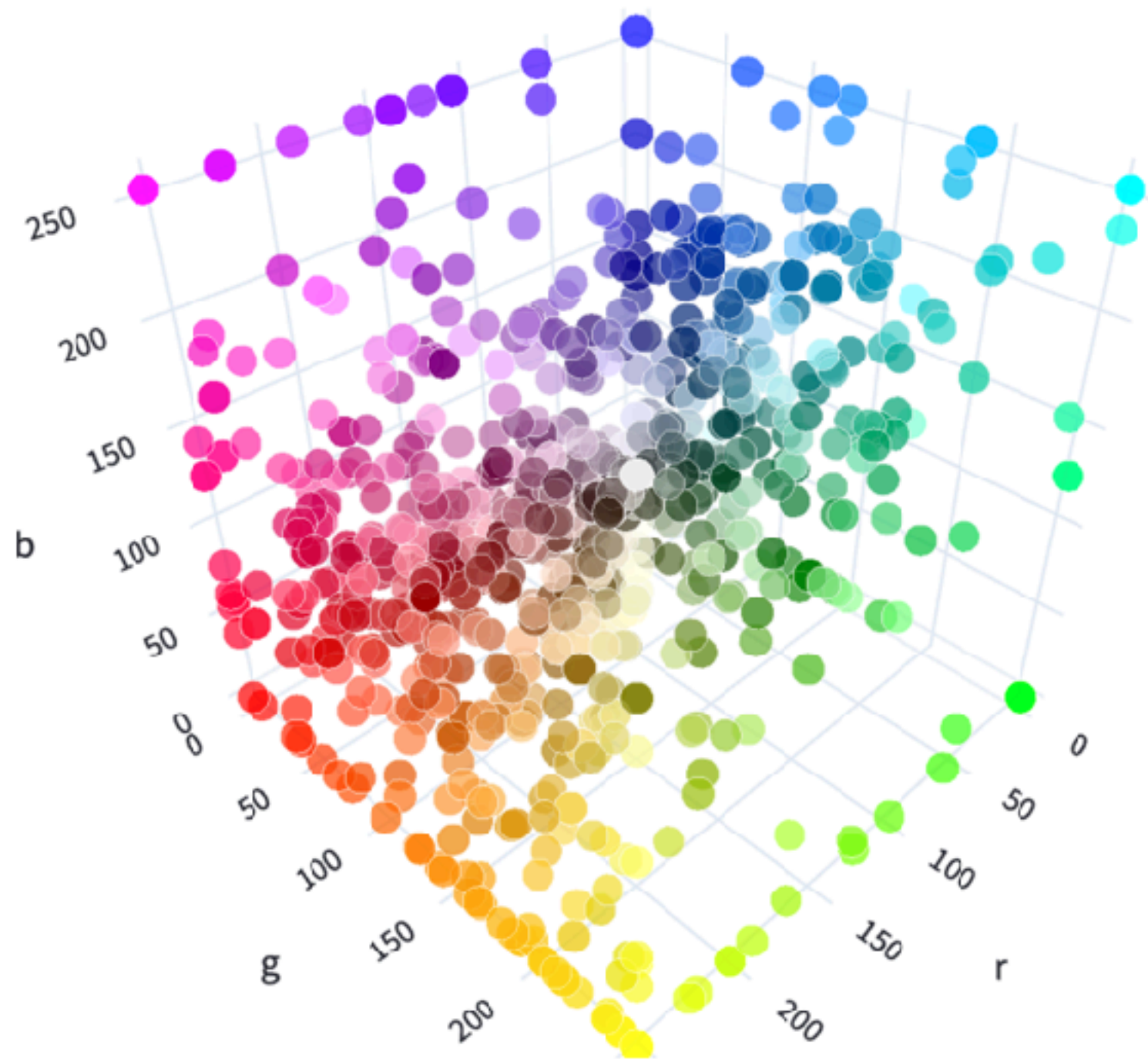
[0, 255, 0]

<https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-rgb>





# Visualize RGB

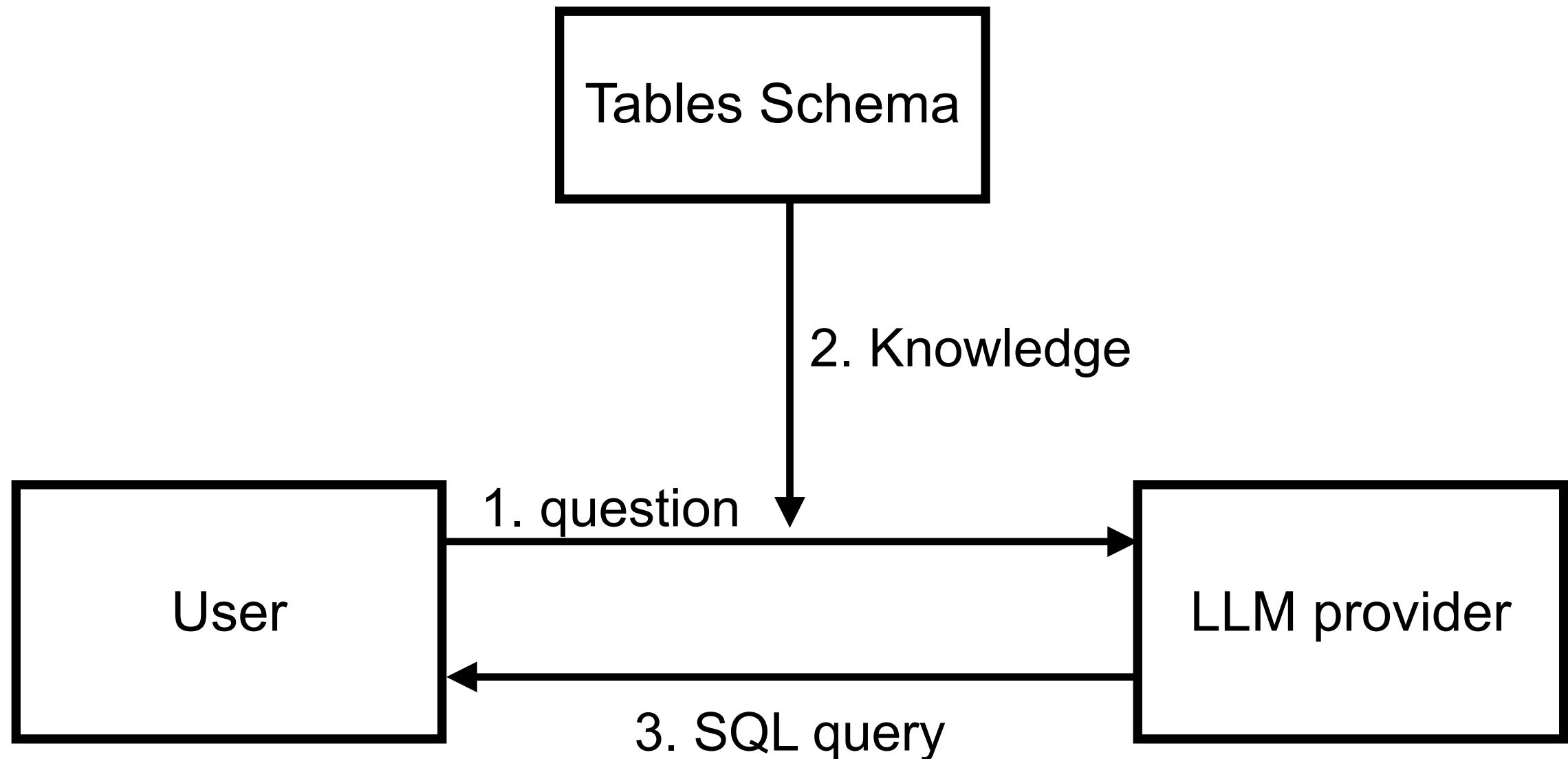


[https://huggingface.co/spaces/jphwang/colorful\\_vectors](https://huggingface.co/spaces/jphwang/colorful_vectors)





# Text-to-SQL



<https://github.com/up1/workshop-ai-with-technical-team/tree/main/workshop/demo-sql>



# Software Development Life Cycle



# SDLC

Requirement

Design

Develop

Testing

Deploy



## Planning

Written planning process

Arch diagrams

## Testing

Automated testing

TDD

Testing environments

Testing in prod

Performance testing

Load testing

Generate test data

## Shipping

FF & experimentation

Logging

Monitoring & alerting

Staged rollouts

## Development

Automated dev env

CI/CD

Prototyping

Code review

Code generation

Templates

Cross-platform dev

Preview env

Post-commit code review

Linting

Static code analysis

Project mgmt

## Maintenance

Debug production

Documentation

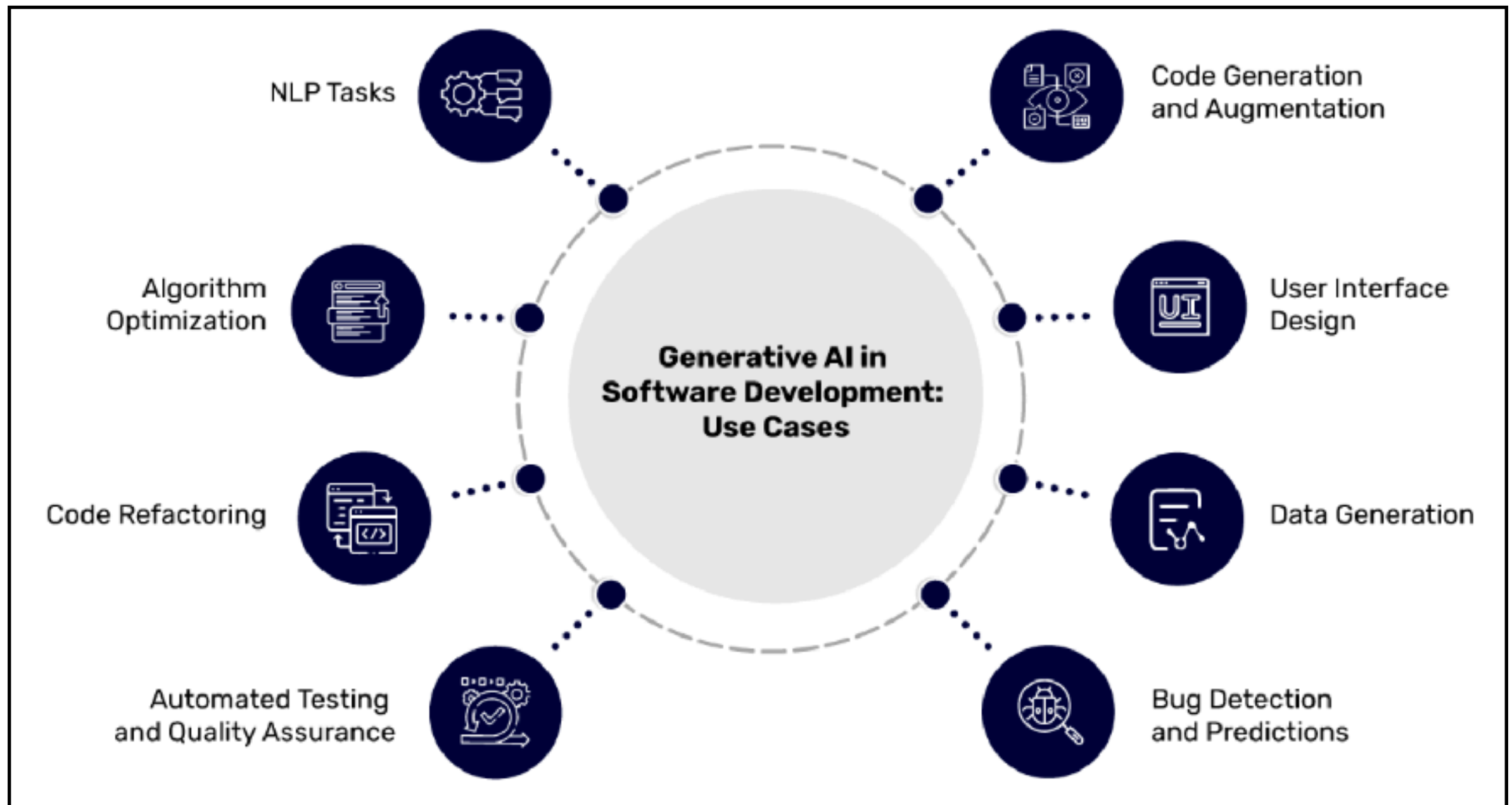
Runbook examples

Mitigation runbook

[pragmaticengineer.com](https://pragmaticengineer.com)



# SDLC



# Impacts with productivity ?

Automated  
simple tasks

Improve quality  
and reliability

Improve  
communication

Faster  
prototype



**Generative AI isn't just a tool**  
**it's your team member**

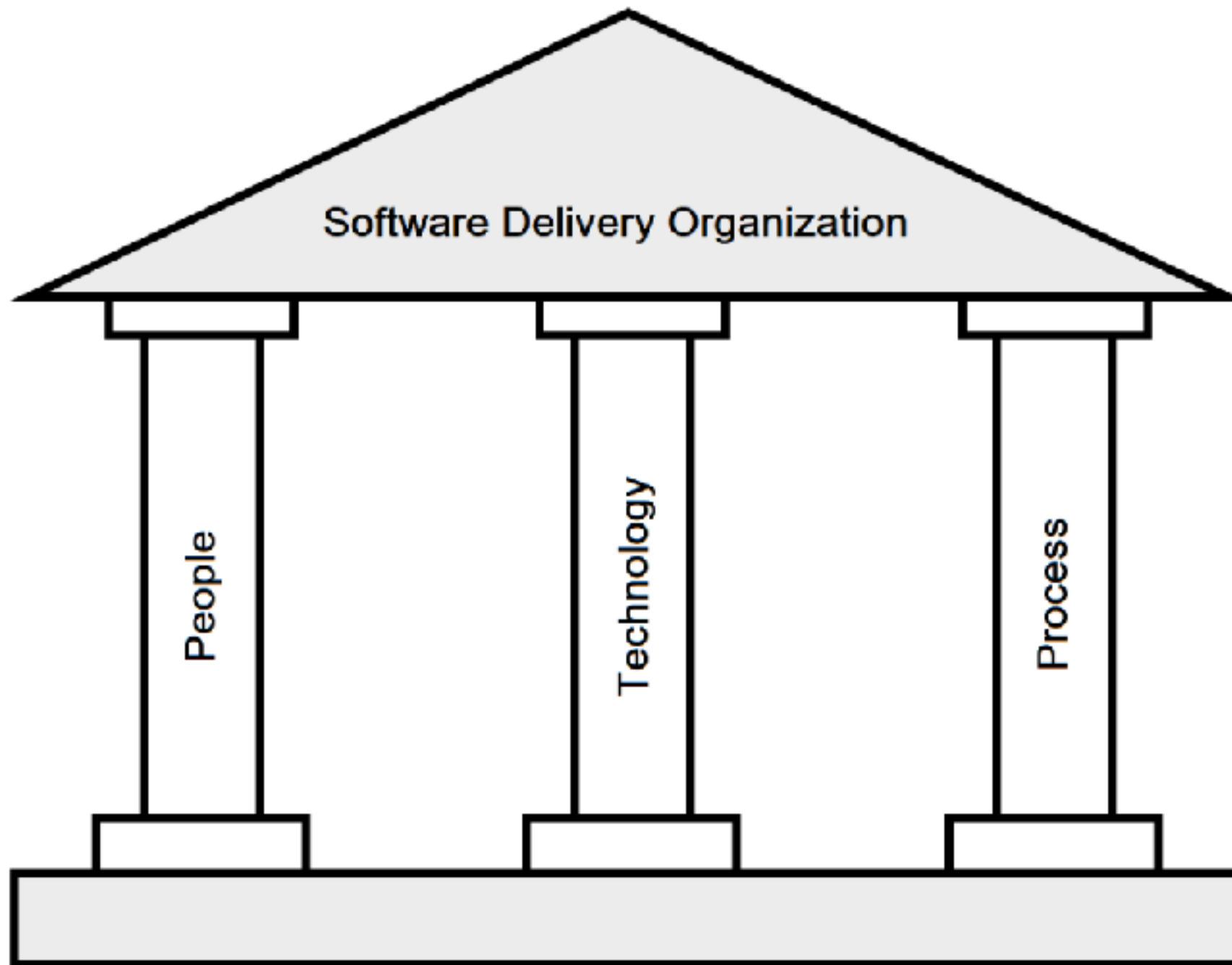




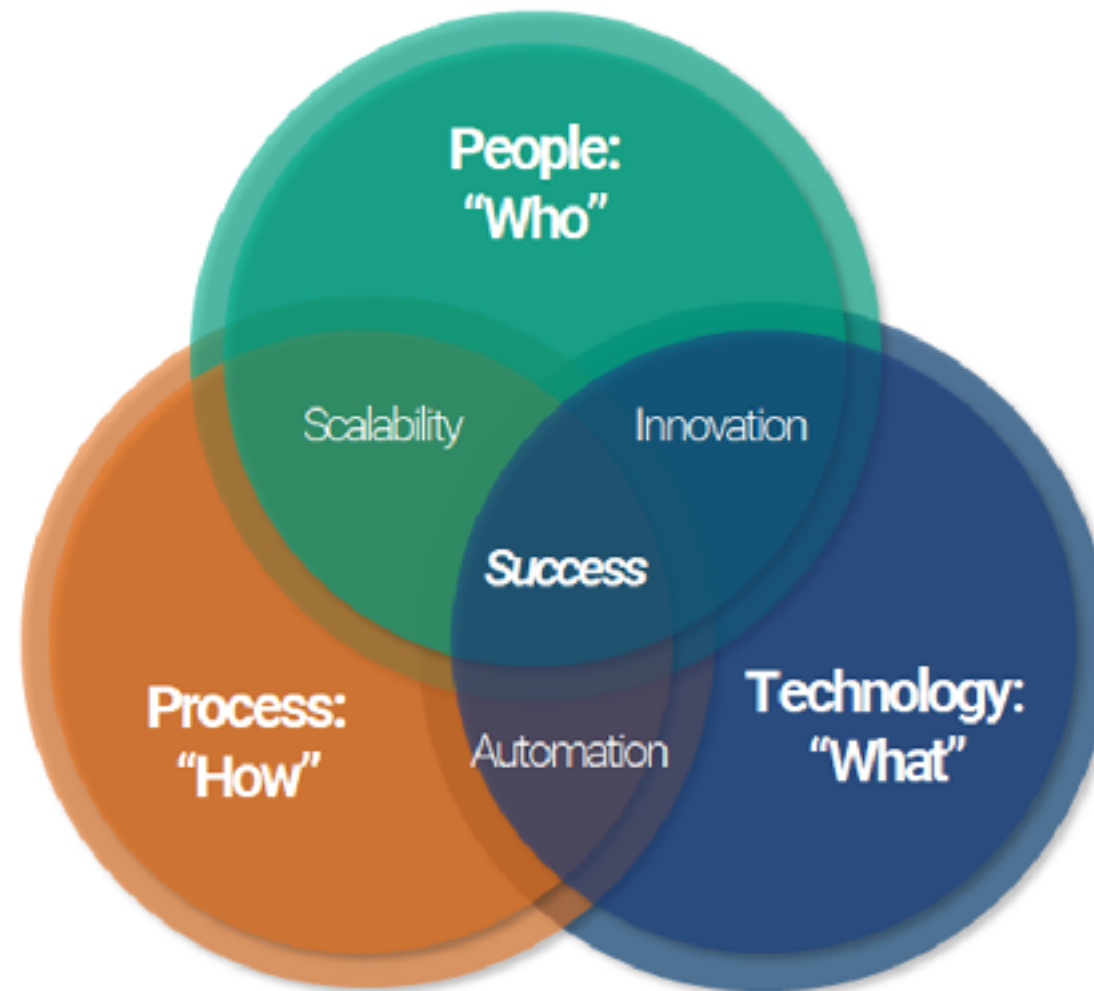
**Trust, but *verify* output !!**



# 3 Pillar of Software Development



# 3 Pillar of Software Development



# Requirement and Analysis



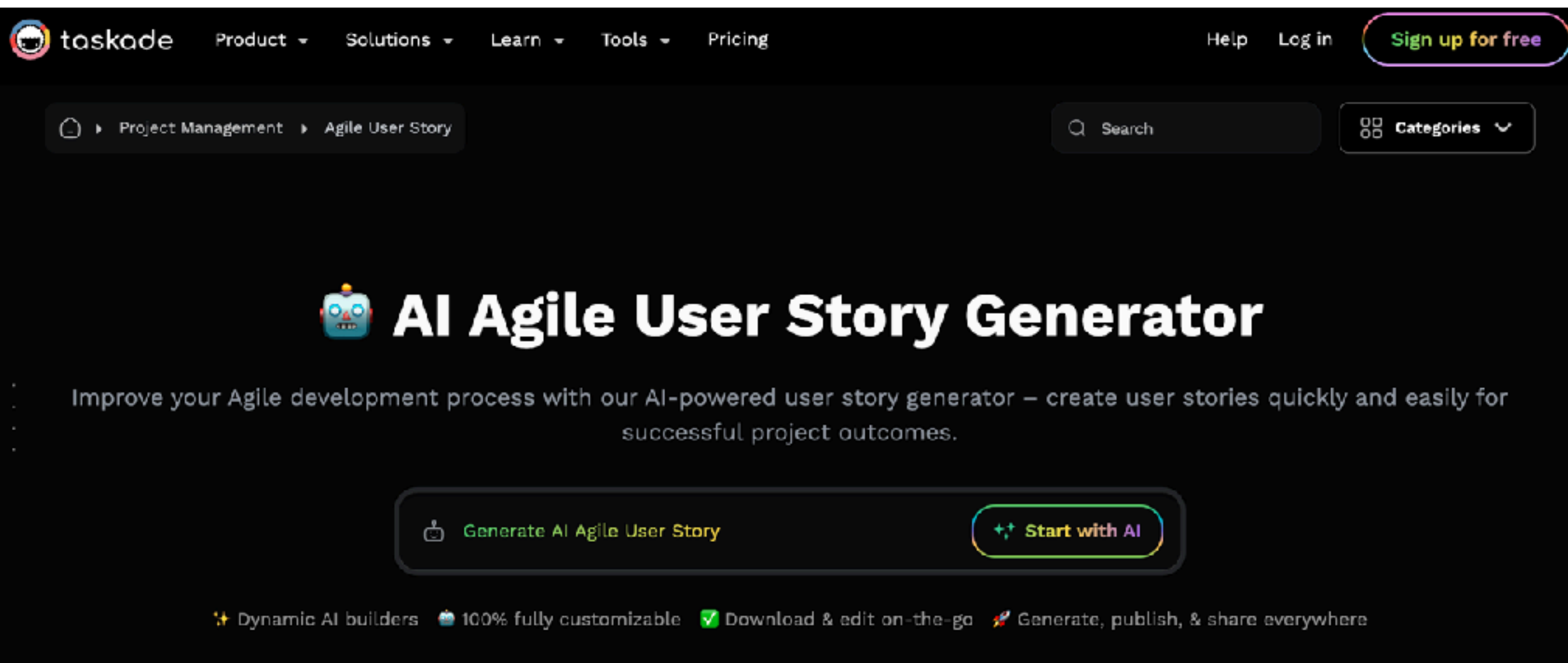
# Requirement and Analysis



Requirements writing and analysis  
User story generation



# Agents AI for Automated tasks



The screenshot shows the Taskade website's navigation bar with links for Product, Solutions, Learn, Tools, and Pricing. A 'Sign up for free' button is in the top right. Below the navigation bar, a breadcrumb trail reads 'Project Management > Agile User Story'. A search bar and a 'Categories' dropdown are also visible. The main heading is 'AI Agile User Story Generator', accompanied by a robot icon. Below this, a descriptive text states: 'Improve your Agile development process with our AI-powered user story generator – create user stories quickly and easily for successful project outcomes.' Two primary action buttons are present: 'Generate AI Agile User Story' and 'Start with AI'. At the bottom, a row of features is listed: 'Dynamic AI builders', '100% fully customizable', 'Download & edit on-the-go', and 'Generate, publish, & share everywhere'.

taskade Product Solutions Learn Tools Pricing Help Log in Sign up for free

Project Management > Agile User Story Search Categories

## AI Agile User Story Generator

Improve your Agile development process with our AI-powered user story generator – create user stories quickly and easily for successful project outcomes.

 Generate AI Agile User Story  Start with AI

✦ Dynamic AI builders 🤖 100% fully customizable ✅ Download & edit on-the-go 🚀 Generate, publish, & share everywhere

<https://www.taskade.com/generate/project-management/agile-user-story>



# Example with food delivery

## Food Delivery Workflow Template

 **Order Processing #order**

- ☐ Check for new orders
  - ☐ Verify customer details
  - ☐ Confirm payment status
- ☐ Prepare order items
  - ☐ Gather ingredients
  - ☐ Cook or prepare food
  - ☐ Package items securely

 **Delivery Management #delivery**

- ☐ Assign delivery driver
- ☐ Plan delivery route
  - ☐ Prioritize multiple deliveries
  - ☐ Use GPS for directions
- ☐ Confirm delivery with customer
  - ☐ Send delivery notification
  - ☐ Obtain customer signature

 **Post-Delivery Tasks #postdelivery**

 What would you like to do next?

 Create project

 Continue writing

 Make longer



<https://www.taskade.com/generate/project-management/agile-user-story>

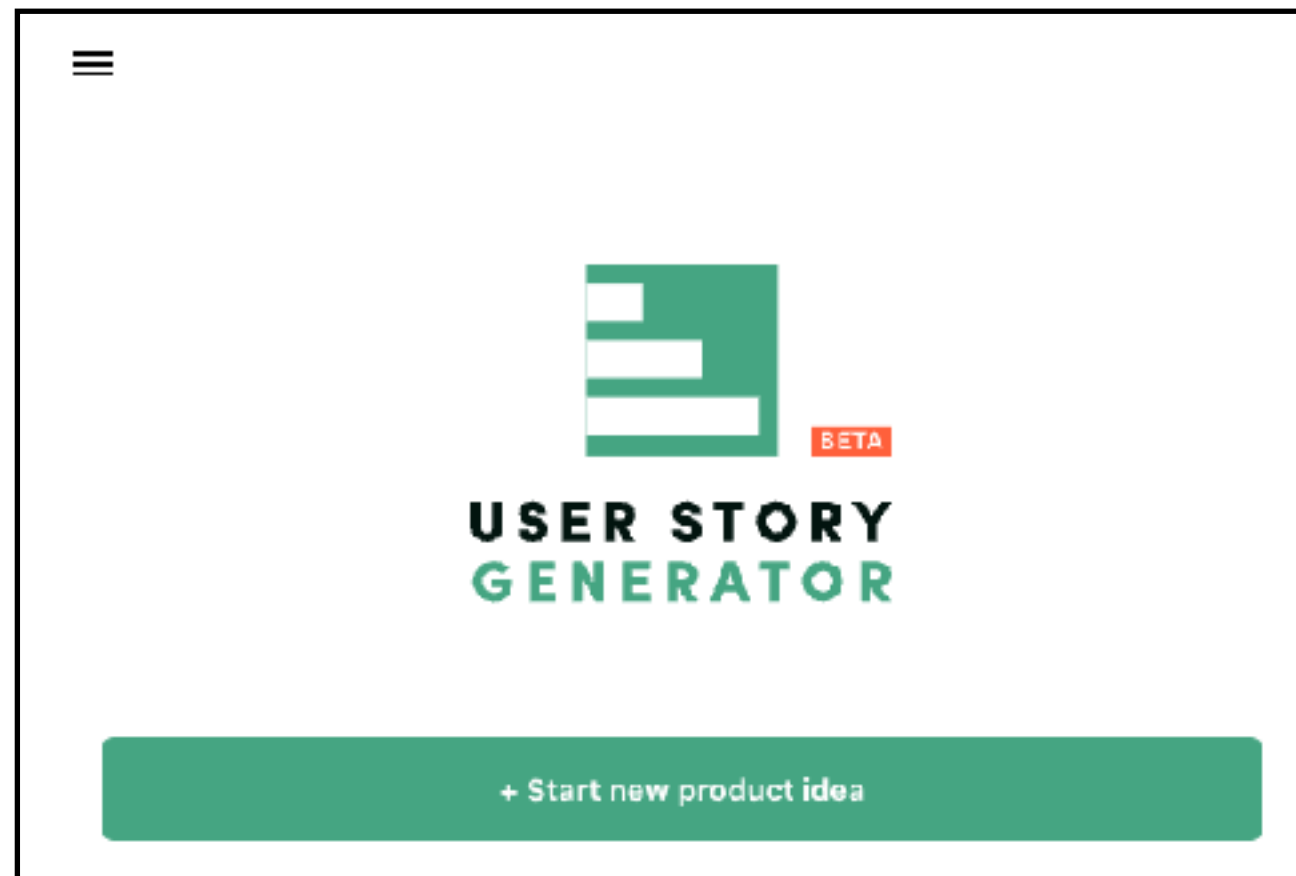
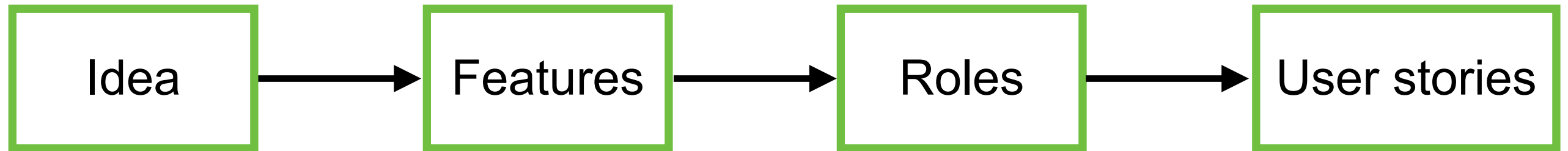
© 2020 - 2026 Siam Chamnankit Company Limited. All rights reserved.

AI for Software Development

72



# User Story Generator

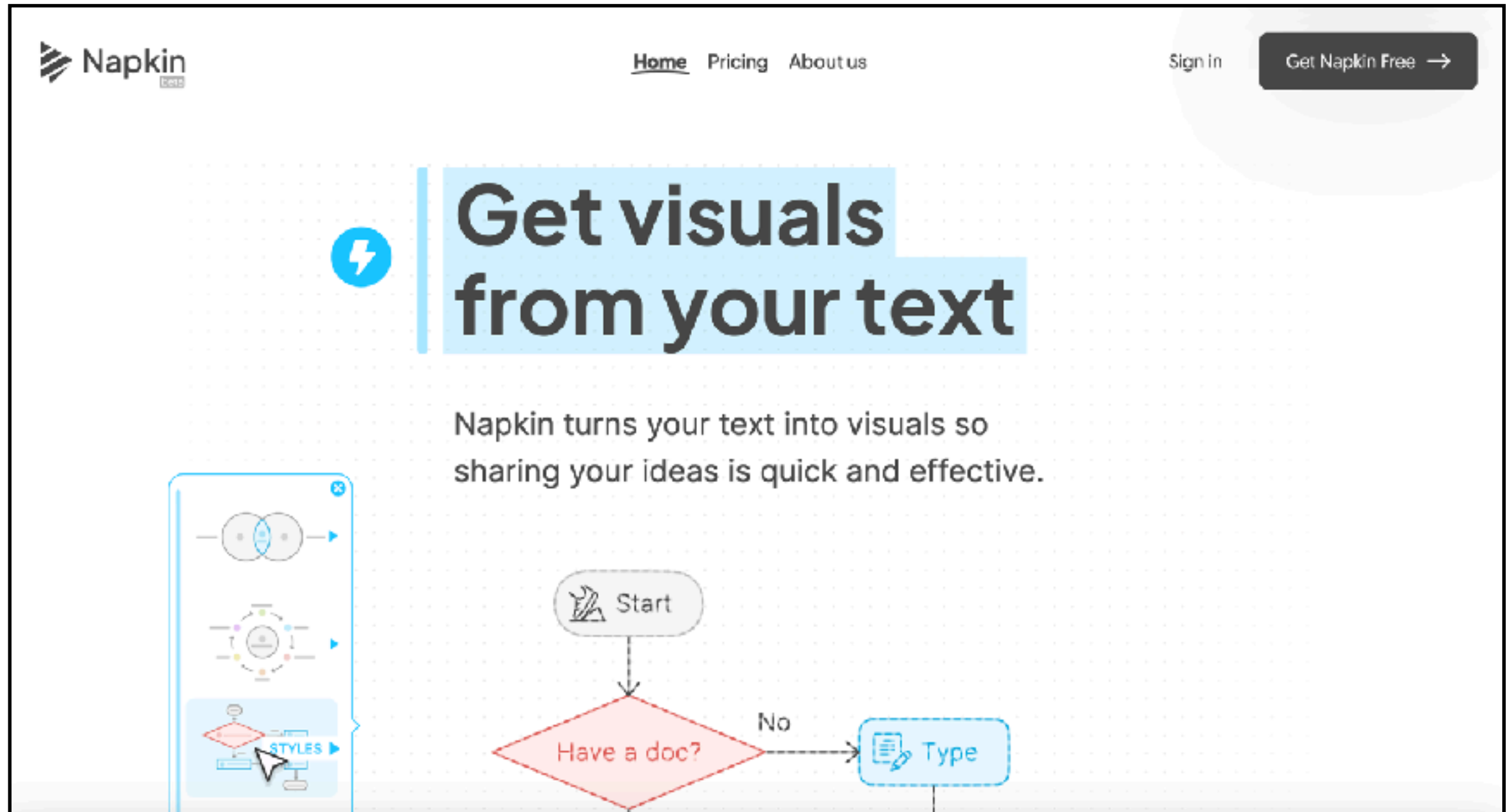


<https://userstorygenerator.ai/>





# Napkin



The image shows the Napkin AI website. At the top left is the Napkin logo. To its right are links for Home, Pricing, and About us. Further right are links for Sign in and a button labeled 'Get Napkin Free' with a right arrow. The main heading is 'Get visuals from your text', preceded by a blue lightning bolt icon. Below this is the text 'Napkin turns your text into visuals so sharing your ideas is quick and effective.' On the left side, there is a vertical panel showing three types of visualizations: a Venn diagram, a circular flow diagram, and a flowchart. The flowchart is highlighted with a blue box and a mouse cursor pointing to it, with the word 'STYLES' next to it. Below the panel, a sample flowchart is displayed on a grid background. The flowchart starts with an oval labeled 'Start', followed by a diamond decision box labeled 'Have a doc?'. An arrow labeled 'No' points from the 'Have a doc?' box to a rounded rectangle labeled 'Type'.

**Get visuals from your text**

Napkin turns your text into visuals so sharing your ideas is quick and effective.

Start

Have a doc?

No

Type

<https://www.napkin.ai/>



# Requirement analysis

Clarify of User requirement ?

<https://github.com/up1/workshop-ai-with-technical-team/wiki/Requirement-analysis>



# Data Analysis

The screenshot shows the Julius AI website. At the top is a navigation bar with links for Product, Use Cases, Resources, Security, Connectors, Community, and Pricing, along with Log In and Sign Up buttons. The main heading is "Add data, ask questions" with the subtext "Ask for what you want and Julius analyzes the data for you". Below this is a three-step process:

- 1 Connect all your data sources**  
Connect with data sources like databases, spreadsheets, and more. The visual shows icons for Google Drive, XLSX, PDF, and a database.
- 2 Ask for analysis**  
You provide the questions, Julius handles the analysis. The visual shows a text box with the example question: "Predict customer churn from purchase history".
- 3 Get results, instantly**  
Choose from charts, tables or full reports tailored to your data. The visual shows various data visualization charts.

At the bottom of the process is a blue button that says "Get started for free >".

<https://julius.ai/>



# Design Process



# Design



Architecture writing assistance

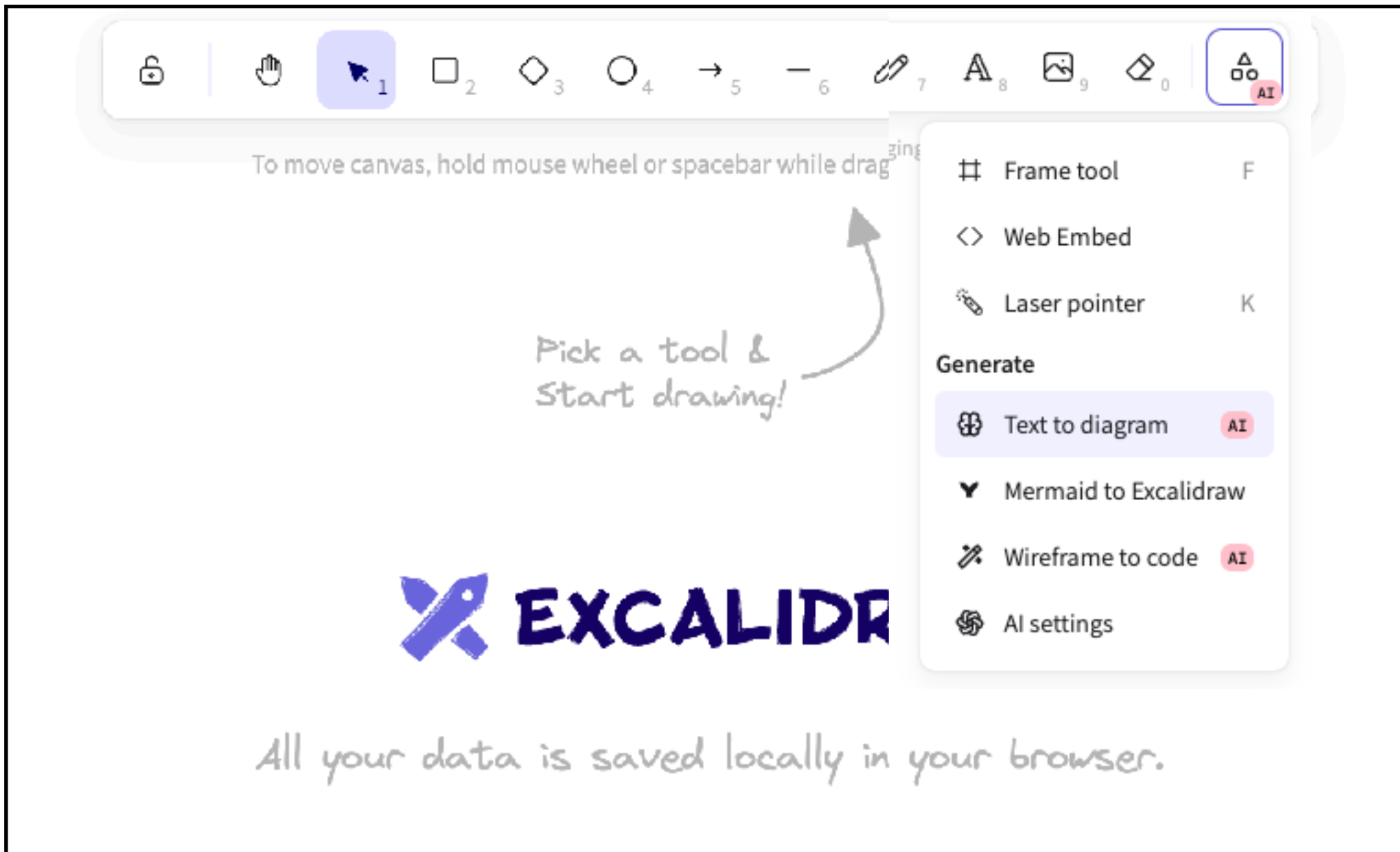
Sequence flow diagram generation

Data modeling

UX/UI design assistance



# Excalidraw with AI



<https://excalidraw.com/>



# Demo

Text to diagram **AI Builder** Mermaid

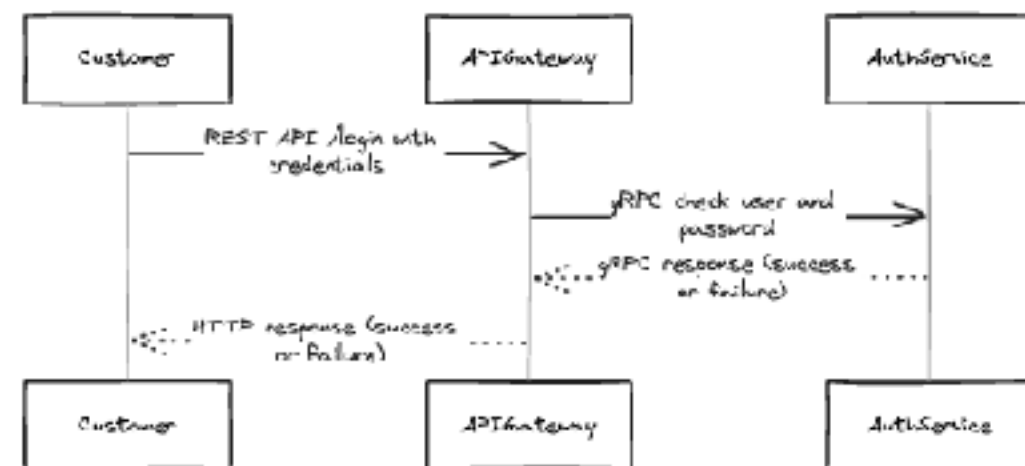
Currently we use Mermaid as a middle step, so you'll get best results if you describe a diagram, workflow, flow chart, and similar.

Prompt

9 requests left today

Try to generate authentication service from below  
1. Customer call api gateway with REST API /login  
2. Api gateway check user and password from auth service via gRPC  
3. API gateway send response to client

Preview



Generate →

Cmd Enter

View as Mermaid →

Insert →





# Mermaid Diagram

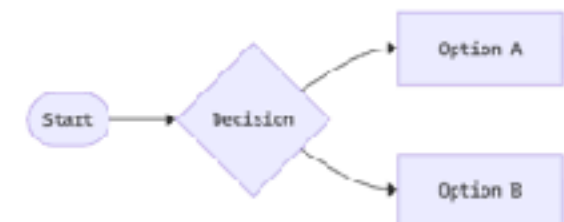
## Mermaid Diagramming and charting tool

JavaScript based diagramming and charting tool that renders Markdown-inspired text definitions to create and modify diagrams dynamically.

Try Editor

Get started

```
1 flowchart LR
2   A(["Start"])
3   A --> B{"Decision"}
4   B --> C["Option A"]
5   B --> D["Option B"]
```

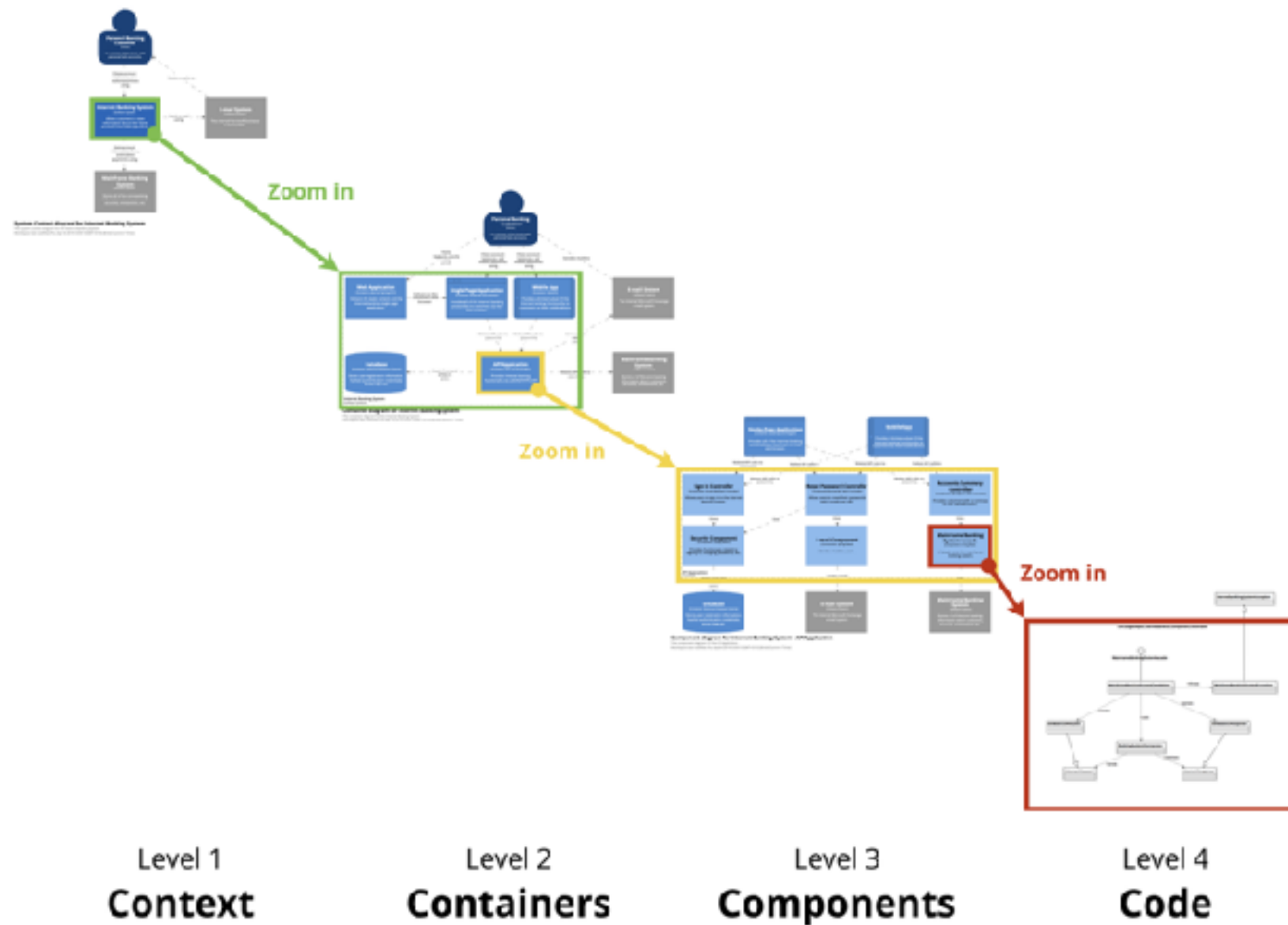


<https://mermaid.js.org/>





# Software Architecture



<https://c4model.com/>



# Database

The screenshot shows the Database Build interface. On the left, there's a sidebar with a 'New database' button and a list of databases: 'User Management Database ...' and 'User and Shipping Address ...'. The main area is titled 'Diagram' and shows a database diagram with two tables: 'shipping\_addresses' and 'users'. The 'shipping\_addresses' table has columns: 'id', 'user\_id', 'address\_line1', 'address\_line2', 'city', 'state', 'postal\_code', and 'country'. The 'users' table has columns: 'id', 'name', and 'email'. A message on the right says: 'Create table of user that have many shipping address'. Below this, there's a section titled 'Executed SQL' with a dropdown arrow. The text below says: 'The tables `users` and `shipping\_addresses` have been created successfully. Here's a brief overview:'. There are two bullet points: '• **Users Table:** Contains user information with columns for `id`, `name`, and `email`.' and '• **Shipping Addresses Table:** Stores multiple addresses for each user, with columns for `id`, `user\_id` (foreign key referencing `users`), `address\_line1`, `address\_line2`, `city`, `state`, `postal\_code`, and `country`.' At the bottom right, there's a message: 'Conversation renamed to User and Shipping Address Management change'. At the bottom left, there's a 'Toggle theme' button. At the bottom center, there's a status bar showing 'PG 18 | Local-only database' and icons for 'Primary key', 'Identity', 'Unique', and 'Nullable'. At the bottom right, there's a button that says 'Message AI or write SQL' with an upward arrow icon.

New database

User Management Database ...

User and Shipping Address ...

Diagram

Migrations

shipping\_addresses

| Column        | Type |
|---------------|------|
| id            | int8 |
| user_id       | int8 |
| address_line1 | text |
| address_line2 | text |
| city          | text |
| state         | text |
| postal_code   | text |
| country       | text |

users

| Column | Type |
|--------|------|
| id     | int8 |
| name   | text |
| email  | text |

Create table of user that have many shipping address

Executed SQL

The tables `users` and `shipping\_addresses` have been created successfully. Here's a brief overview:

- **Users Table:** Contains user information with columns for `id`, `name`, and `email`.
- **Shipping Addresses Table:** Stores multiple addresses for each user, with columns for `id`, `user\_id` (foreign key referencing `users`), `address\_line1`, `address\_line2`, `city`, `state`, `postal\_code`, and `country`.

Conversation renamed to User and Shipping Address Management change

Toggle theme

PG 18 | Local-only database

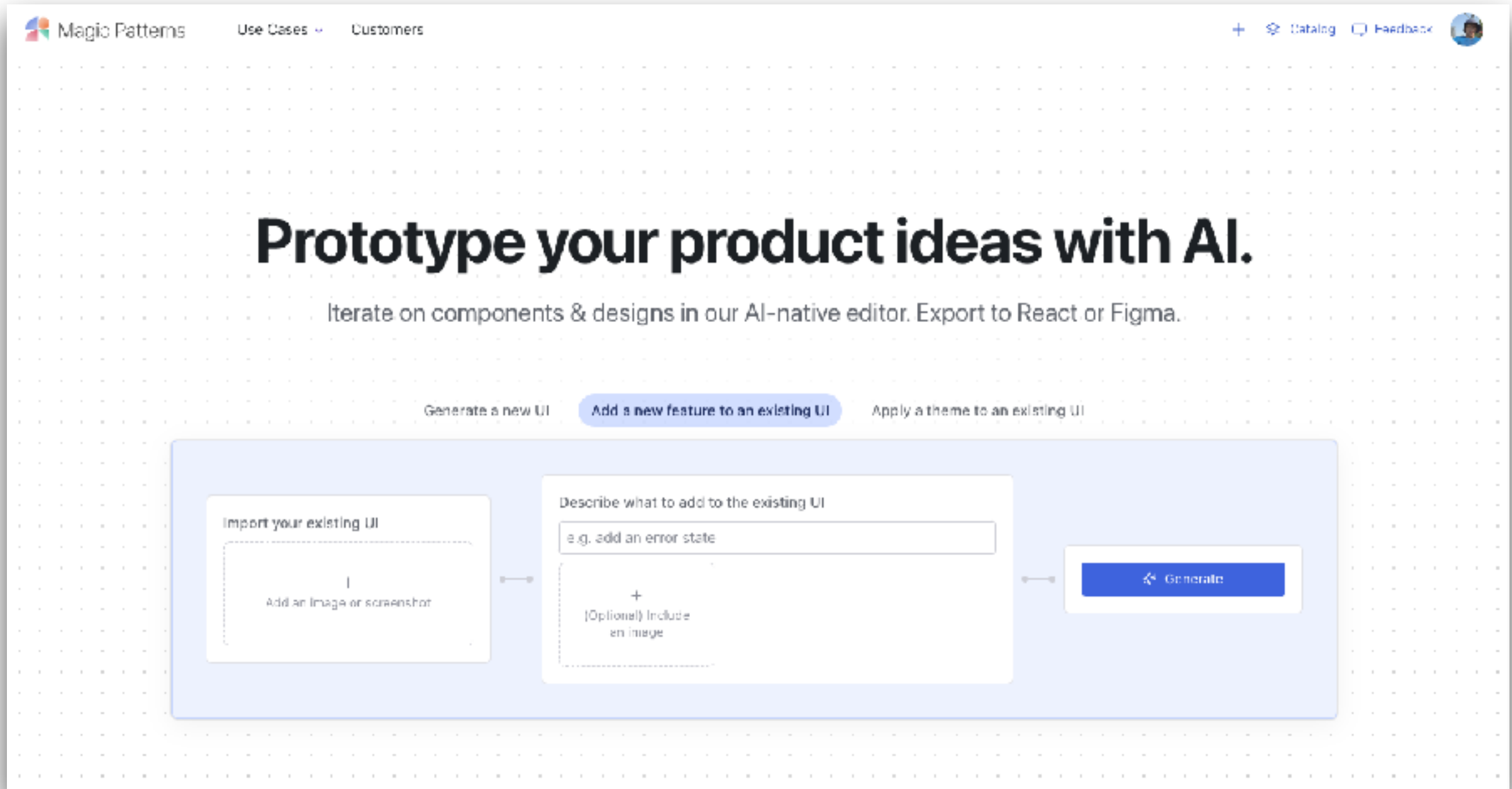
Primary key Identity Unique Nullable

Message AI or write SQL

<https://database.build/>



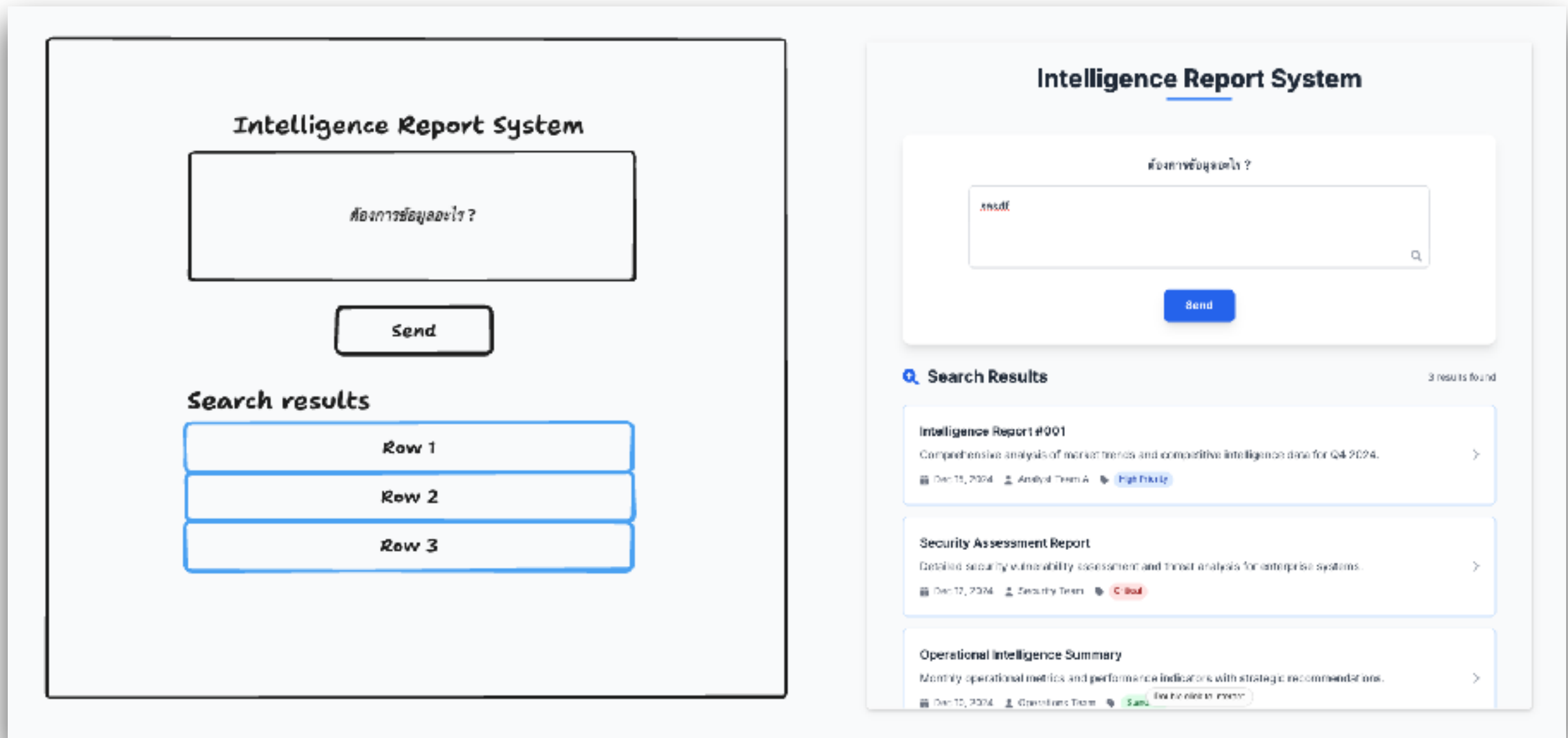
# Magic Pattern



<https://www.magicpatterns.com/>



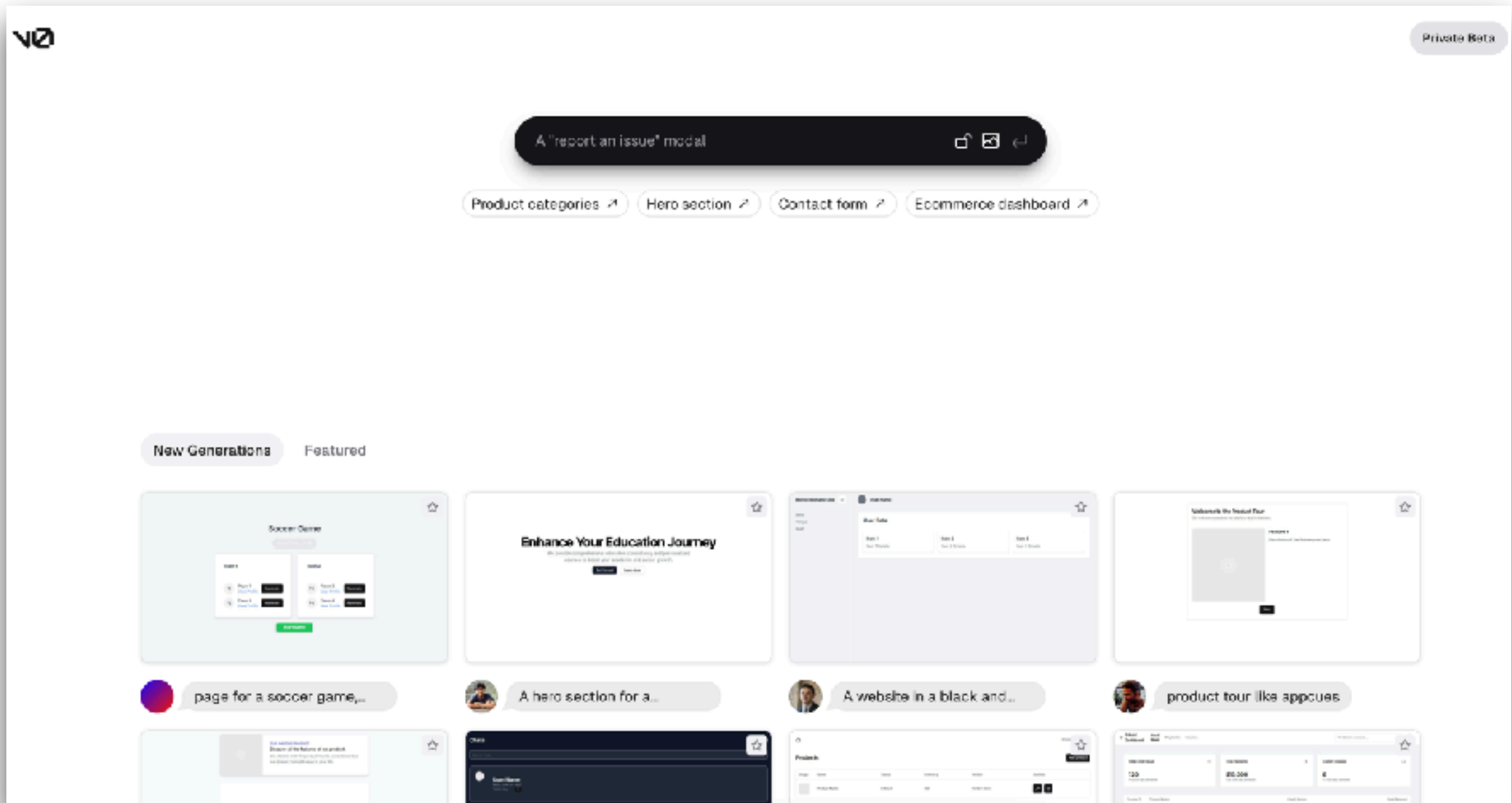
# Make Real



<https://github.com/tldraw/make-real>



# V0.dev



<https://v0.dev/>



# Bolt.new

## What do you want to build?

Prompt, run, edit, and deploy full-stack web apps.

How can Bolt help you today?



Start a blog with Astro

Build a mobile app with NativeScript

Create a docs site with Vitepress

Scaffold UI with shadcn

Draft a presentation with Slidex

Code a video with Remotion

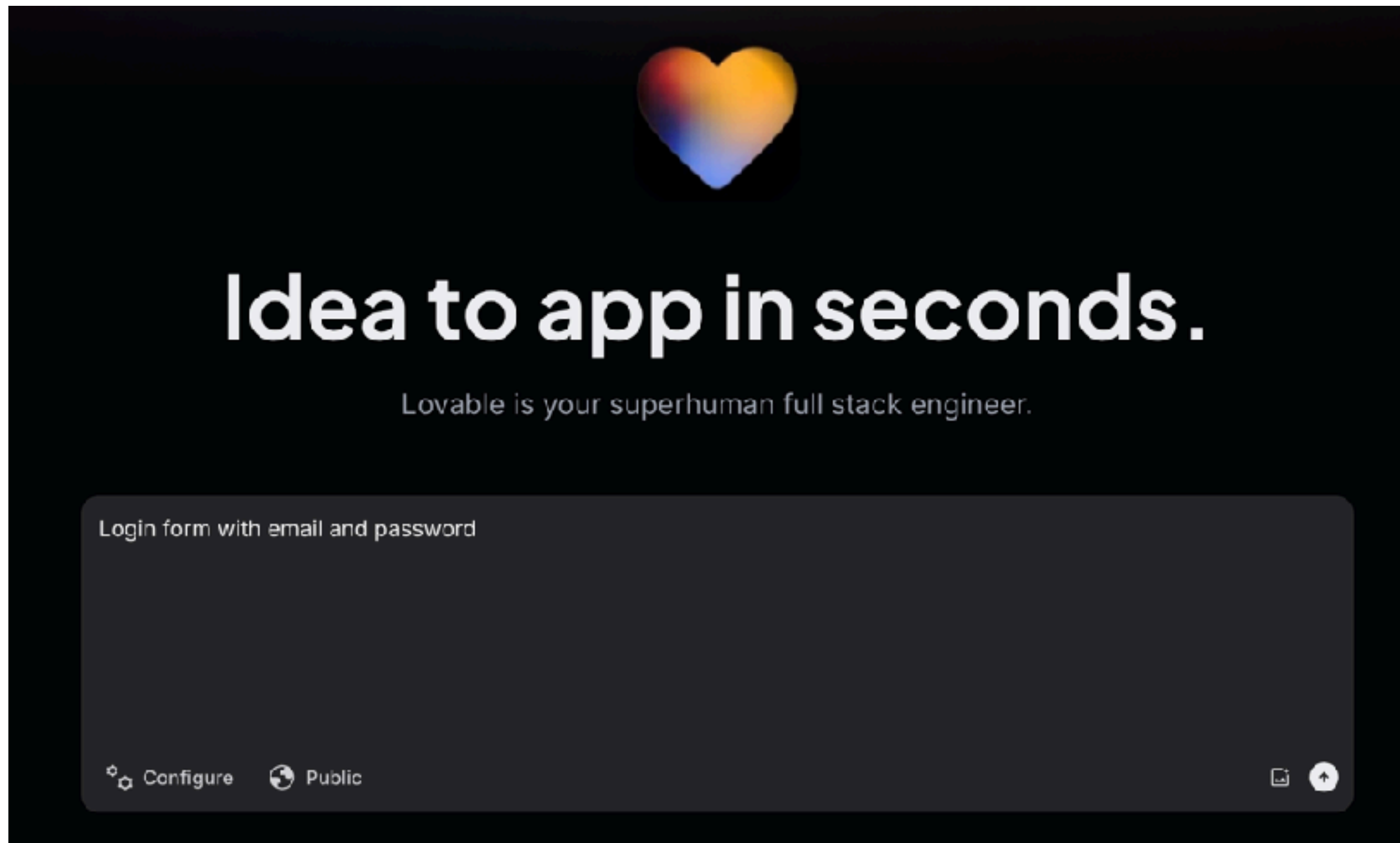
or start a blank app with your favorite stack



<https://bolt.new/>



# Lovable



<https://lovable.dev/>





# SuperDesign

## Vibe it. Design it.

Explore freely, iterate fast. Your design, AI-powered.

Design on web

Integrate with   

Describe what you want to create...

  /   Gemini 3 Flash 

 Use AI Designer  

 Recreate Screenshot

 Import from Site

 Explore Effects

<https://app.superdesign.dev/>

AI for Software Development  
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89



# Development Process



# Develop



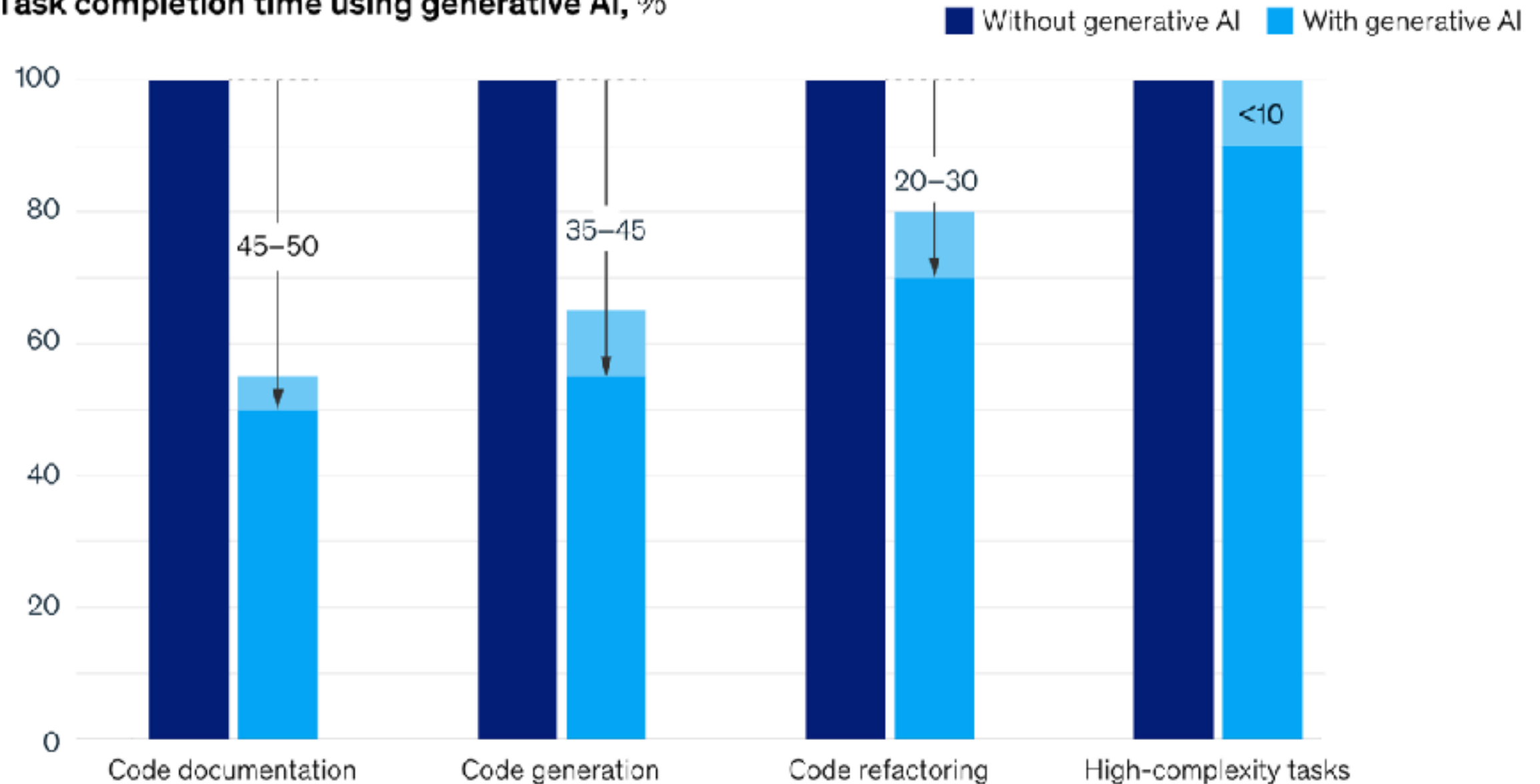
Code generation  
Review and explain code  
Debugging code  
Improve consistency  
Code translation



# Development

**Generative AI can increase developer speed, but less so for complex tasks.**

Task completion time using generative AI, %



<https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/unleashing-developer-productivity-with-generative-ai>



# Category of Tools

## Chat AI

ChatGPT  
Gemini  
Claude.ai  
DeepSeek

## Code AI

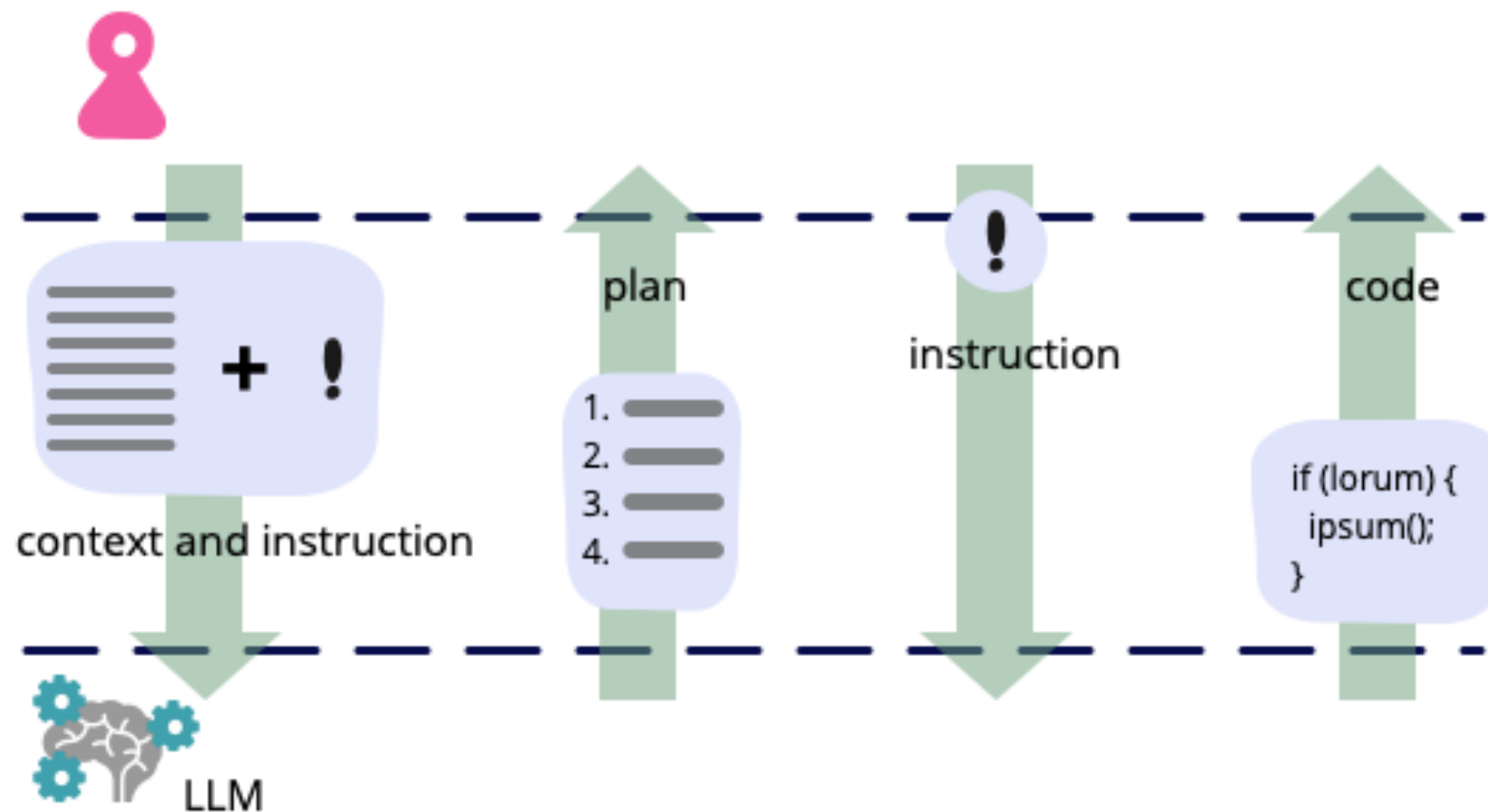
Copilot in VSCode  
Cursor IDE  
Windsurf  
Zed

## AI Agent Public and Local

Aider  
Claude code  
Gemini CLI  
Codex CLI  
Qwen3-coder



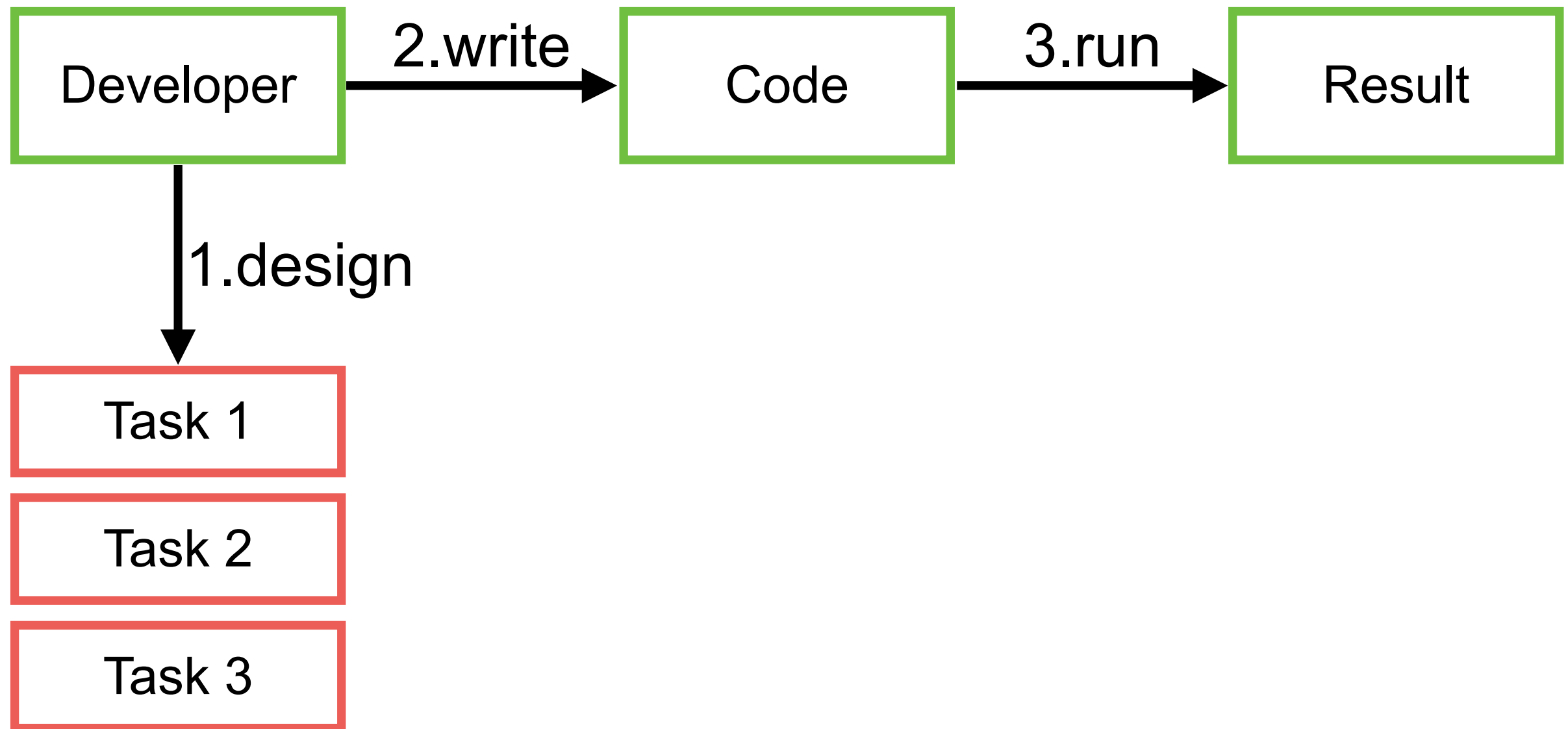
# Development



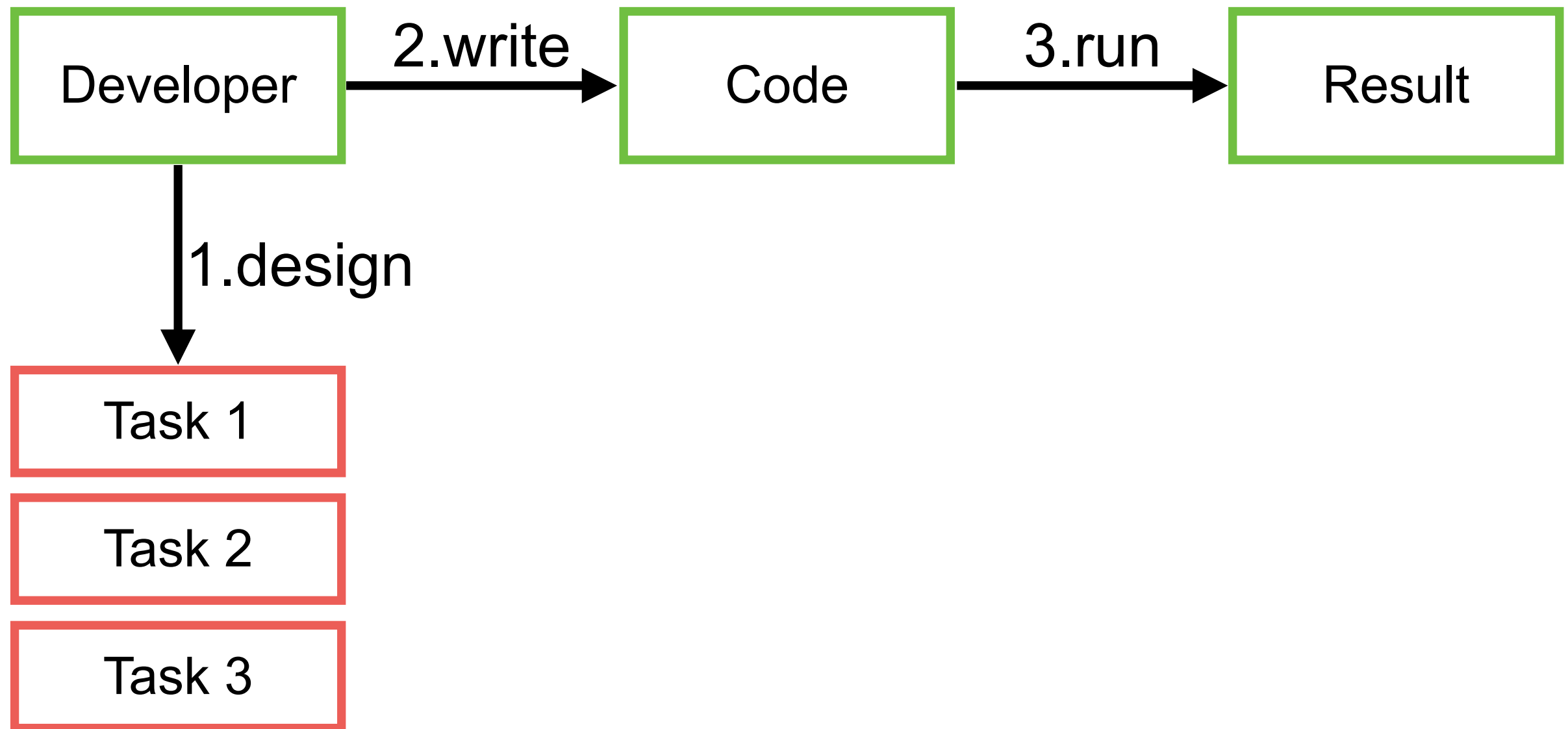
<https://martinfowler.com/articles/2023-chatgpt-xu-hao.html>



# Development

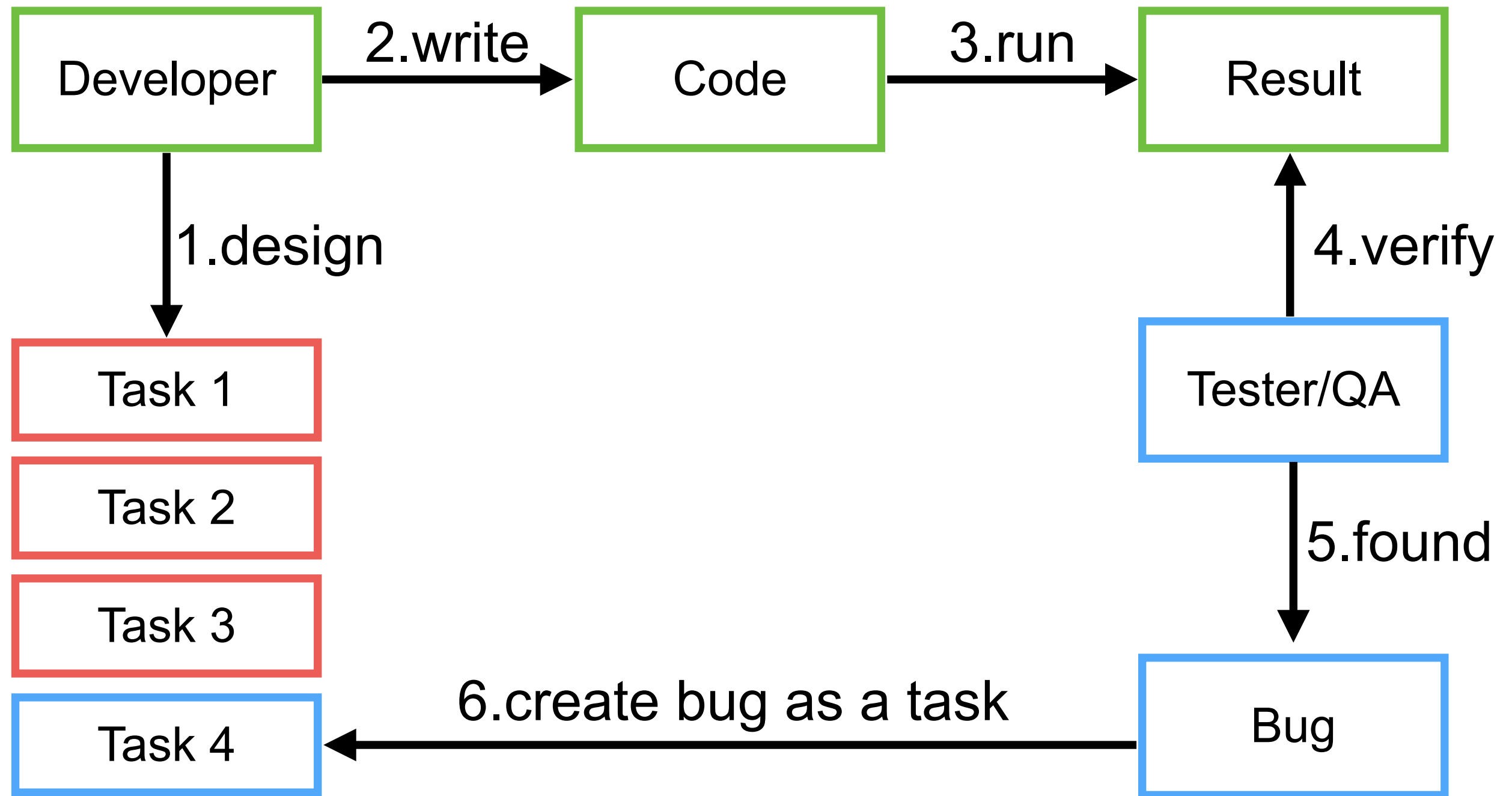


# Development





# Development + Testing





# Main Features

Ask question

Generate code

Refactor code

Document code

Find problems in your code



# Pair programming with AI



# AI Pair Programming

## Aider is AI pair programming in your terminal

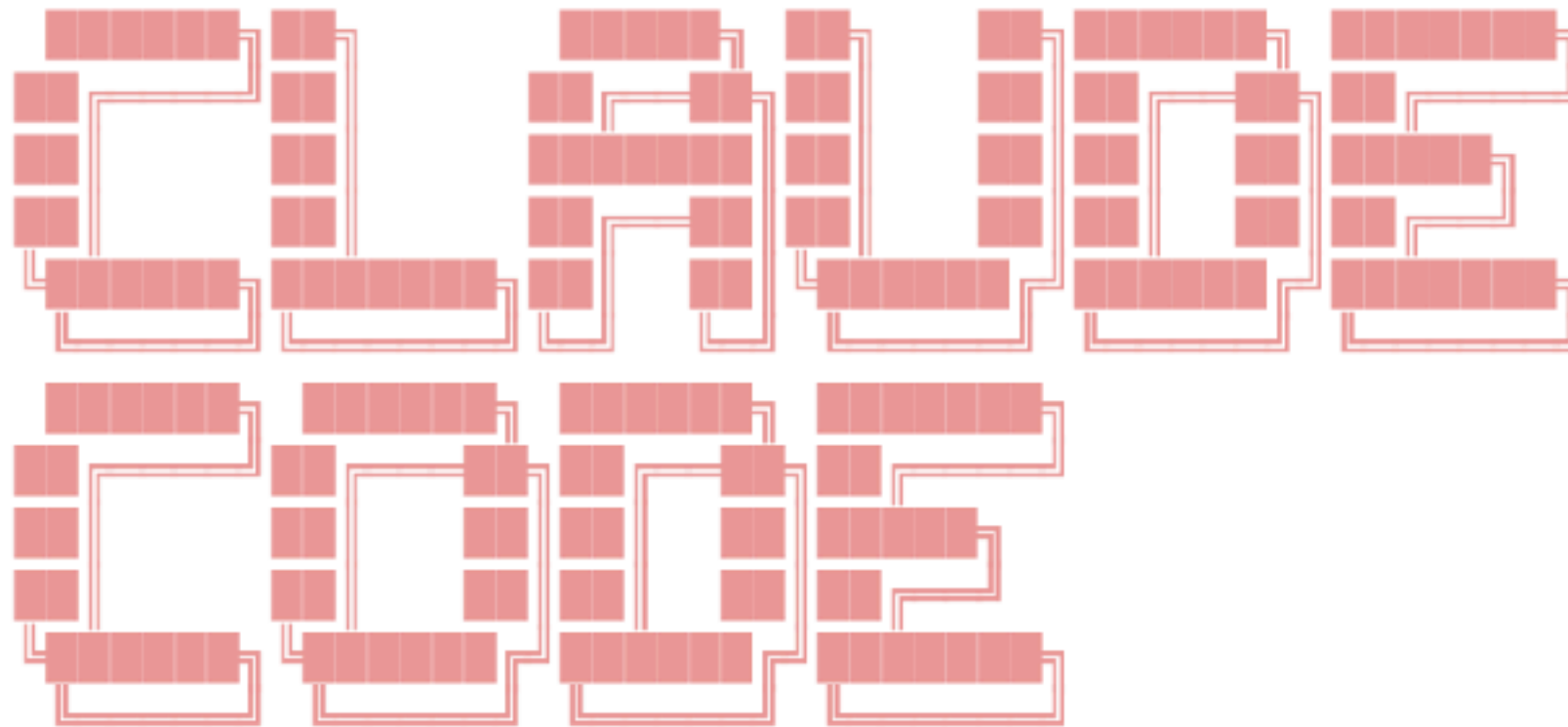
Aider lets you pair program with LLMs, to edit code in your local git repository. Start a new project or work with an existing git repo. Aider works best with GPT-4o & Claude 3.5 Sonnet and can [connect to almost any LLM](#).



<https://github.com/paul-gauthier/aider>



\* Welcome to **Claude Code** research preview!



**Claude Code is billed based on API usage through your Anthropic Console account.**

Pricing may evolve as we move towards general availability.

Press **Enter** to login to your Anthropic Console account...

2025/02

<https://www.anthropic.com/news/claude-3-7-sonnet>





2025/06

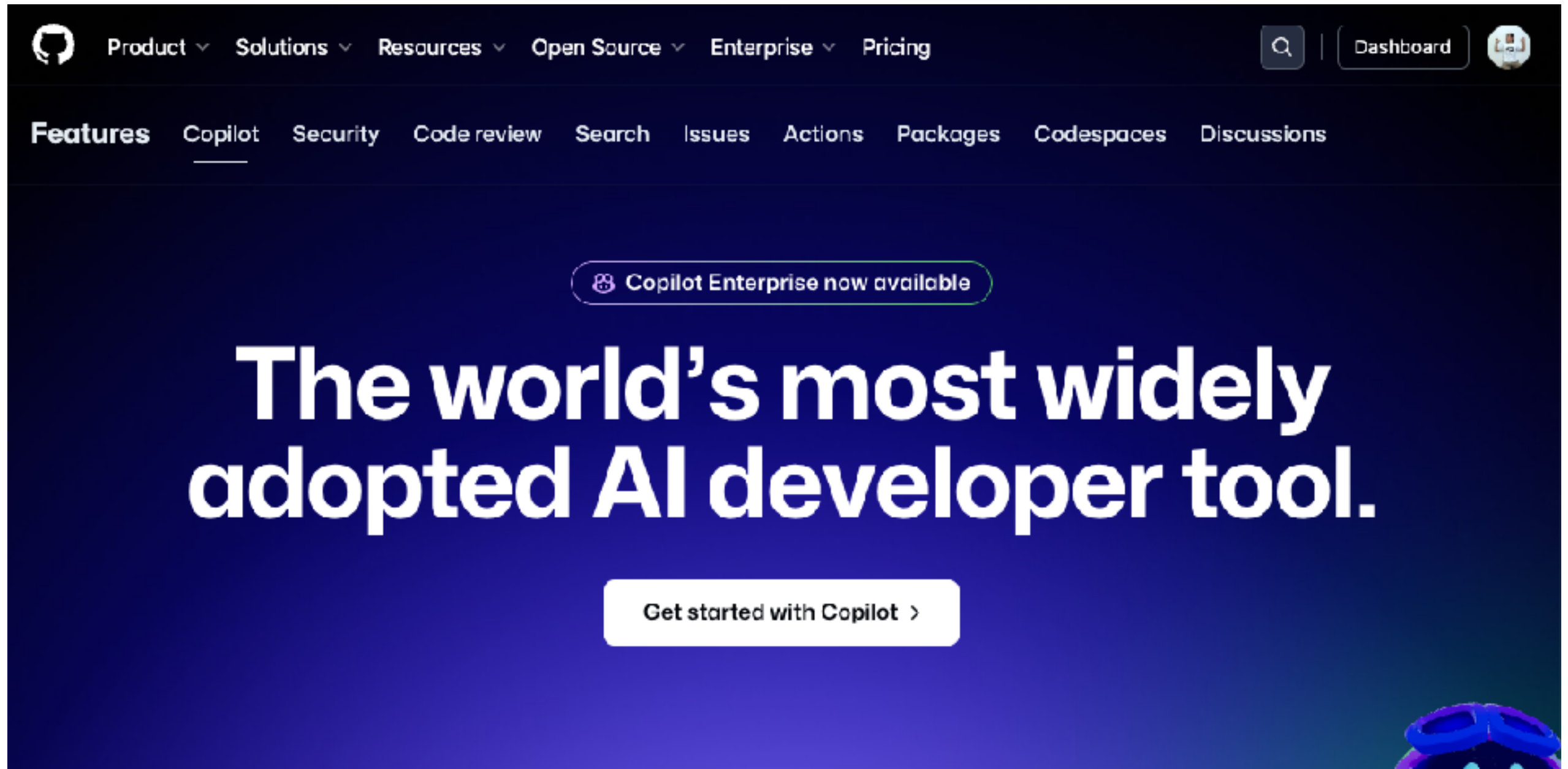
<https://github.com/google-gemini/gemini-cli>



# AI in IDE



# GitHub Copilot

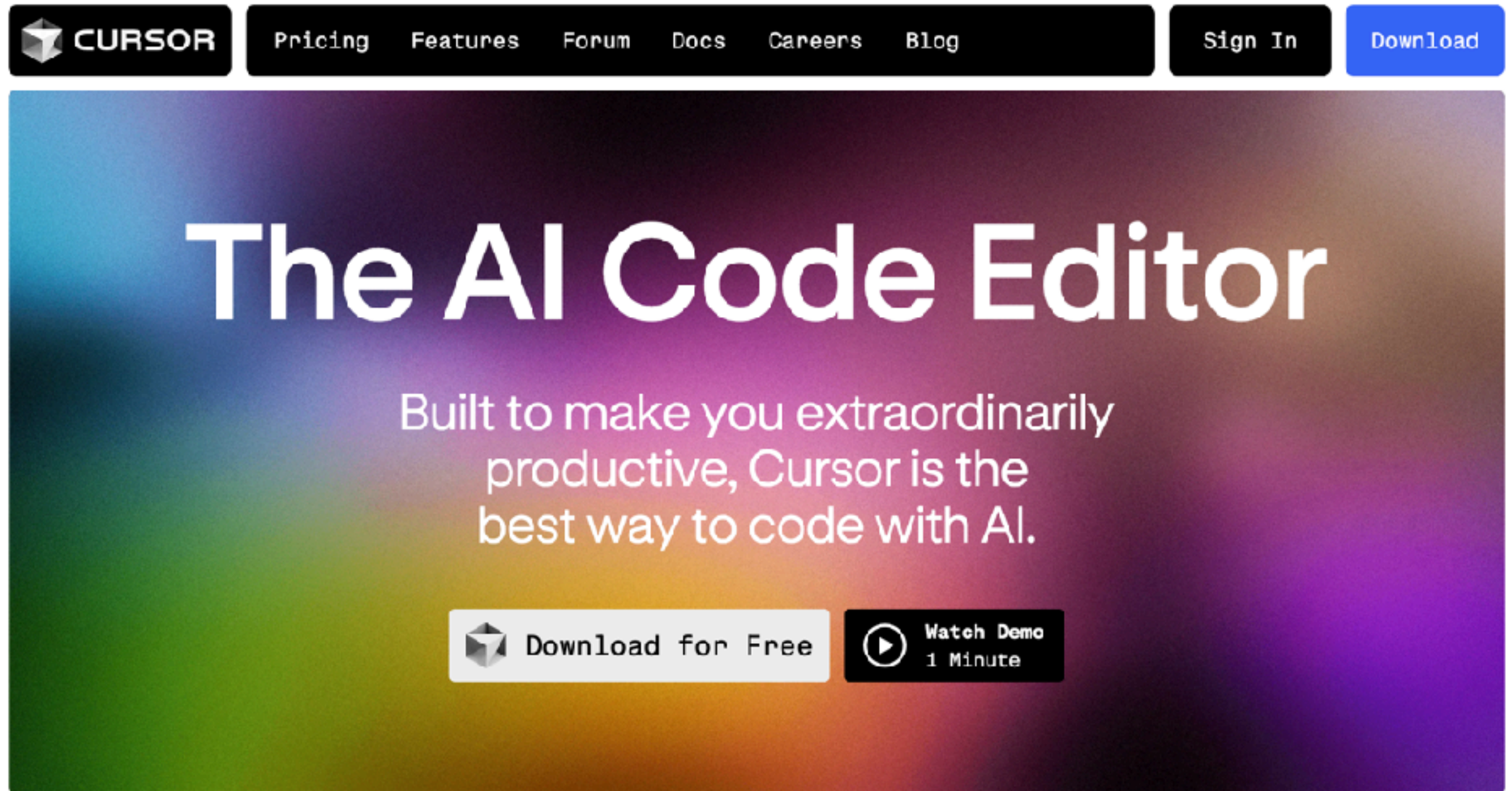


<https://github.com/features/copilot>





# Cursor.sh



The image shows the top portion of the Cursor.sh website. At the top is a dark navigation bar with the Cursor logo on the left and links for Pricing, Features, Forum, Docs, Careers, and Blog in the center. On the right of the navigation bar are 'Sign In' and 'Download' buttons. Below the navigation bar is a large hero section with a colorful gradient background. The main heading 'The AI Code Editor' is centered in large white text. Below it, a subheading states: 'Built to make you extraordinarily productive, Cursor is the best way to code with AI.' At the bottom of the hero section are two buttons: 'Download for Free' (with the Cursor logo) and 'Watch Demo 1 Minute' (with a play button icon).

**CURSOR** Pricing Features Forum Docs Careers Blog Sign In Download

## The AI Code Editor

Built to make you extraordinarily productive, Cursor is the best way to code with AI.

 Download for Free  Watch Demo 1 Minute

<https://www.cursor.com/>





# Cursor Directory

The screenshot shows the Cursor Directory website. At the top, there's a navigation bar with 'cursor.directory', 'Get latest updates', 'Subscribe', 'Live', 'Learn', and 'About'. Below this is a search bar and a filter for 'All' and 'Popular'. A sidebar on the left lists various technologies with their respective counts: TypeScript (15), Python (9), Next.js (9), React (9), PHP (5), C# (4), Expo (4), React Native (4), Tailwind (4), Supabase (4), Web Development (3), Game Development (3), JavaScript (3), and Laravel (3). The main content area displays a grid of developer profiles. Each profile includes a bio, a list of skills, and a 'Submit +' button. The developers shown are Krish Kalaria, Mohammadali Karimi, and Nathan Brachette. Their bios and skills are as follows:

- Krish Kalaria**: You are an expert in TypeScript, React Native, Expo, and Mobile UI development. Skills: Code Style and Structure, Write concise, technical TypeScript code, Use functional and declarative programming patterns, Prefer iteration and modularization over duplication, Use descriptive variable names with auxiliary verbs (e.g., isLoading, hasError), Structure files: exported component, subcomponents, helpers, static content, Follow Expo's official documentation for component structure.
- Mohammadali Karimi**: You are a Senior Front-End Developer and an Expert in ReactJS, NextJS, JavaScript, TypeScript, HTML, CSS and modern UI/UX frameworks (e.g., TailwindCSS, Shadcn, Radix). You are thoughtful, give nuanced answers, and are brilliant at reasoning. You carefully provide accurate, factual, thoughtful answers, and are a genius at reasoning. Skills: Follow the user's requirements carefully & to the letter, First think step-by-step - describe your plan for what to build in pseudocode, written out in great detail.
- Nathan Brachette**: You are an expert in TypeScript, Gatsby, React and Tailwind. Skills: Code Style and Structure, Write concise, technical TypeScript code, Use functional and declarative programming patterns, Prefer iteration and modularization over code duplication, Use descriptive variable names with auxiliary verbs (e.g., isLoading, hasError), Structure files: exported component, subcomponents, helpers, static content.

<https://cursor.directory/>



# Windsurf by Codeium



## Built to keep you in *flow state*

The first agentic IDE, and then some. The Windsurf Editor is where the work of developers and AI truly flow together, allowing for a coding experience that feels like literal magic.

 Download the Windsurf Editor

[See all download options](#)

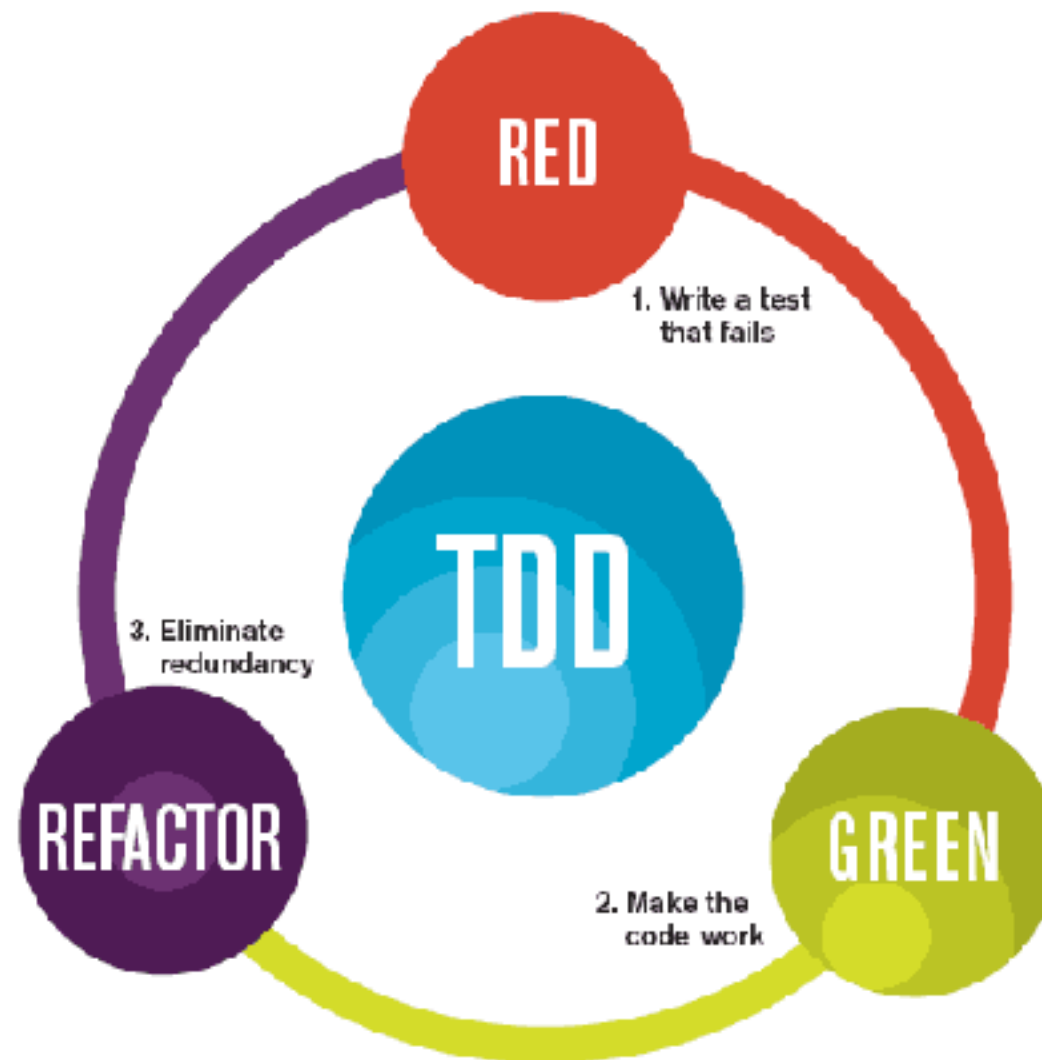
<https://codeium.com/windsurf>



# Test-Driven Development



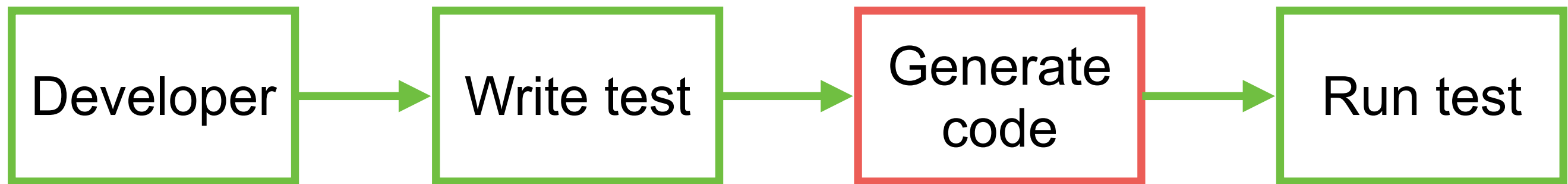
# Test-Driven-Development



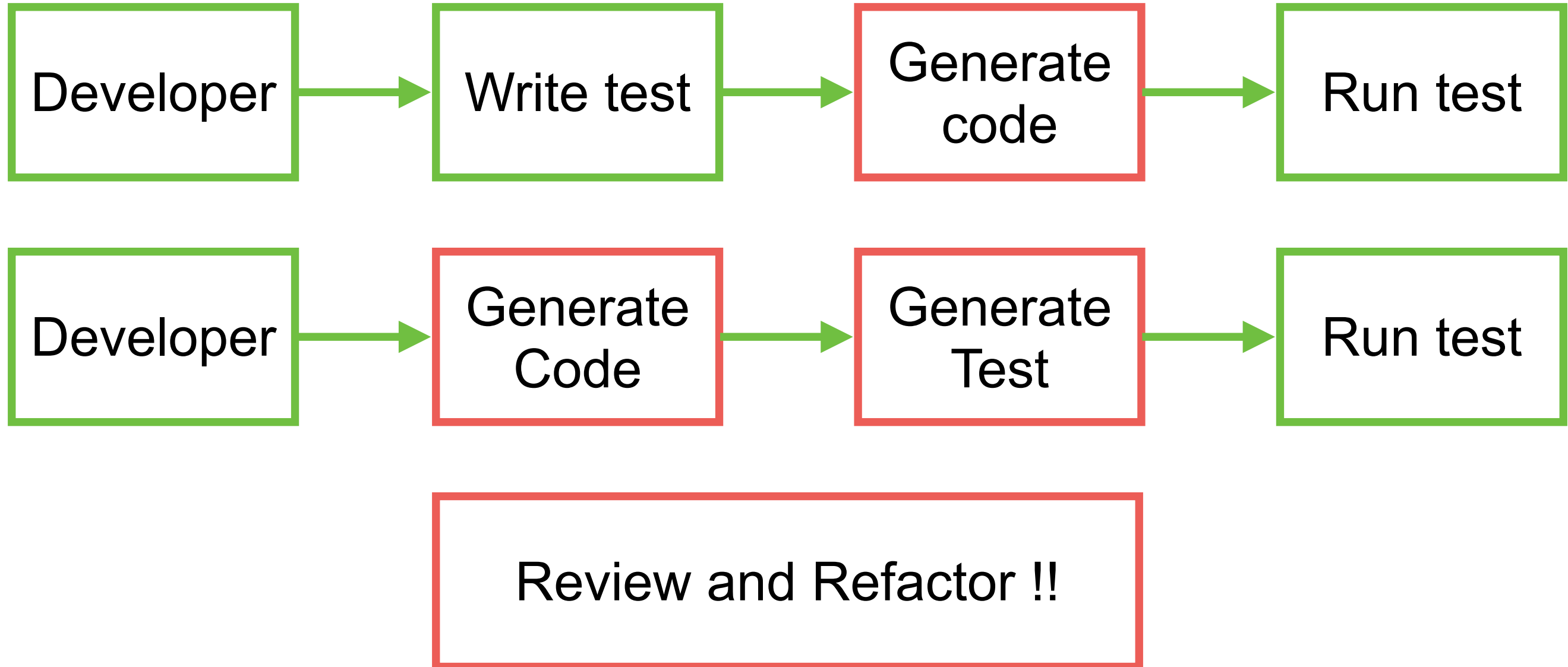
The mantra of Test-Driven Development (TDD) is "red, green, refactor."



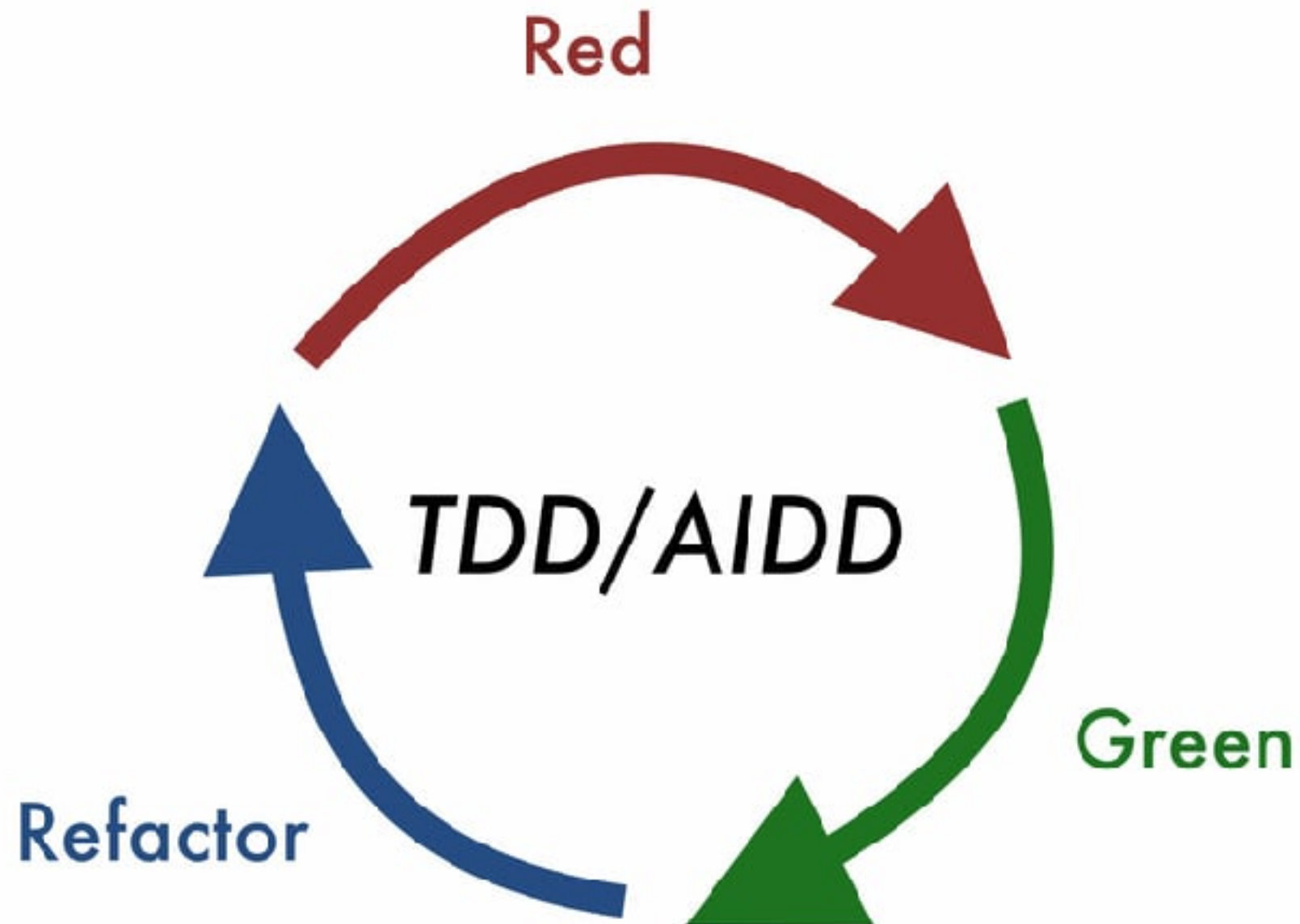
# Test-Driven-Development



# Test-Driven-Development



# AI-DD

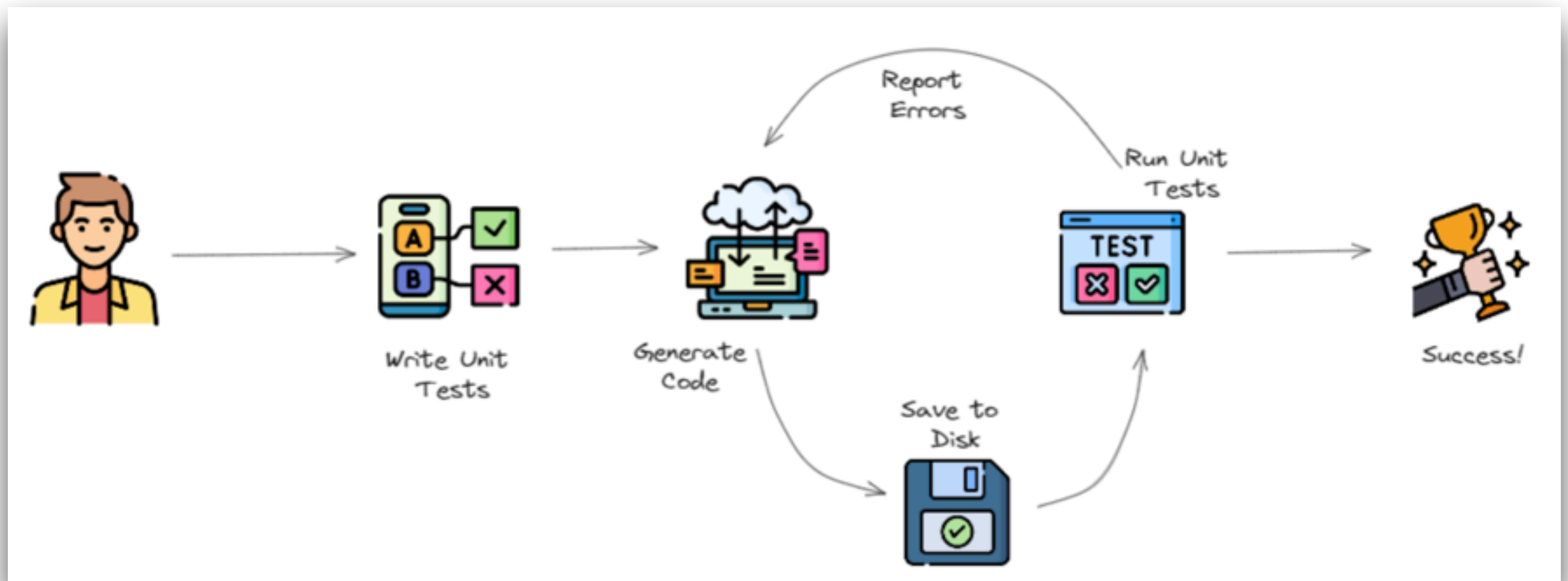


<https://dev.to/dawiddahl/ai-is-changing-the-way-we-code-ai-driven-development-aid-2ngo>





# TDD with AI

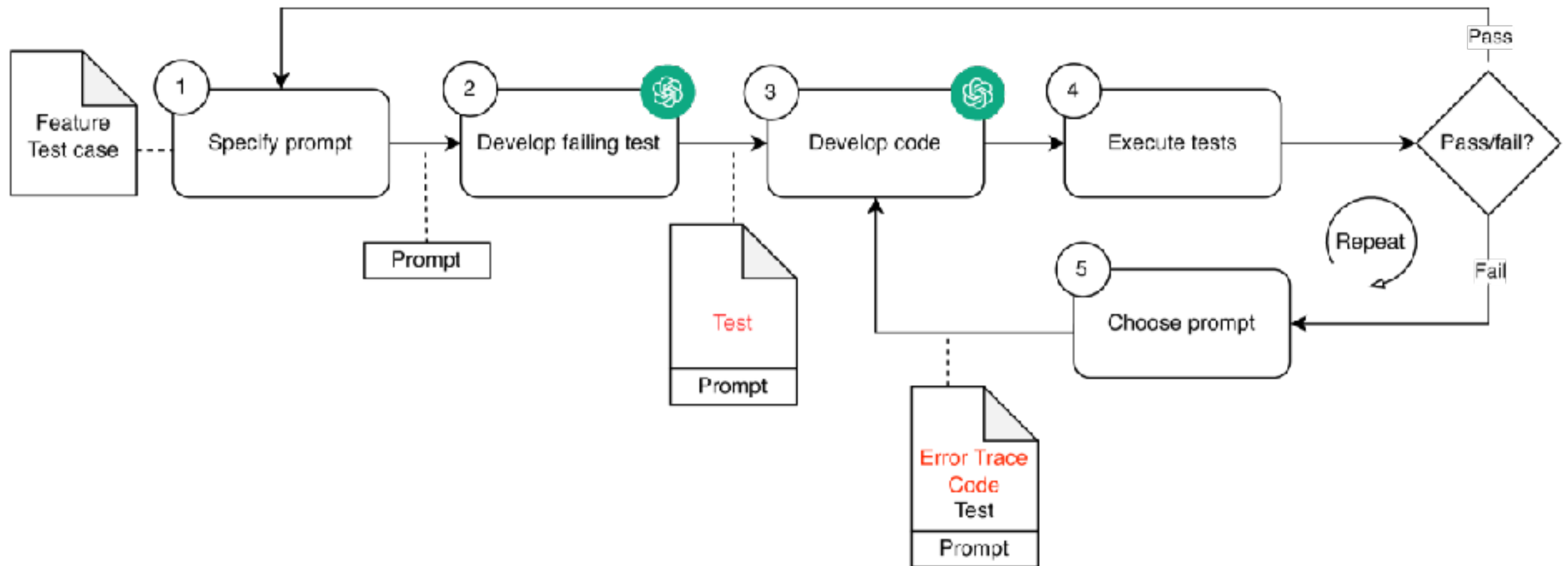


<https://github.com/allenheltondev/tdd-ai>





# TDD with AI



<https://arxiv.org/html/2405.10849v1>



# Test-Driven-Generation (TDG)



# Test-Driven-Generation (TDG)

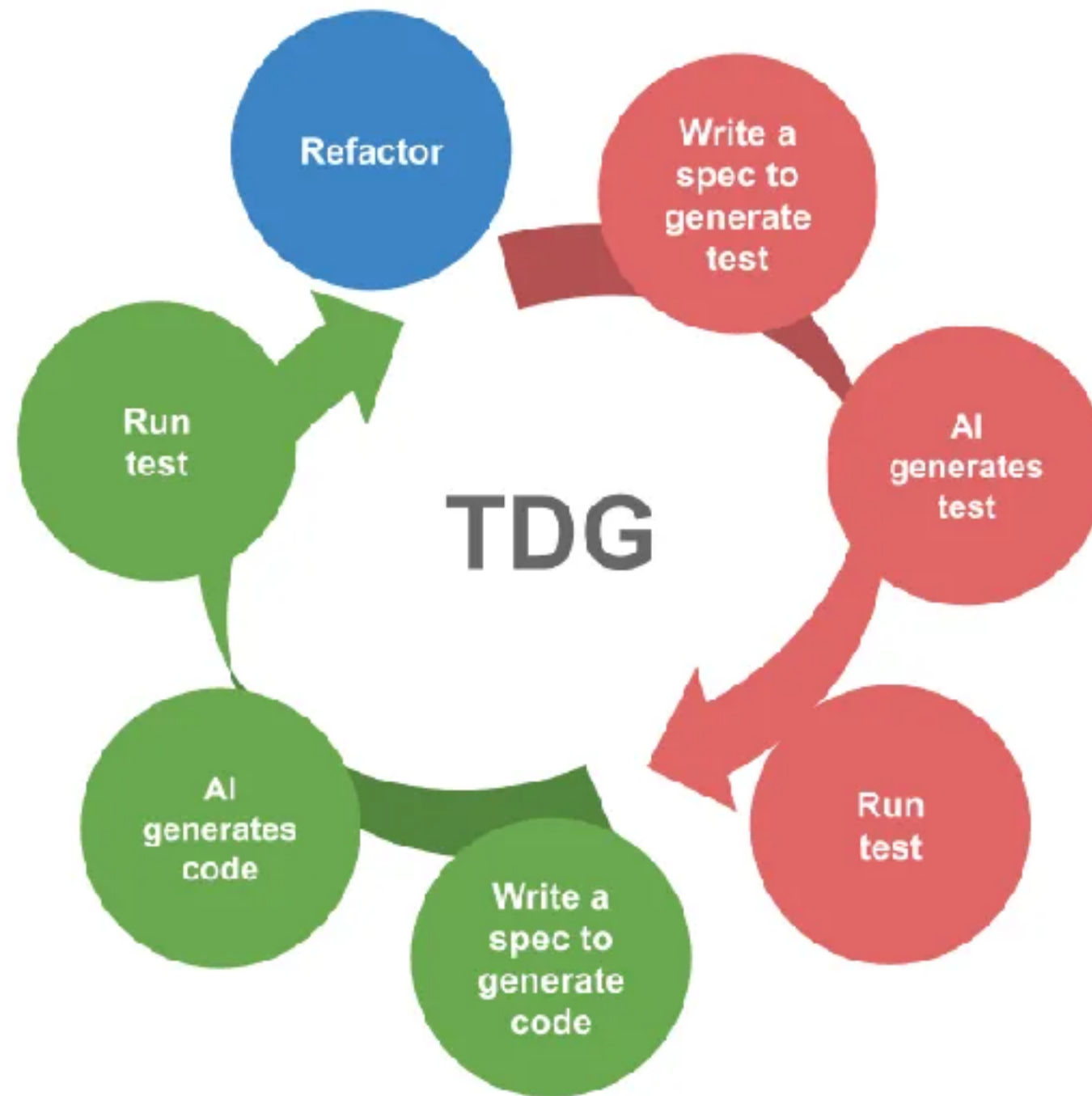
Development practice that integrate Generative AI into the development life-cycle



<https://chanwit.medium.com/test-driven-generation-tdg-adopting-tdd-again-this-time-with-gen-ai-27f986bed6f8>



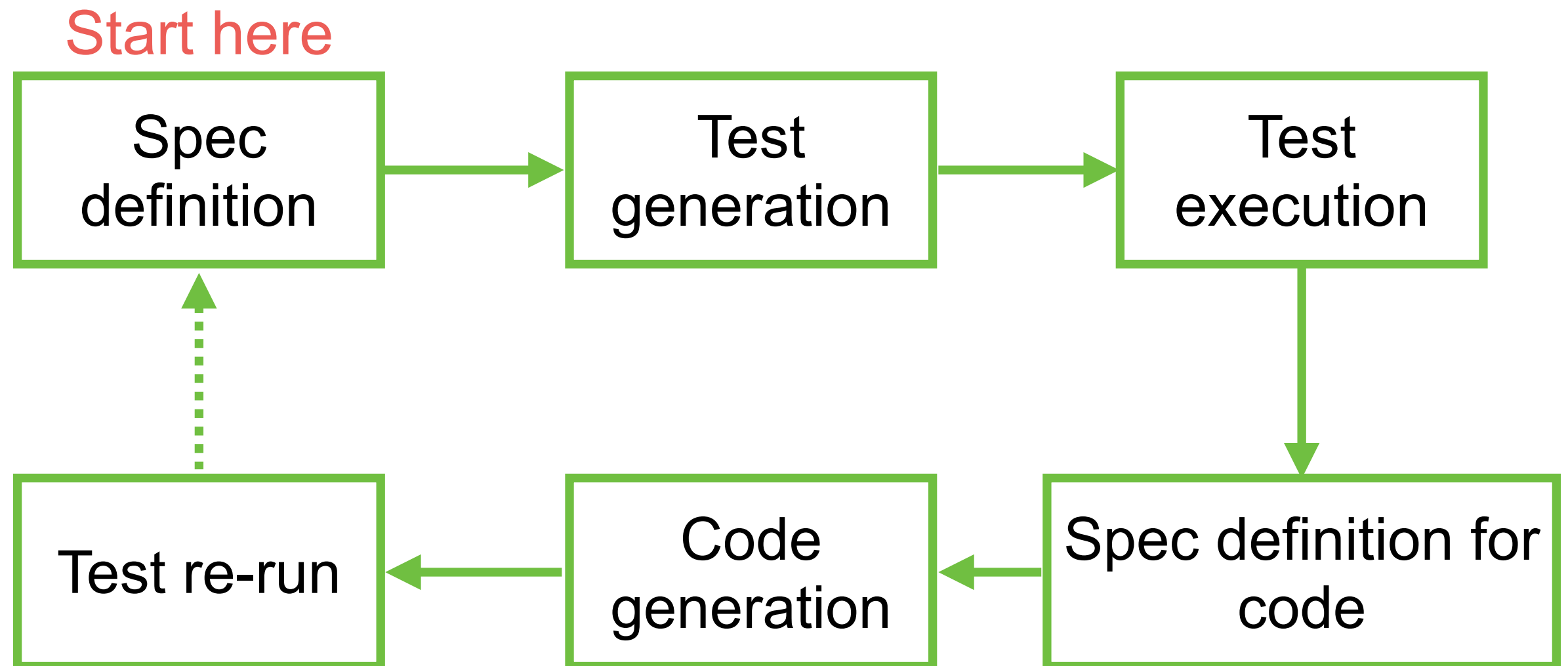
# Test-Driven-Generation (TDG)



<https://chanwit.medium.com/test-driven-generation-tdg-adopting-tdd-again-this-time-with-gen-ai-27f986bed6f8>



# Test-Driven-Generation (TDG)



# Tips and Techniques with AI

Scope of prompt

Error in result

Setup and config  
project

Latest information

Use technical  
keywords



# Specification-Driven Development (SDD)



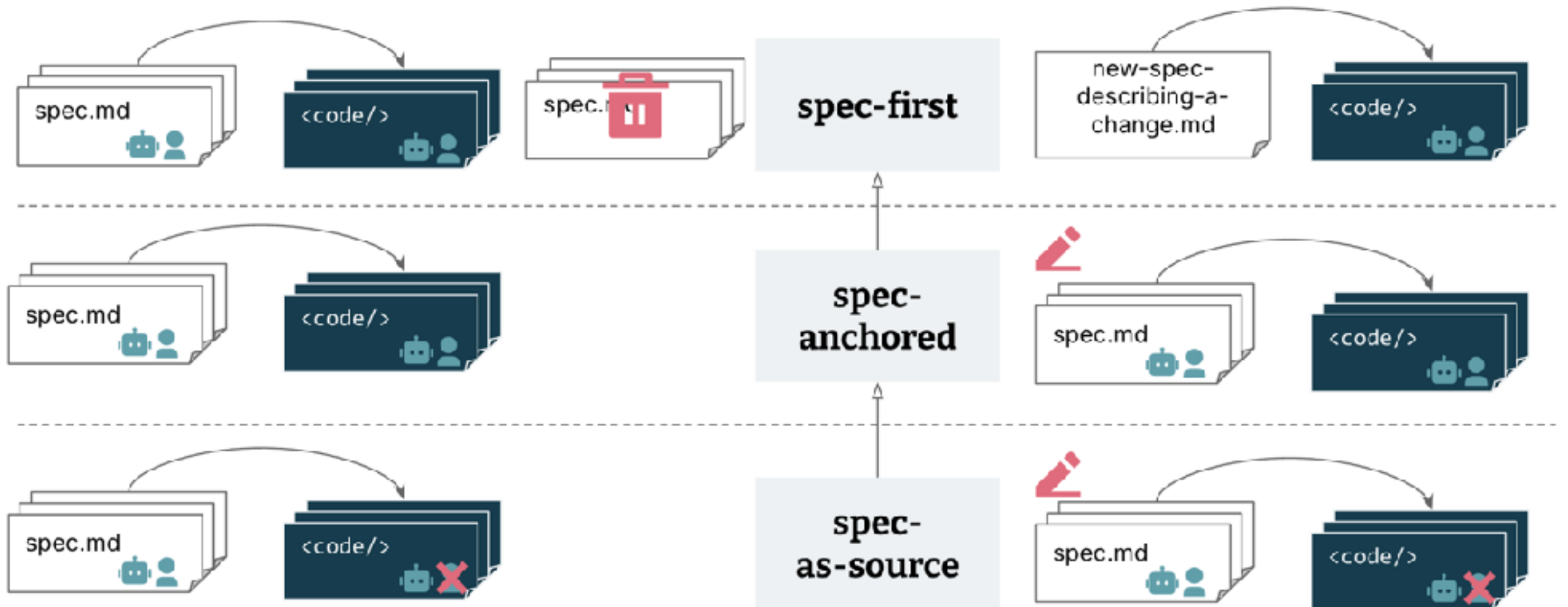


# Workflow SDD

## Levels of “spec-driven”

Creation of feature

Evolution and maintenance of feature



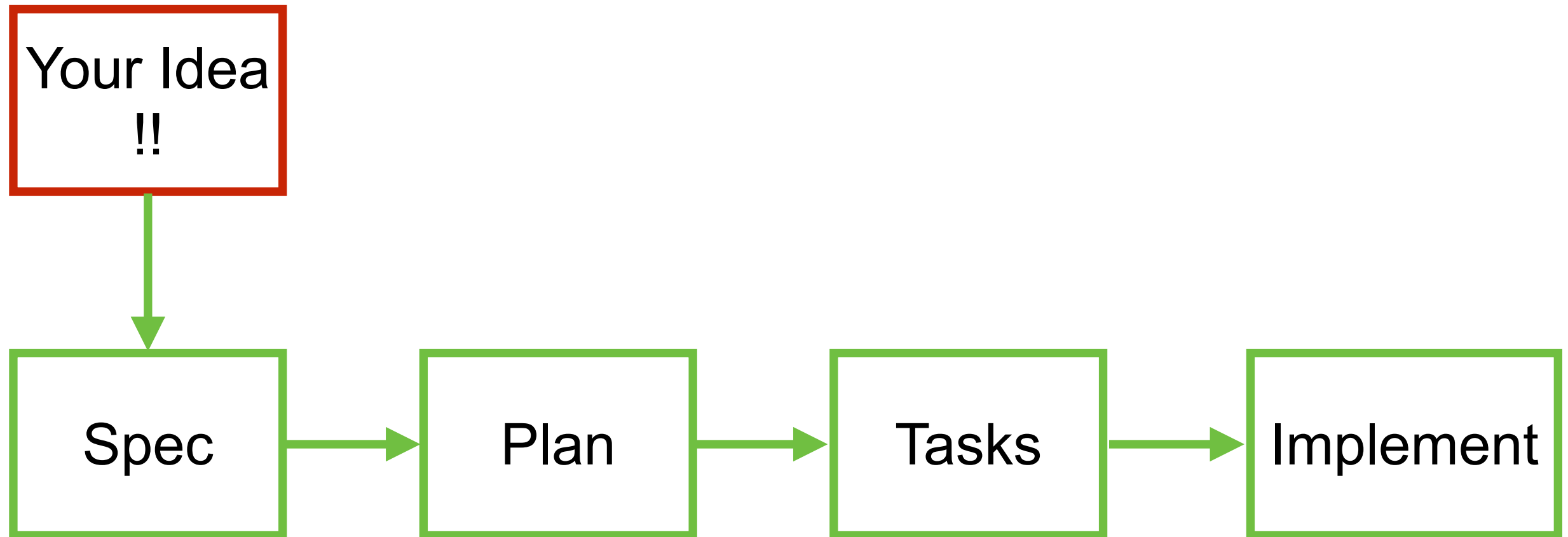
<https://martinfowler.com/articles/exploring-gen-ai.html>

<https://martinfowler.com/articles/exploring-gen-ai/sdd-3-tools.html>

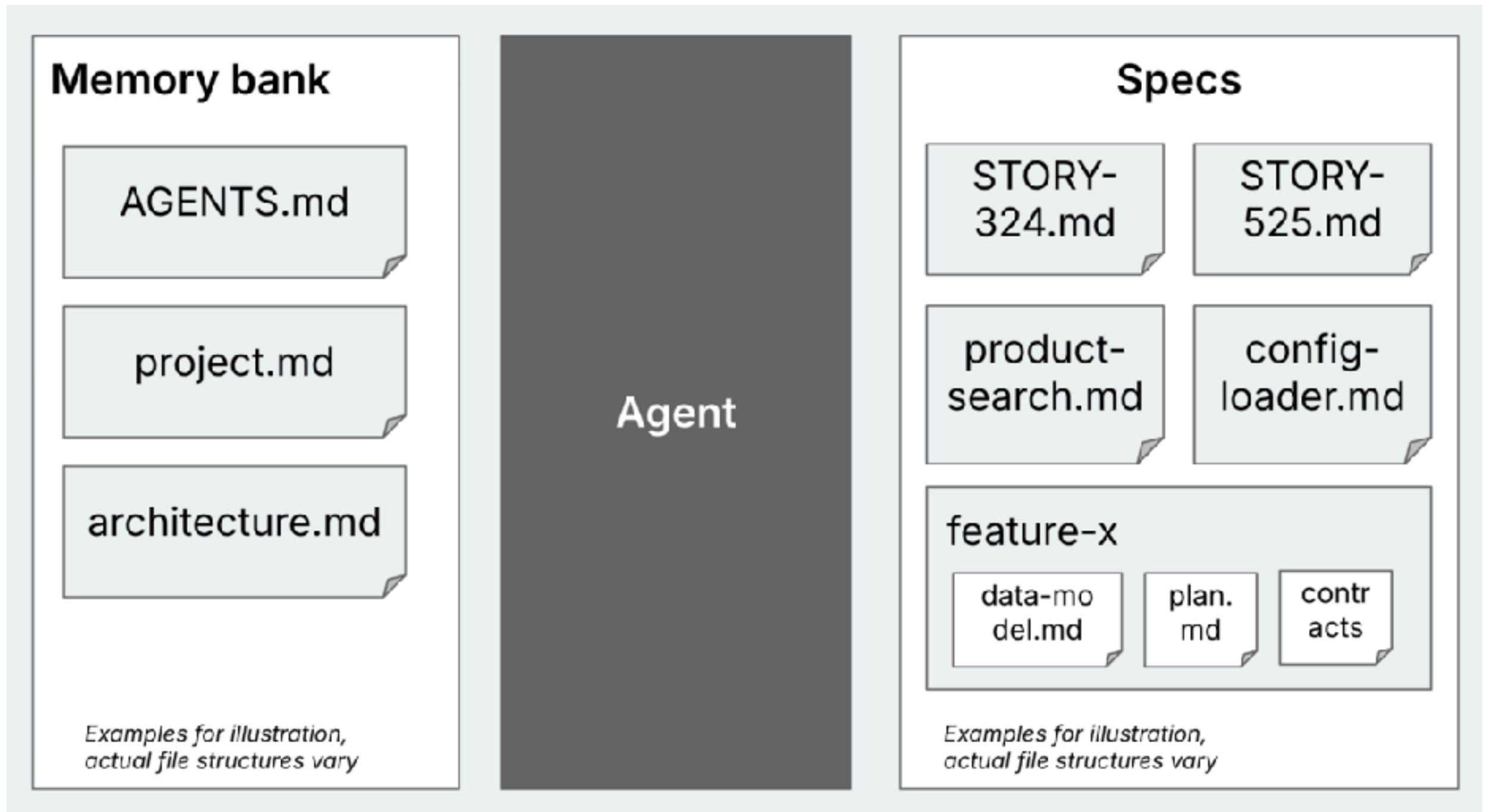




# Workflow SDD



# Memory Bank



# Workflow Tools

AWS Kiro

Requirement  
Design  
Tasks

Spec-kit

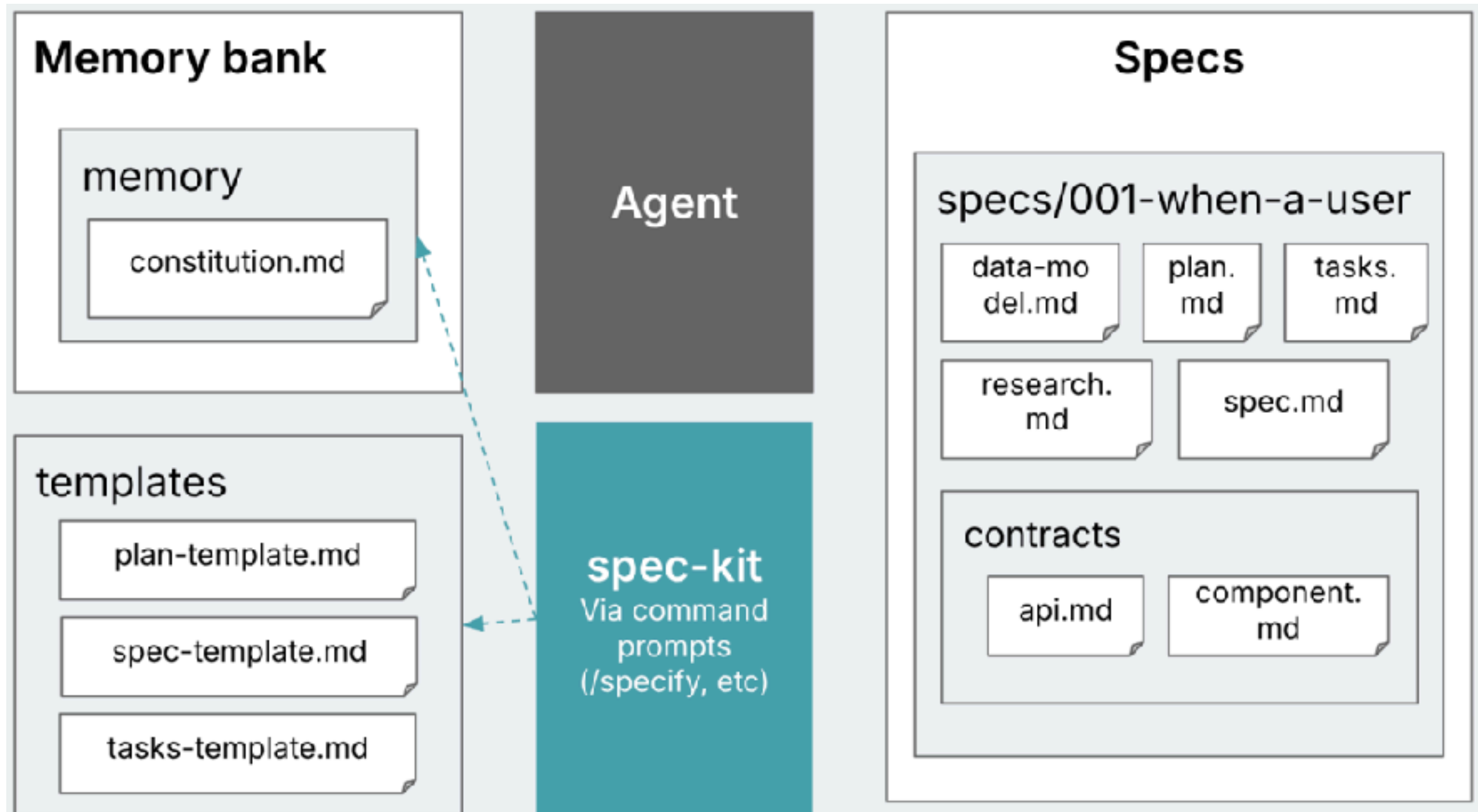
Specify  
Plan  
Tasks

Custom by IDE

VSCode rules  
Cursor rules  
Github rules



# Spec-Kit



# Workshop with Coding

Chat

Text Editor with AI

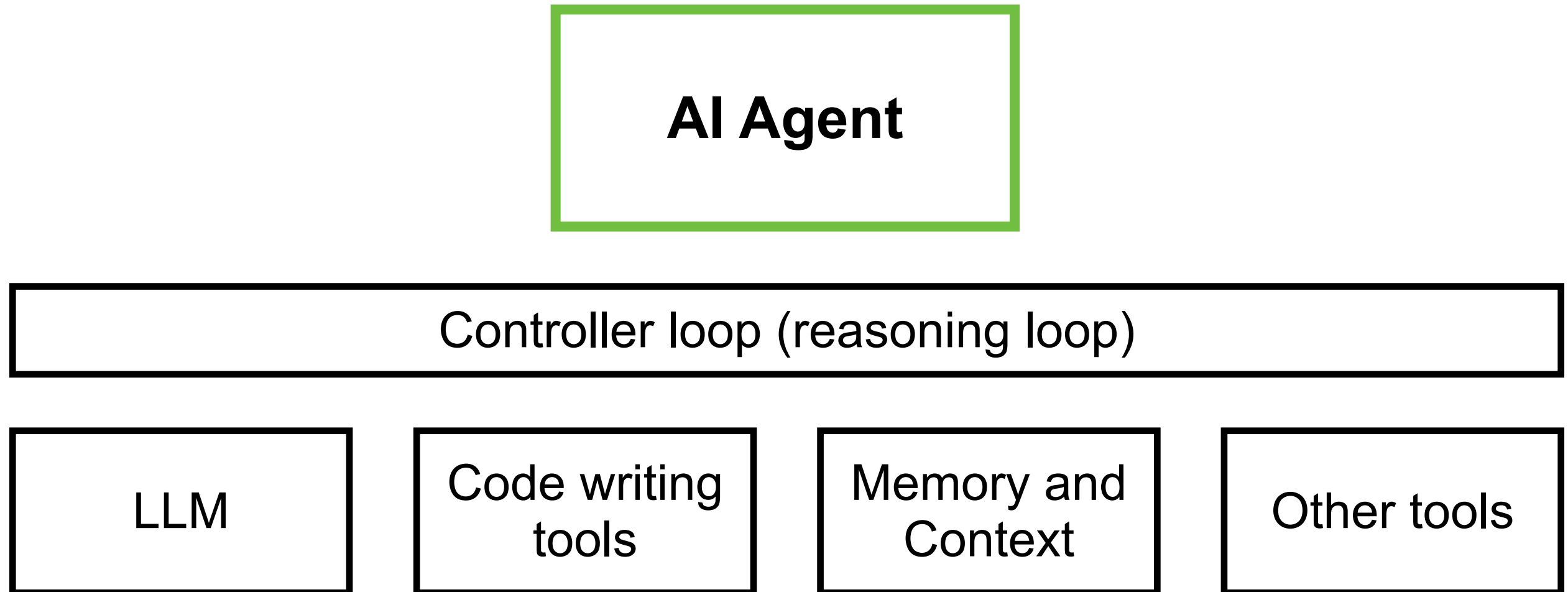
Pair programming  
with AI

SDD and Rules

BMAD-METHOD



# AI Agent for Coding



# AI Agent for Coding

Claude Code  
Github Copilot Chat  
Cline  
Qwen code  
Gemini CLI  
Pi coding agent





# Testing Process with High quality process

Functional

Non-Functional





# Testing



Test cases writing  
Test code generation  
Bug detection  
Test planning  
Data test generation



# 6 Keys Software Quality

Defect density

Code duplication

Hardcode token/key

Security  
vulnerabilities

Outdated package

Non-permissive  
opensource libraries

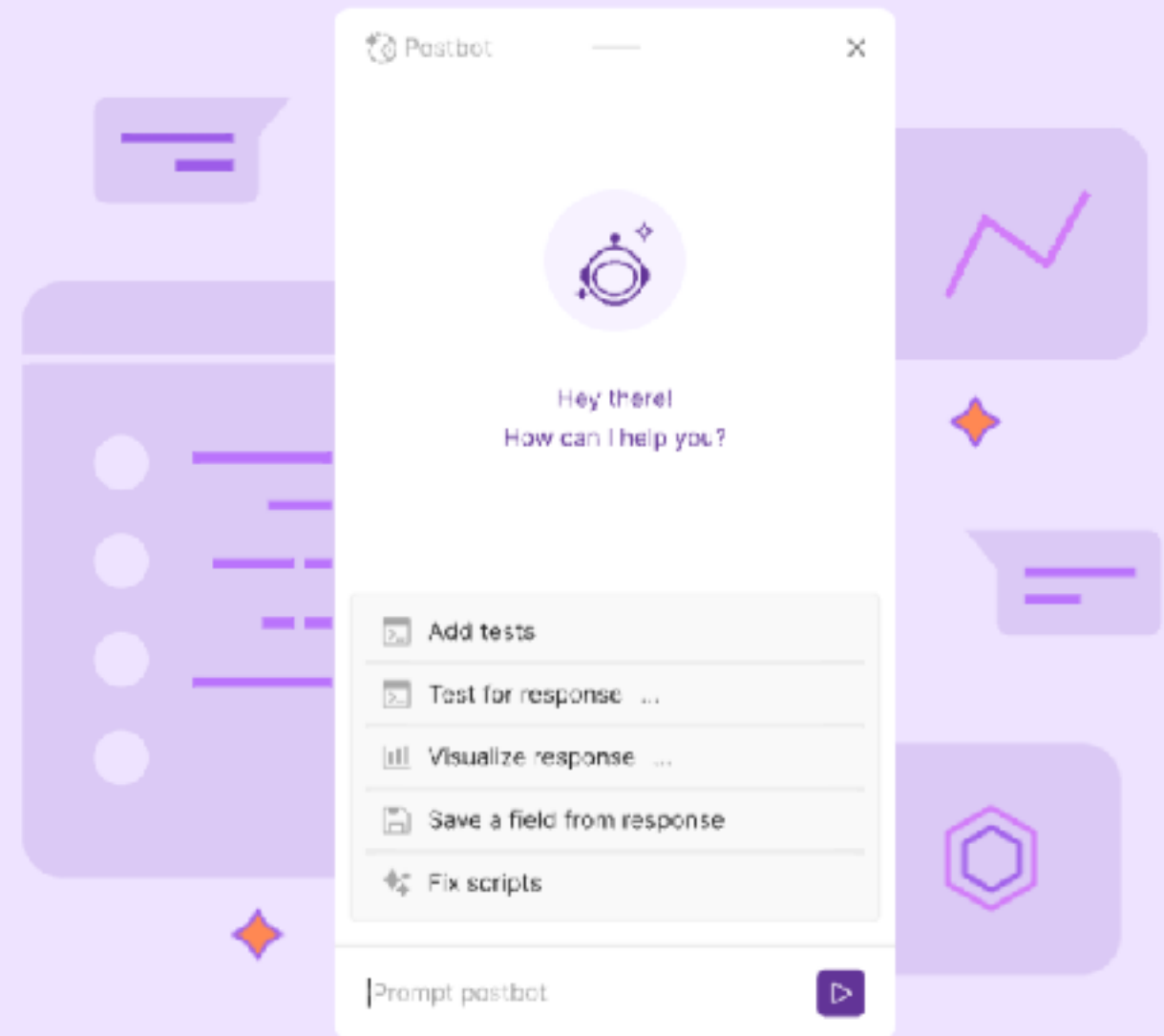


# Testing with Postman

**Postbot, our AI-powered assistant, will supercharge your API development.**

Speed up your most common API development workflows with natural-language input, conversational interactions, and contextual suggestions.

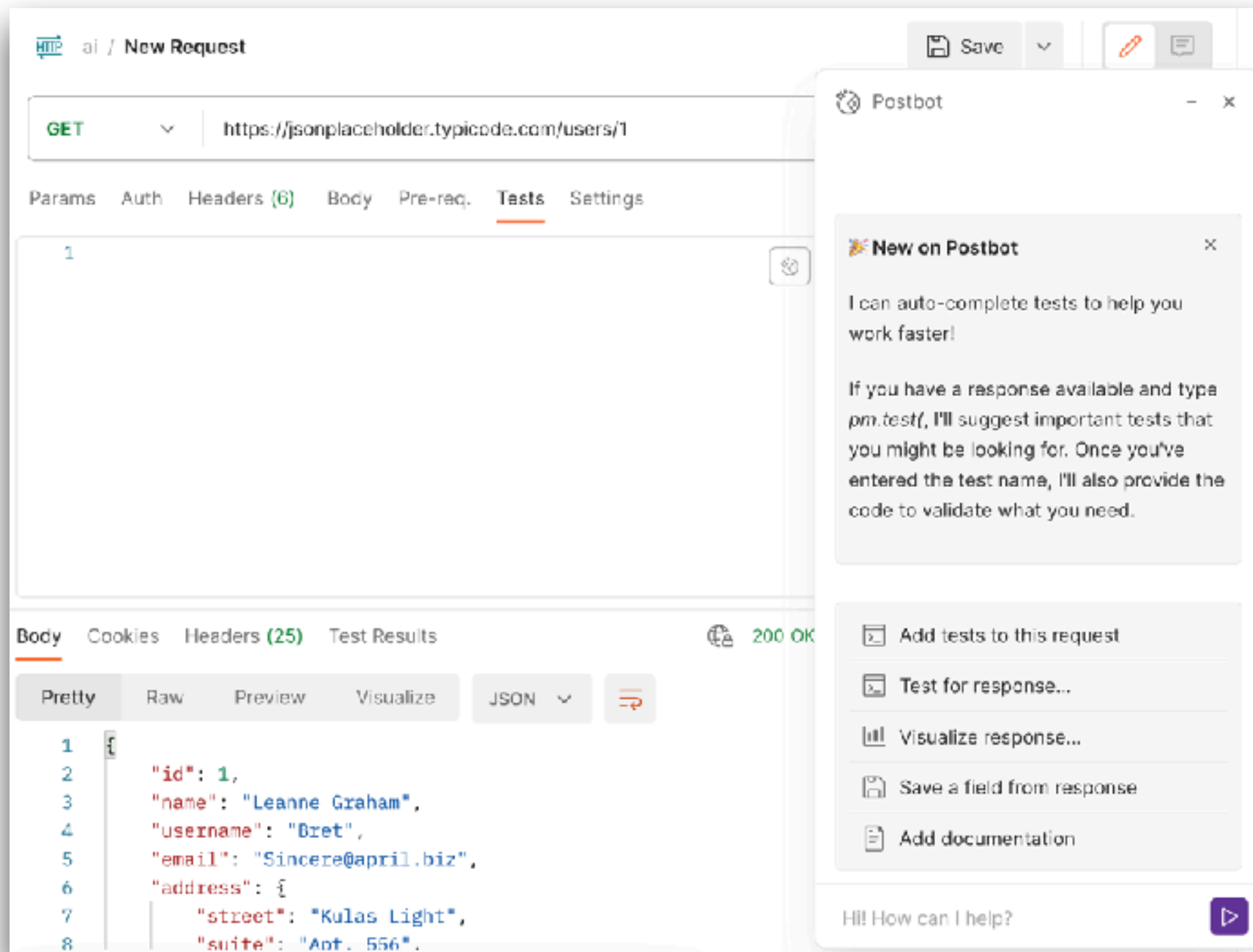
Get Started



<https://www.postman.com/product/postbot/>



# PostBot



<https://www.postman.com/product/postbot/>



# Browser Use

[FEATURES](#)[PRICING](#)[BLOG](#)[DOCUMENTATION](#) 72,343 26.3K 23.5K[CLOUD](#)

[BETA] THE MOST STEALTH BROWSER INFRASTRUCTURE

## The AI browser agent

Repetitive work is dead. Browser Use empowers anyone to automate repetitive online tasks, no code required. No barriers. Simply tell it what you want done.

<https://browser-use.com/>



# Playwright Test Agent



<https://playwright.dev/docs/test-agents>



# AI Testing Tools

The screenshot displays the homepage of the AI Testing Tools website. At the top, the title "AI Testing Tools" is prominently displayed in purple. Below it, a subtitle reads: "Your one-stop destination for AI-powered tools for software testing, test automation, test management, and more." A navigation bar contains four buttons: "All" (highlighted in purple), "Test Automation", "Test Management", and "MCP Servers".

Below the navigation bar, there is a search bar with the placeholder text "Search among 75 items". To the right of the search bar are several filter options: "Category", "Pricing", "AI Capabilities", "Automation Capabilities", and a "Reset Filters" link.

The main content area features three featured tool cards:

- Test Collab**: Labeled "Featured" with a checkmark icon. The card shows a screenshot of the Test Collab interface, which includes a table of test cases with columns for name, status, and priority. Below the screenshot, the text reads: "Test Collab is AI-powered test management software for planning, executing, and tracking".
- Lynqa**: Labeled "New" and "Featured" with a checkmark icon. The card has a dark blue background with the Lynqa logo (a stylized blue fox head) and the text: "Execute your Xray tests, as-is, with AI". Below the card, it says: "Lynqa is an AI Agent for Manual Test Execution."
- Test Management By Testsigma**: Labeled "Featured" with a checkmark icon. The card shows a screenshot of the Test Management By Testsigma interface, which displays a list of test cases and their execution status. Below the screenshot, it says: "Test Management by Testsigma is a fully".

<https://www.testingtools.ai/>





# Deployment Process





# Deploy and manage



CI/CD pipeline

Infrastructure as a code

Automated script

Performance and monitoring suggestion

Document generation

AI-assist support

ChatOps, AIOps



# K8sGPT



CLOUD NATIVE  
SANDBOX

## K8sGPT joins the CNCF Sandbox

K8sGPT is a tool for scanning your kubernetes clusters, diagnosing and triaging issues in simple english. It has SRE experience codified into its analyzers and helps to pull out the most relevant information to enrich it with AI.

Get it now!

<https://k8sgpt.ai/>



# PromptOps



Solutions ▾

Resources ▾

Contact us

Log In

## ChatGPT for your DevOps Teams

Turn DevOps tasks into automated workflows  
with a single prompt straight from Slack

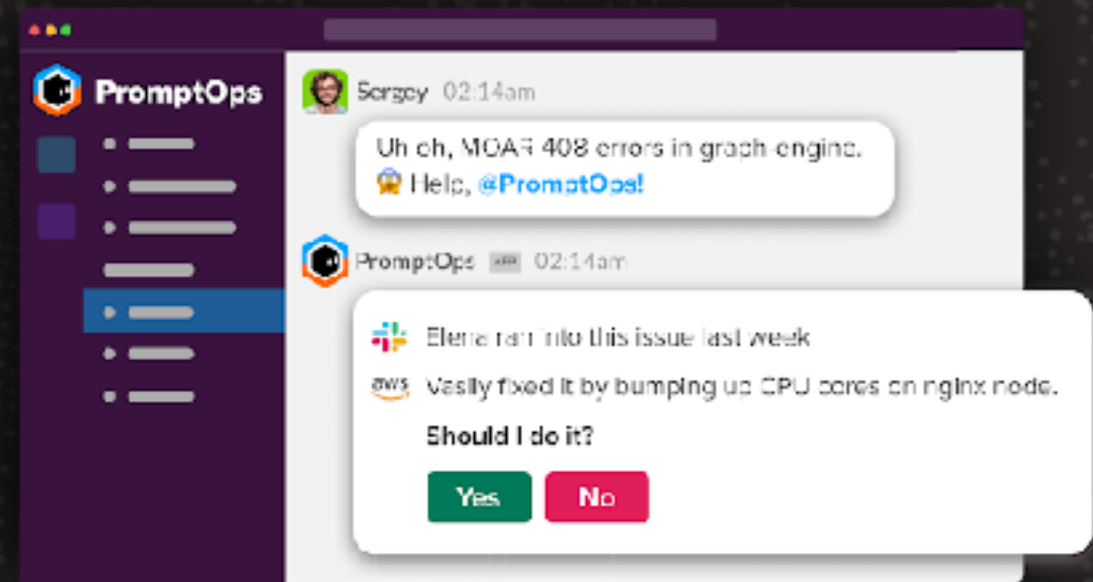
Get started

Learn More



PromptOps APP

Hello, I'm PromptOps, your DevOps virtual assistant for managing, troubleshooting, and running DevOps tasks directly from Slack!



Acknowledge

Resolve

Run a play ▾

Edit code

<https://www.promptops.com/devops/>



# ChatOps for DevOps

[Product](#)[How It Works](#)[Learn](#)[Company](#)[uptime](#)[Sign In](#)[Book a demo](#)[Sign Up](#)

## > ChatGPT for DevOps

Converse with your engineering platforms, powered by LLM.  
A virtual teammate to handle DevOps requests so you can handle the rest.

[Add Kubi to Slack](#)

Kubi (DevOps)

@Alerts I got an alert from Prometheus:

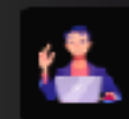


Deployment 'alert-manager' on namespace 'Openfaas' is experiencing high traffic



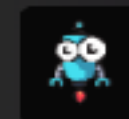
Kubi (DevOps)

@Alerts Should I increase the number of replicas on 'alert-manager'?



Jeff (R&D)

Yes



Kubi (DevOps)



The following deployment has been updated:

Deployment: alert-manager

Namespace: Openfaas

Replicas: 3

<https://www.kubiya.ai/>



# Risks when using Generative AI



# Risks

Quality of output generated

Explainability of decisions

Security policy !!

Sensitive data !!



# Tips

Understand what you want

Modular approach

Clear and Precise inputs

Make sure you understand the code





# Local LLM





# Local LLM

Run LLM on local machine/device  
Try to customize with your requirement

Reduce cost

Data privacy

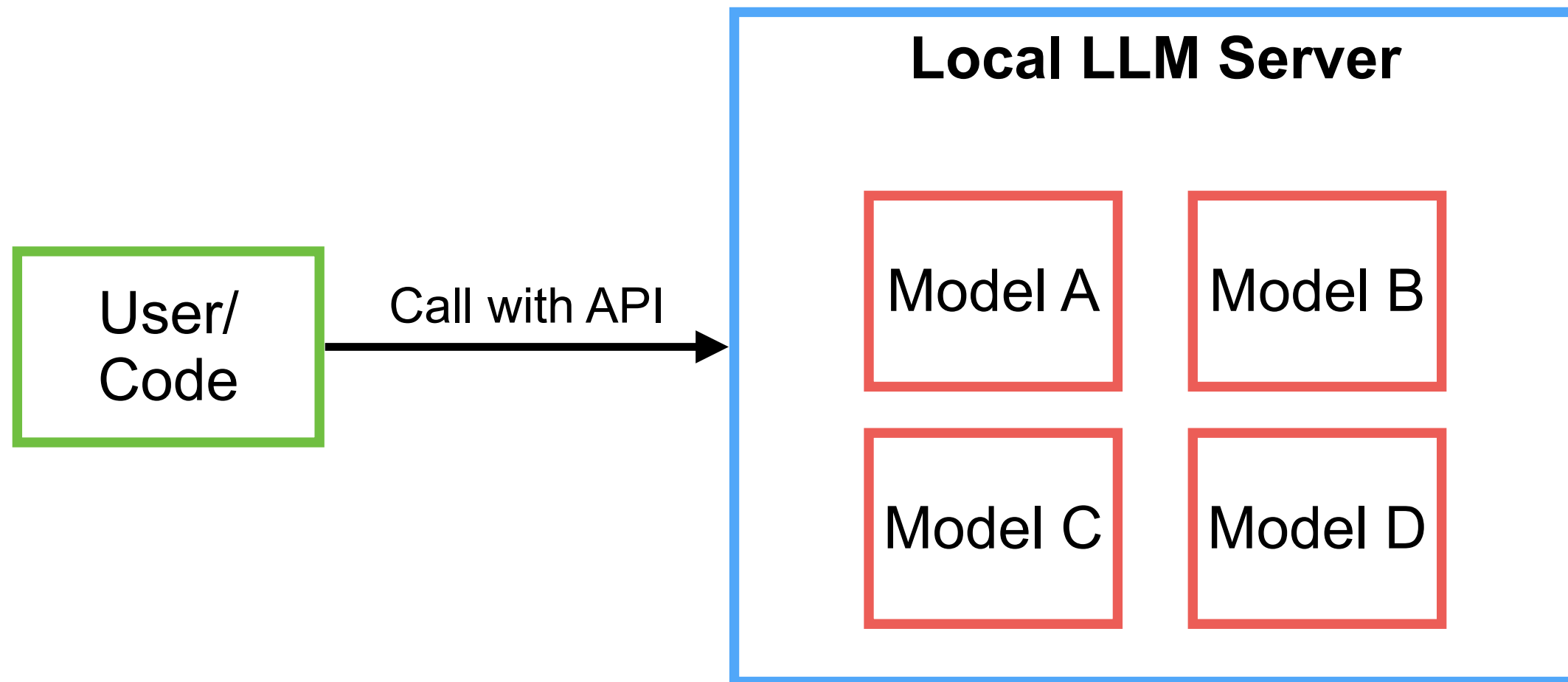
Responsive

Offline mode



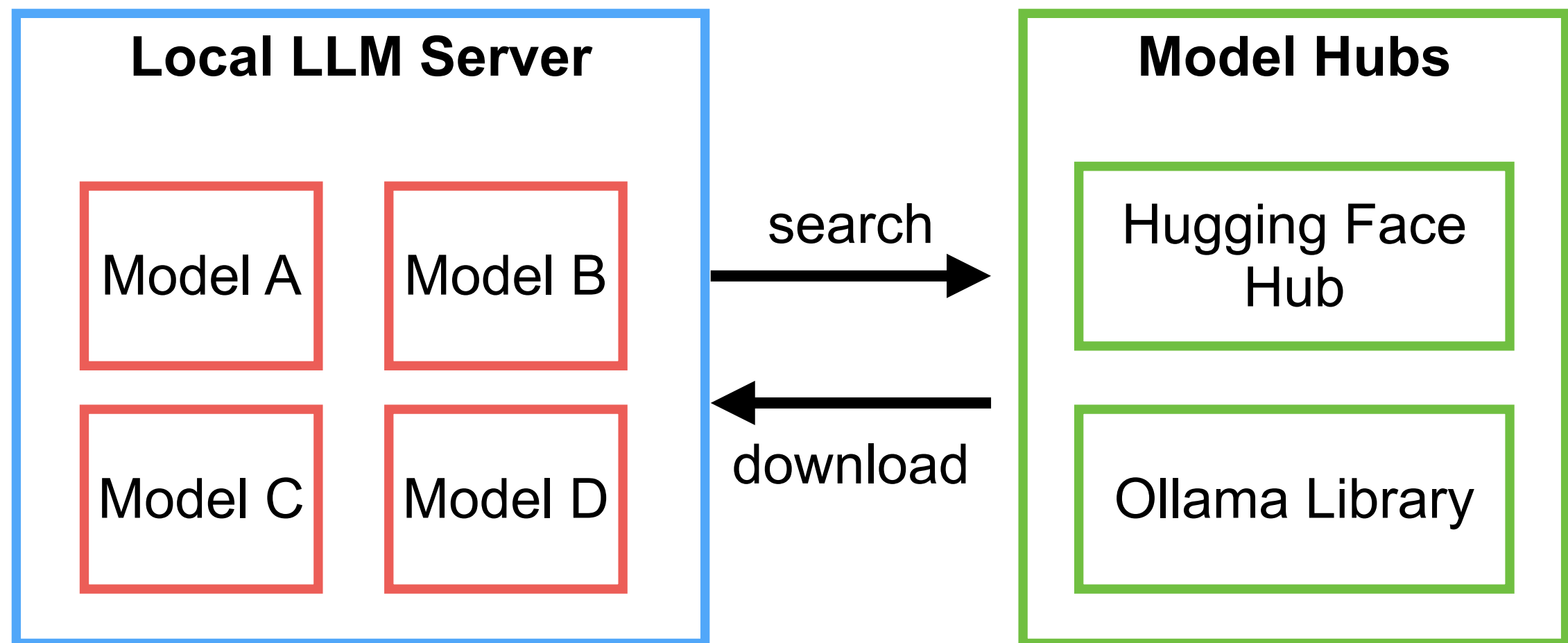
# Local LLM

Improve your LLM models, more accurate answer



# Models ?

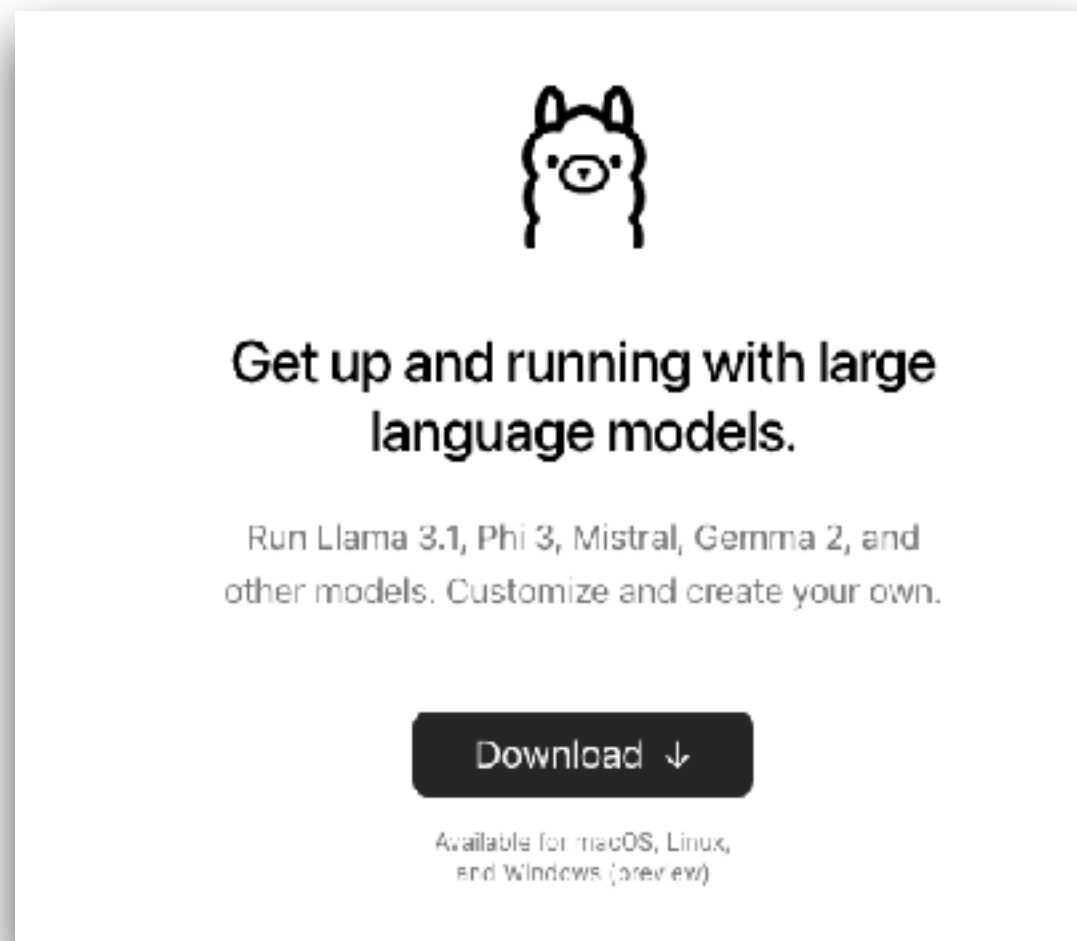
How to download models ?



# Local LLM with Ollama

\$ollama run **llama3.2**

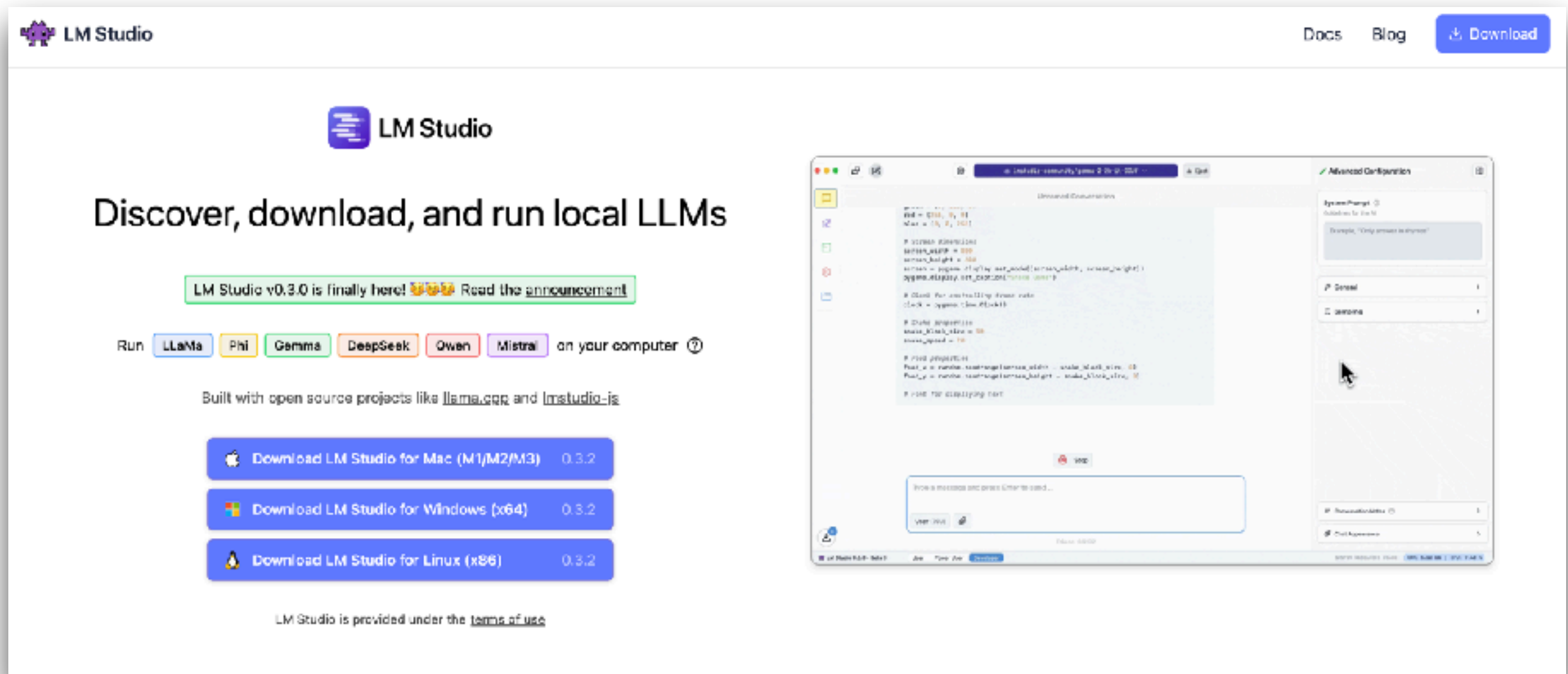
LLM model



<https://ollama.com/>



# Local LLM with LM Studio



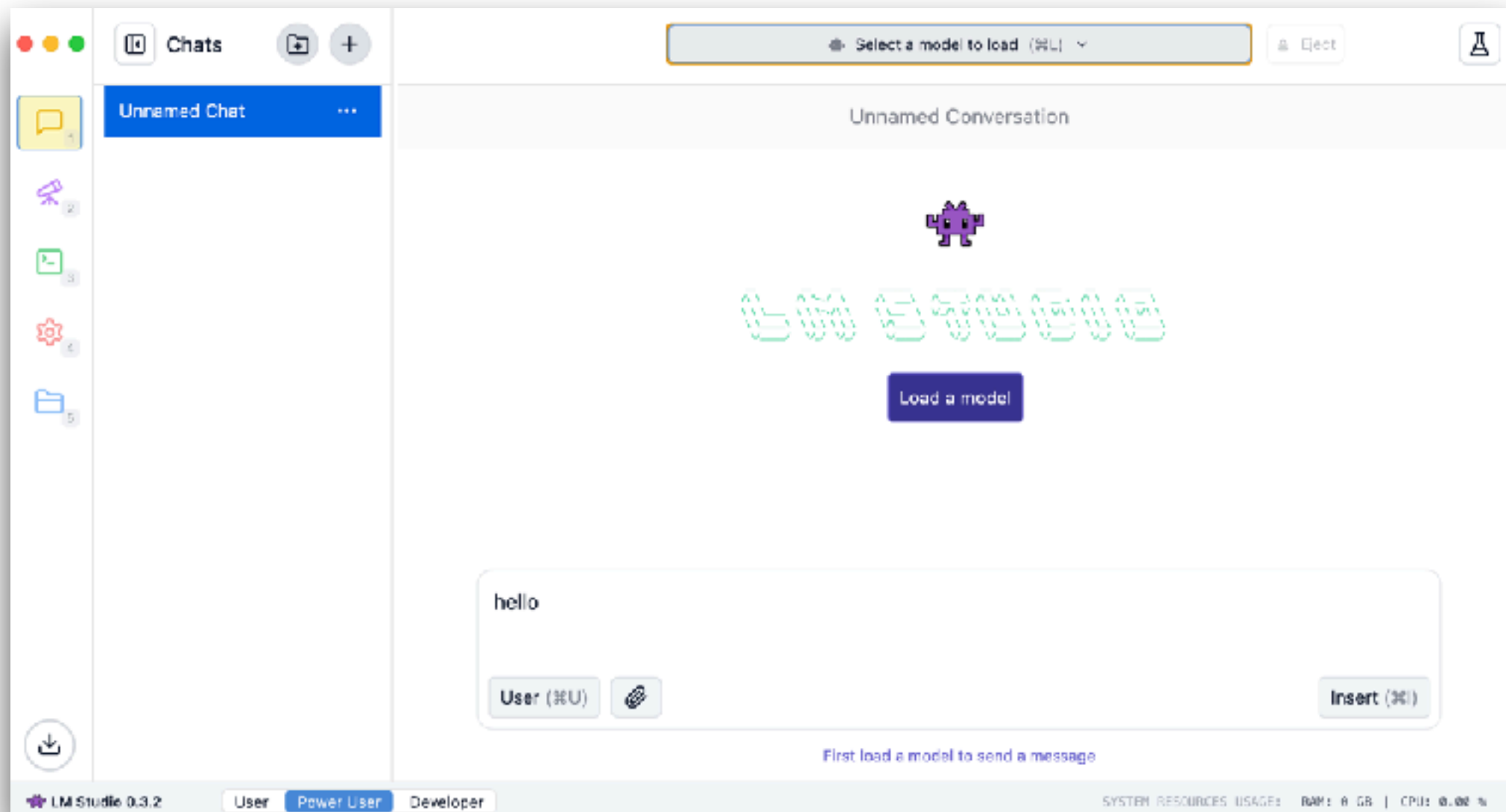
The image shows the LM Studio website and a preview of the application. The website header includes the LM Studio logo, navigation links for 'Docs', 'Blog', and a 'Download' button. The main heading is 'Discover, download, and run local LLMs'. Below this, a green box contains the text 'LM Studio v0.3.0 is finally here! 🎉🎉🎉 Read the announcement'. A section titled 'Run' lists various LLM models: LLaMa, Phi, Gemma, DeepSeek, Qwen, and Mistral. Below this, it states 'Built with open source projects like llama.cpp and lmstudio-is'. Three download buttons are provided for Mac (M1/M2/M3), Windows (x64), and Linux (x86), all version 0.3.2. At the bottom, it mentions 'LM Studio is provided under the terms of use'. The application preview on the right shows a chat interface with a code editor on the left displaying Lua-like code for a game, and a chat window on the right with a text input and a 'Send' button. The application title bar indicates it is running on a Mac.

<https://lmstudio.ai/>



# Local LLM with LM Studio

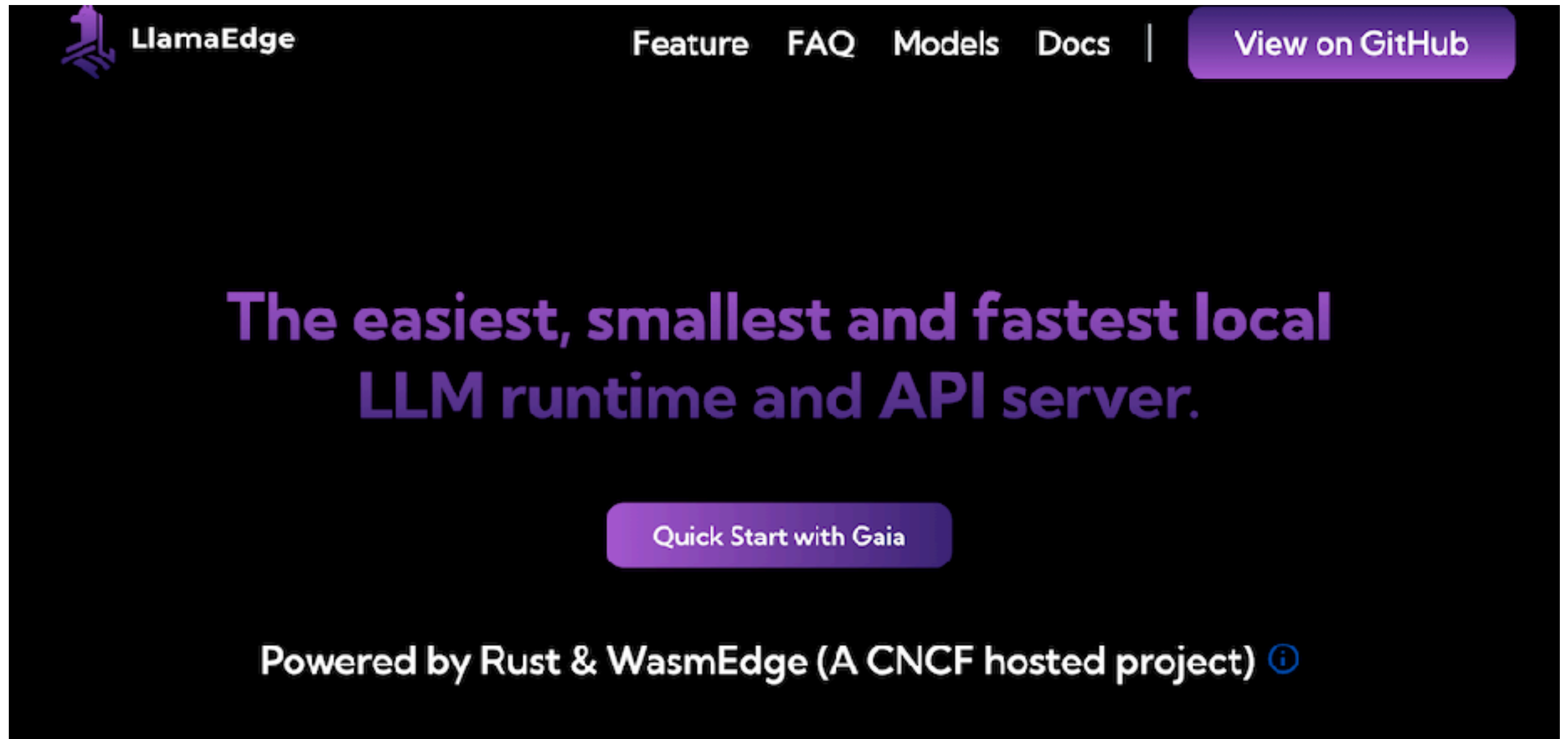
Load model from Hugging Face



<https://lmstudio.ai/>



# Local LLM with LlamaEdge

The image is a screenshot of the LlamaEdge website. At the top left is the LlamaEdge logo, which consists of a stylized purple bird-like icon and the text "LlamaEdge". To the right of the logo are navigation links: "Feature", "FAQ", "Models", and "Docs", separated by a vertical line. Further right is a purple button with the text "View on GitHub". The main body of the page has a dark background with the text "The easiest, smallest and fastest local LLM runtime and API server." in a large, bold, purple font. Below this text is a purple button with the text "Quick Start with Gaia". At the bottom of the page, it says "Powered by Rust & WasmEdge (A CNCF hosted project)" in white text, followed by a small blue information icon.

**LlamaEdge**

[Feature](#) [FAQ](#) [Models](#) [Docs](#) | [View on GitHub](#)

**The easiest, smallest and fastest local LLM runtime and API server.**

[Quick Start with Gaia](#)

Powered by Rust & WasmEdge (A CNCF hosted project) ⓘ

<https://llamaedge.com/>





# Local LLM with LocalAI



<https://localai.io/>





# More

GPT4AI

LlamaFile

Jan.ai

NextChat

Anything LLM

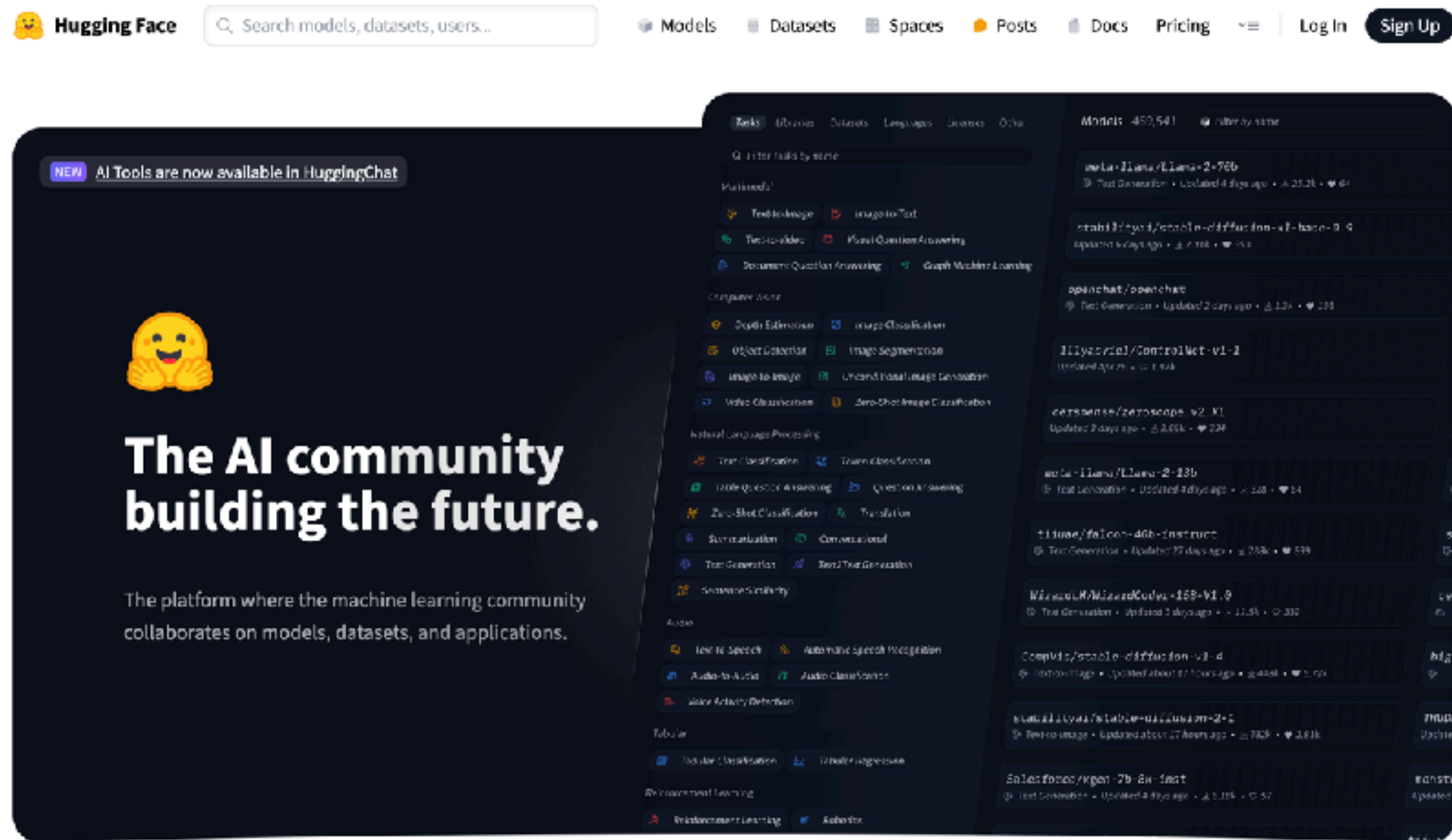
<https://github.com/Hannibal046/Awesome-LLM>



# LLM Models



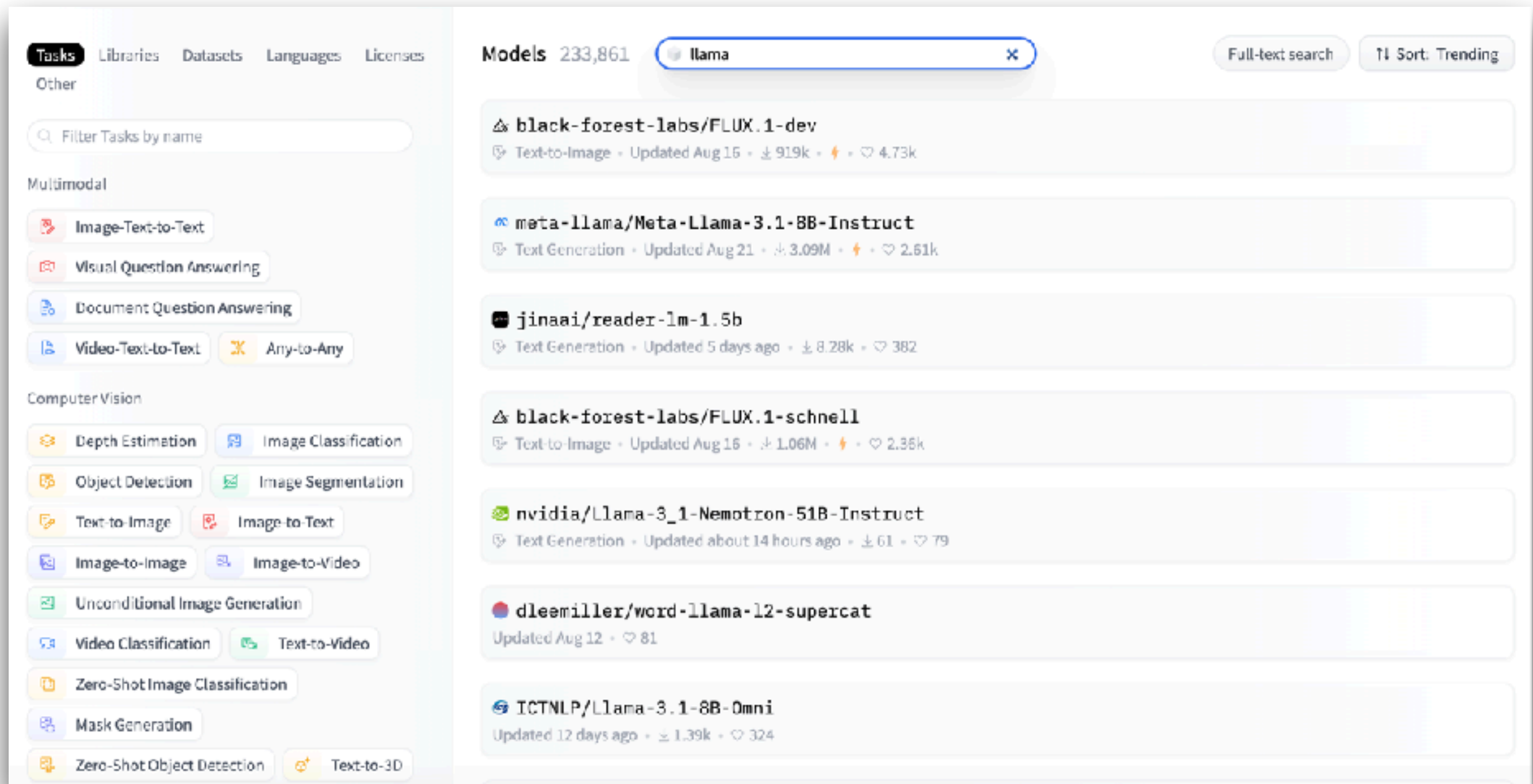
# Hugging Face Model Hub



<https://huggingface.co/>



# Hugging Face :: Model



The screenshot shows the Hugging Face Models interface. On the left, there's a sidebar with navigation tabs: Tasks, Libraries, Datasets, Languages, Licenses, and Other. Under 'Tasks', there are filters for Multimodal (Image-Text-to-Text, Visual Question Answering, Document Question Answering, Video-Text-to-Text, Any-to-Any) and Computer Vision (Depth Estimation, Image Classification, Object Detection, Image Segmentation, Text-to-Image, Image-to-Text, Image-to-Image, Image-to-Video, Unconditional Image Generation, Video Classification, Text-to-Video, Zero-Shot Image Classification, Mask Generation, Zero-Shot Object Detection, Text-to-3D). The main area displays a list of models under the 'llama' search filter, sorted by 'Trending'. The models listed are:

- black-forest-labs/FLUX.1-dev**: Text-to-Image, Updated Aug 16, 919k downloads, 4.73k likes.
- meta-llama/Meta-Llama-3.1-8B-Instruct**: Text Generation, Updated Aug 21, 3.09M downloads, 2.61k likes.
- jinaai/reader-lm-1.5b**: Text Generation, Updated 5 days ago, 8.28k downloads, 382 likes.
- black-forest-labs/FLUX.1-schnell**: Text-to-Image, Updated Aug 16, 1.06M downloads, 2.35k likes.
- nvidia/Llama-3\_1-Nemotron-51B-Instruct**: Text Generation, Updated about 14 hours ago, 61 downloads, 79 likes.
- dleemiller/word-llama-12-supercat**: Updated Aug 12, 81 likes.
- ICTNLP/Llama-3.1-8B-Omni**: Updated 12 days ago, 1.39k downloads, 324 likes.

<https://huggingface.co/>



# Big Code model leaderboard

★ Big Code Models Leaderboard

Inspired from the 🤖 Open LLM Leaderboard and 🤖 Open LLM-Perf Leaderboard, we compare performance of base multilingual code generation models on HumanEval benchmark and MultiPL-E. We also measure throughput and provide information about the models. We only compare open pre-trained multilingual code models, that people can start from as base models for their trainings.

🔍 Evaluation table 📊 Performance Plot 📄 About 🚀 Submit results

🔍 See All Columns

🔍 Search for your model and press ENTER...

🏷️ Filter model types

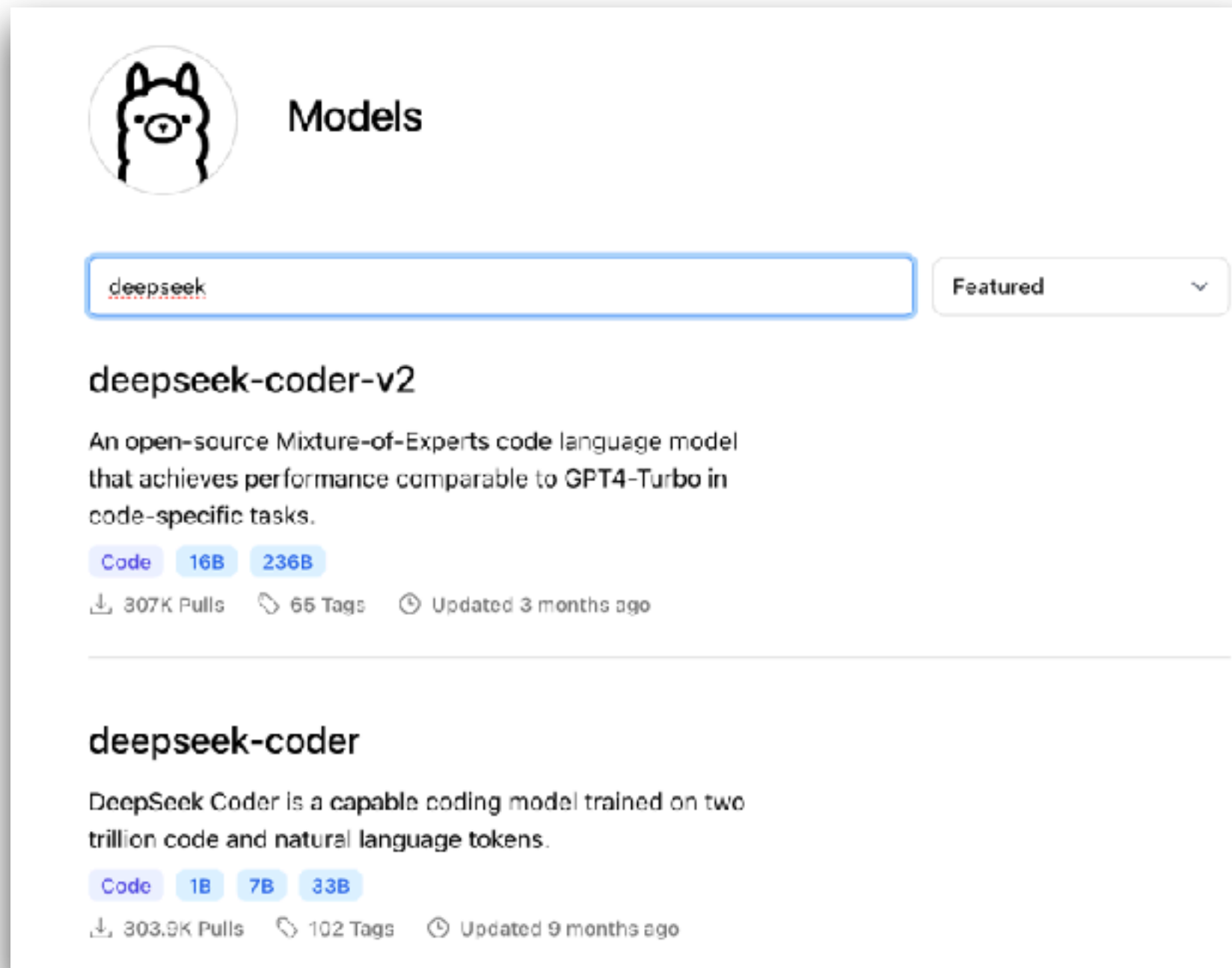
☒ all ☐ base ☐ instruction-tuned ☐ EXT external-evaluation

| T     | Model                                       | Win Rate | humaneval-python | java  | javascript | cpp   |
|-------|---------------------------------------------|----------|------------------|-------|------------|-------|
| 🔍 EXT | <a href="#">OpenCodeInterpreter-DS-33B</a>  | 55.83    | 75.23            | 54.8  | 69.06      | 64.47 |
| 🔍 EXT | <a href="#">Nxcode-CO-7B-orpo</a>           | 55.42    | 87.23            | 60.91 | 71.69      | 68.04 |
| 🔍     | <a href="#">CodeQwen1.5-7B-Chat</a>         | 55.08    | 87.2             | 61.04 | 70.31      | 67.85 |
| 🔍 EXT | <a href="#">CodeFuse-DeepSeek-33b</a>       | 54.33    | 76.83            | 60.76 | 66.46      | 65.22 |
| 🔍 EXT | <a href="#">DeepSeek-Coder-33b-instruct</a> | 52       | 80.02            | 52.03 | 65.13      | 62.36 |
| 🔍 EXT | <a href="#">Artigonz-Coder-DS-6.7B</a>      | 51.5     | 70.89            | 56.84 | 66.16      | 59.75 |
| 🔍 EXT | <a href="#">DeepSeek-Coder-7b-instruct</a>  | 50.33    | 80.22            | 53.34 | 65.8       | 59.66 |
| 🔍 EXT | <a href="#">OpenCodeInterpreter-DS-6.7B</a> | 49.67    | 73.2             | 51.41 | 63.85      | 60.81 |

<https://huggingface.co/spaces/bigcode/bigcode-models-leaderboard>



# Model in Ollama

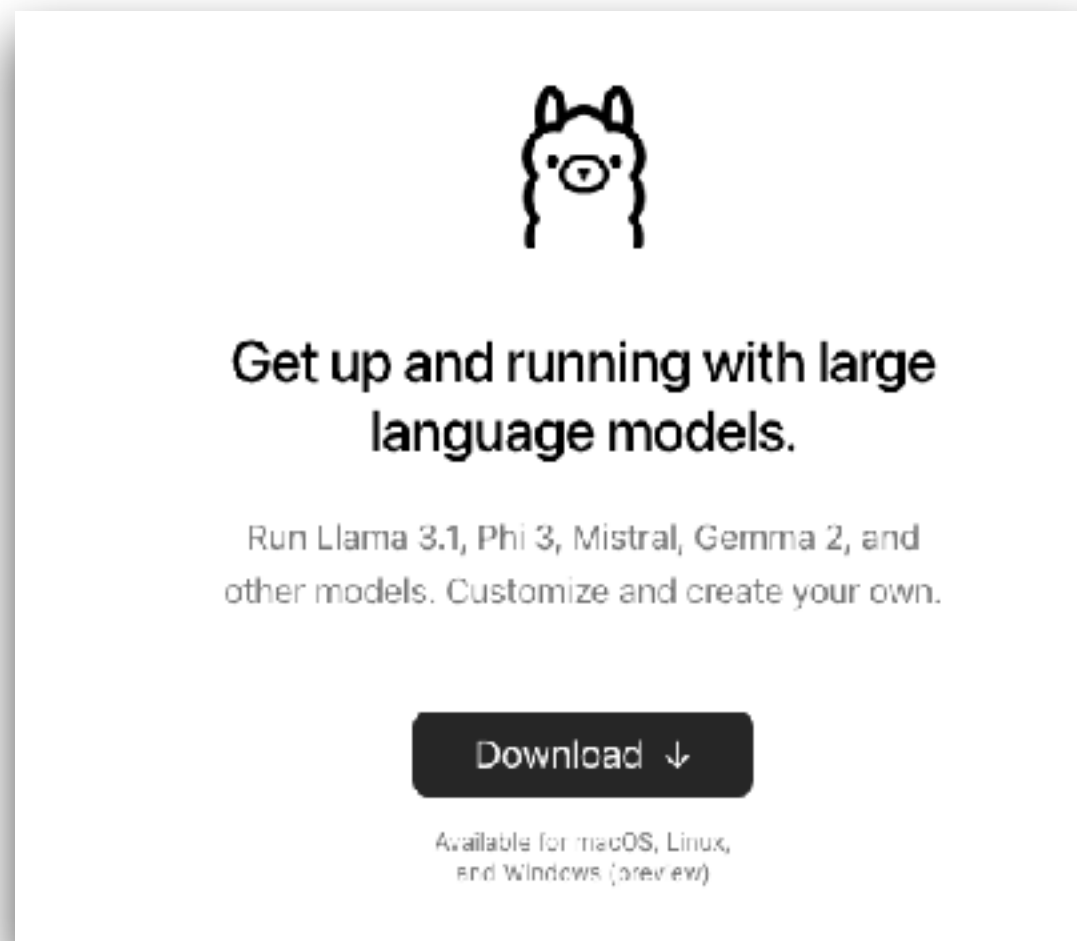


<https://ollama.com/library>



# Workshop with Ollama

\$ollama run **llama3.1**



<https://github.com/up1/workshop-ai-with-technical-team/wiki/Local-LLM-with-Ollama>



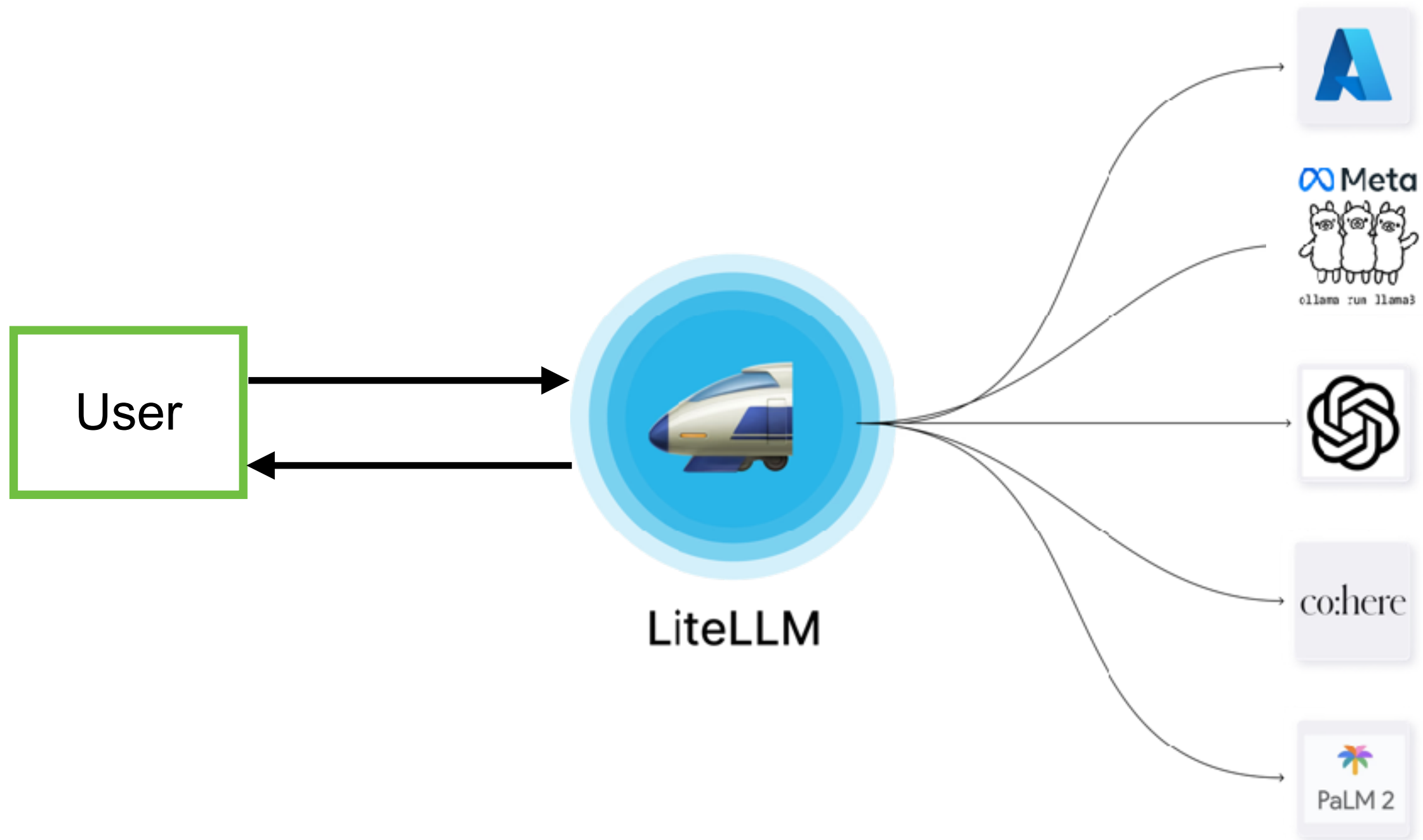


# LiteLLM as a Proxy





# LiteLLM as a Proxy

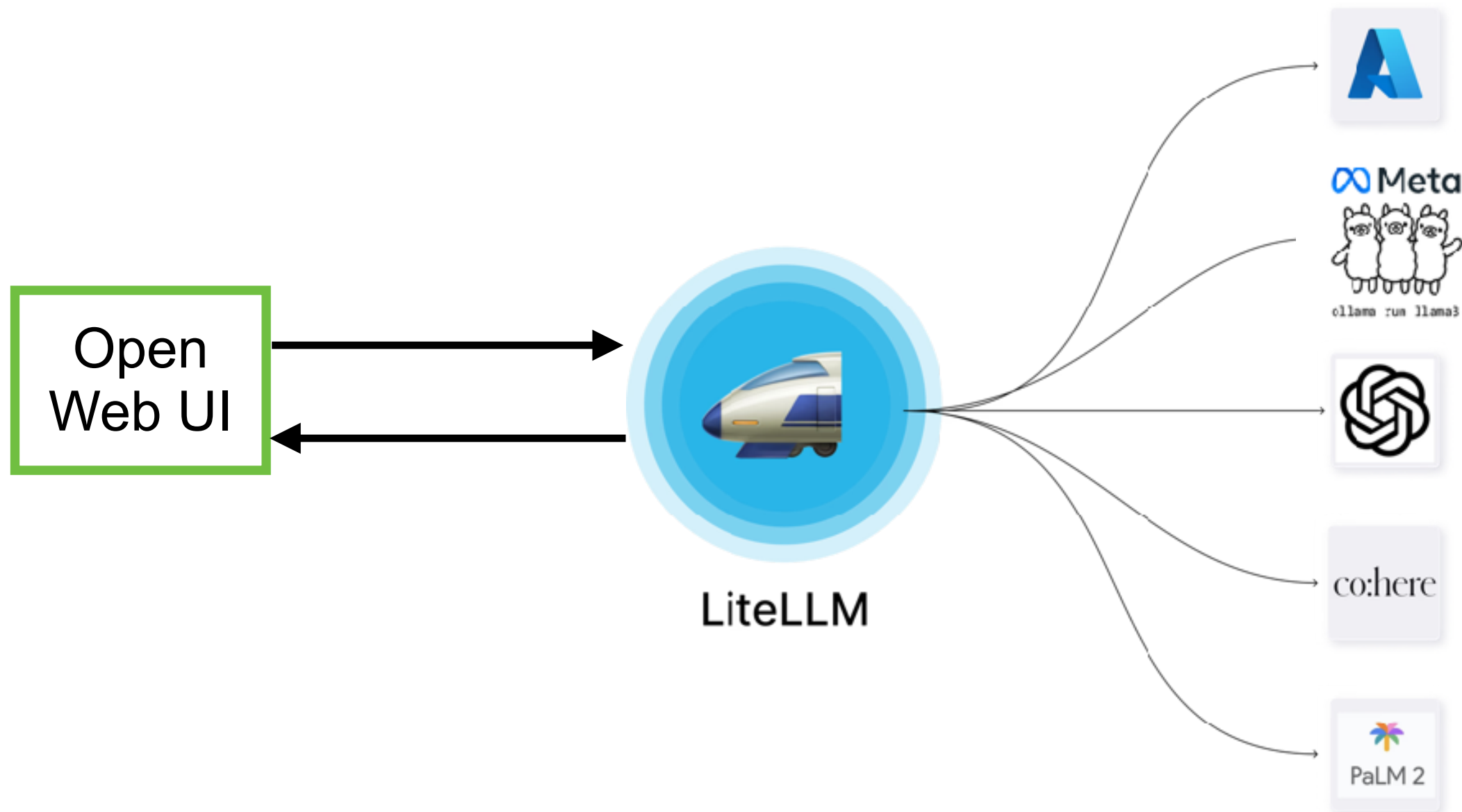


<https://www.litellm.ai/>



# Workshop

Use docker compose to build and run



<https://github.com/up1/workshop-ai-with-technical-team/wiki/LiteLLM-and-WebUI>

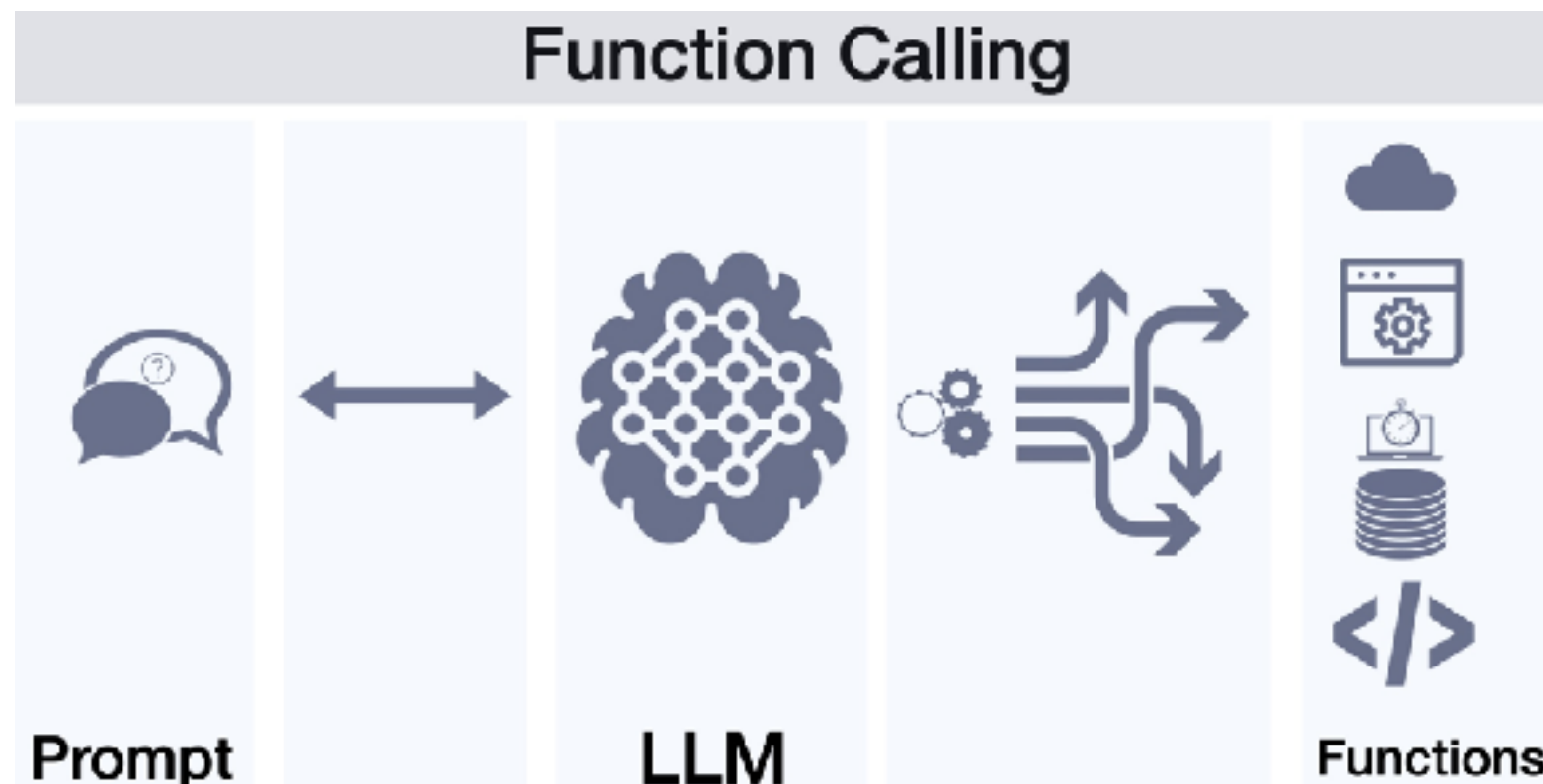


# Function Calling ?

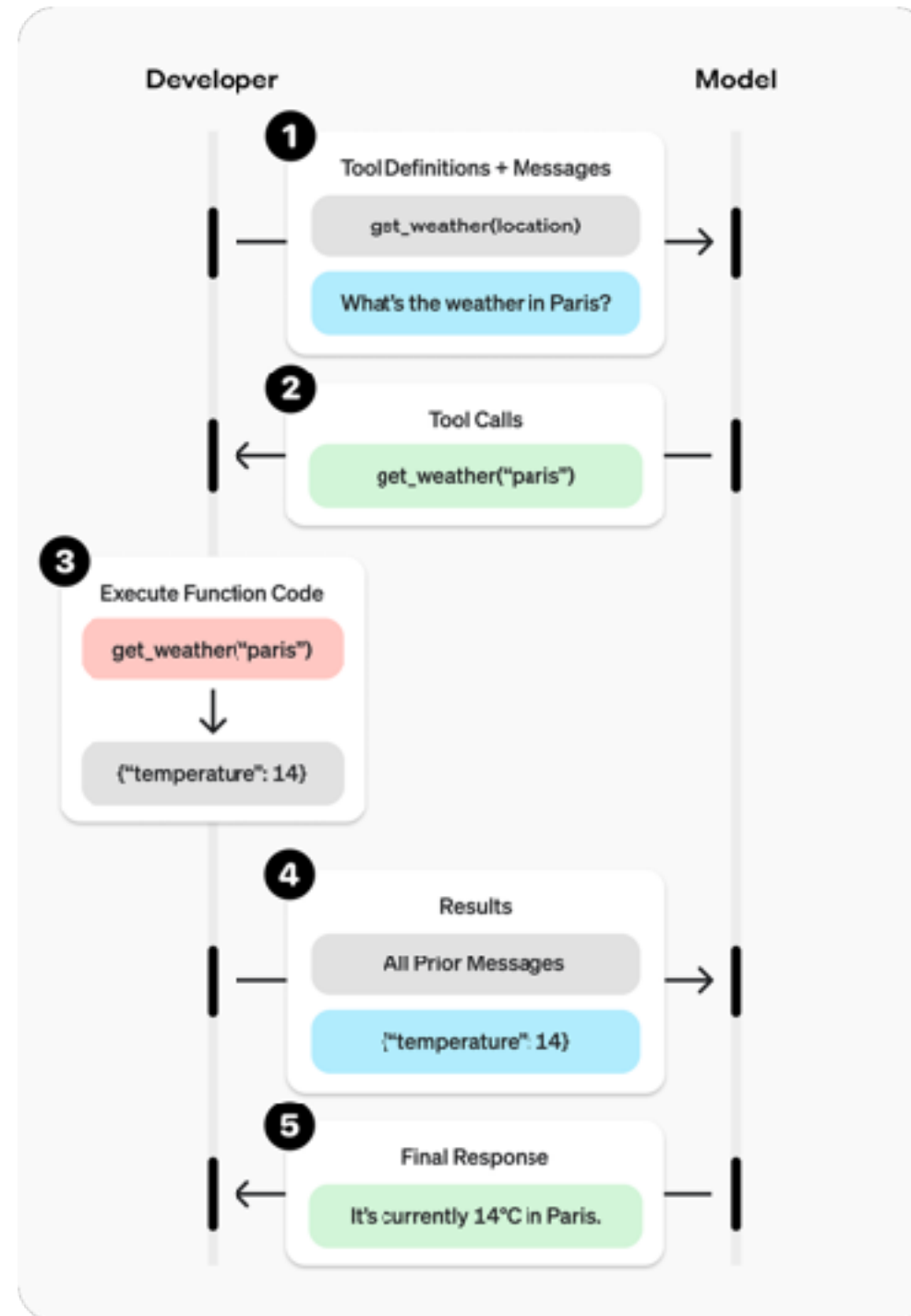


# Function Calling ?

Allow LLM to recognize what tool it need based on user's input and when to invoke it.



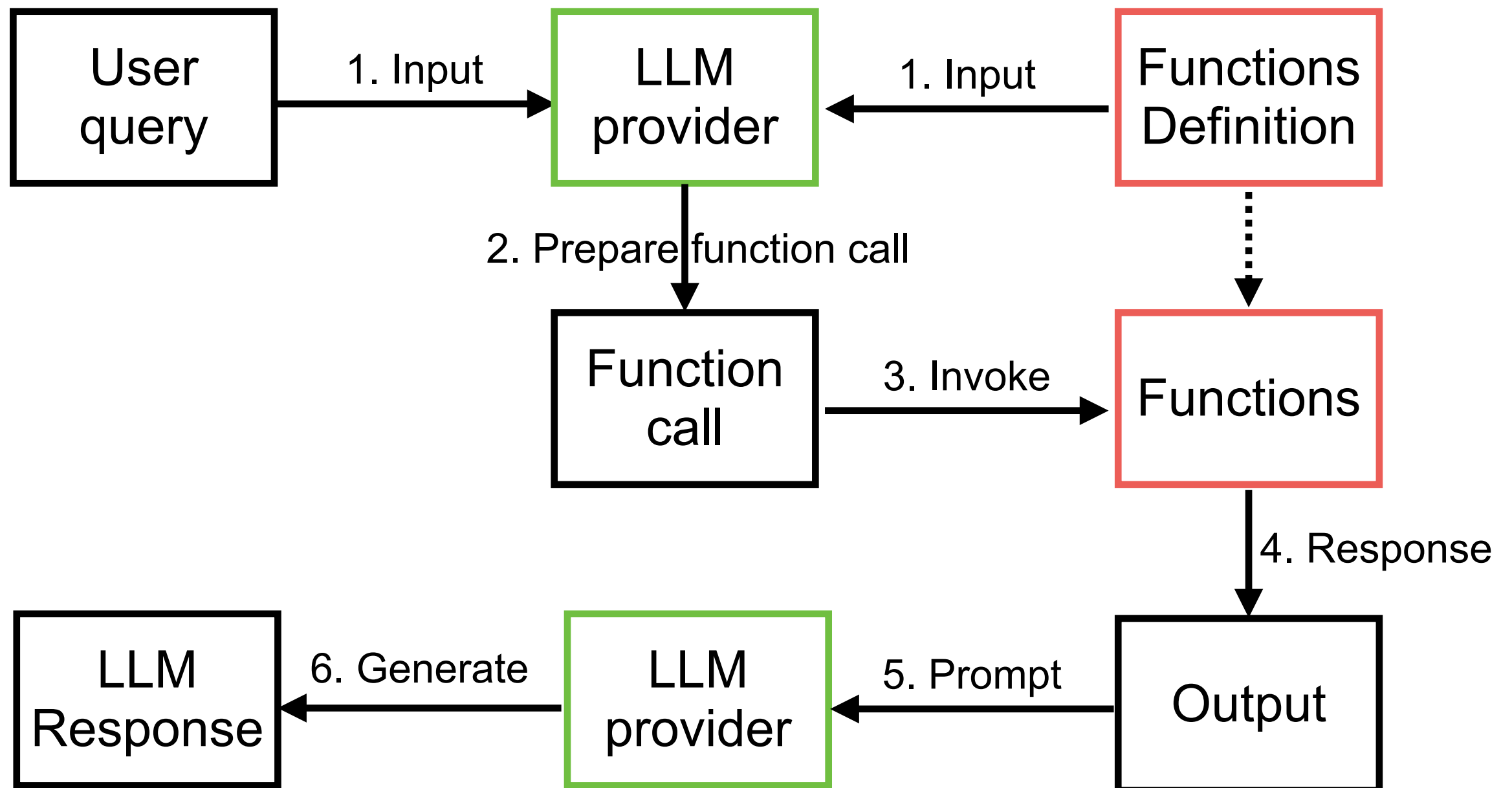
# Function calling ?



<https://platform.openai.com/docs/guides/function-calling?api-mode=responses>



# Function Calling



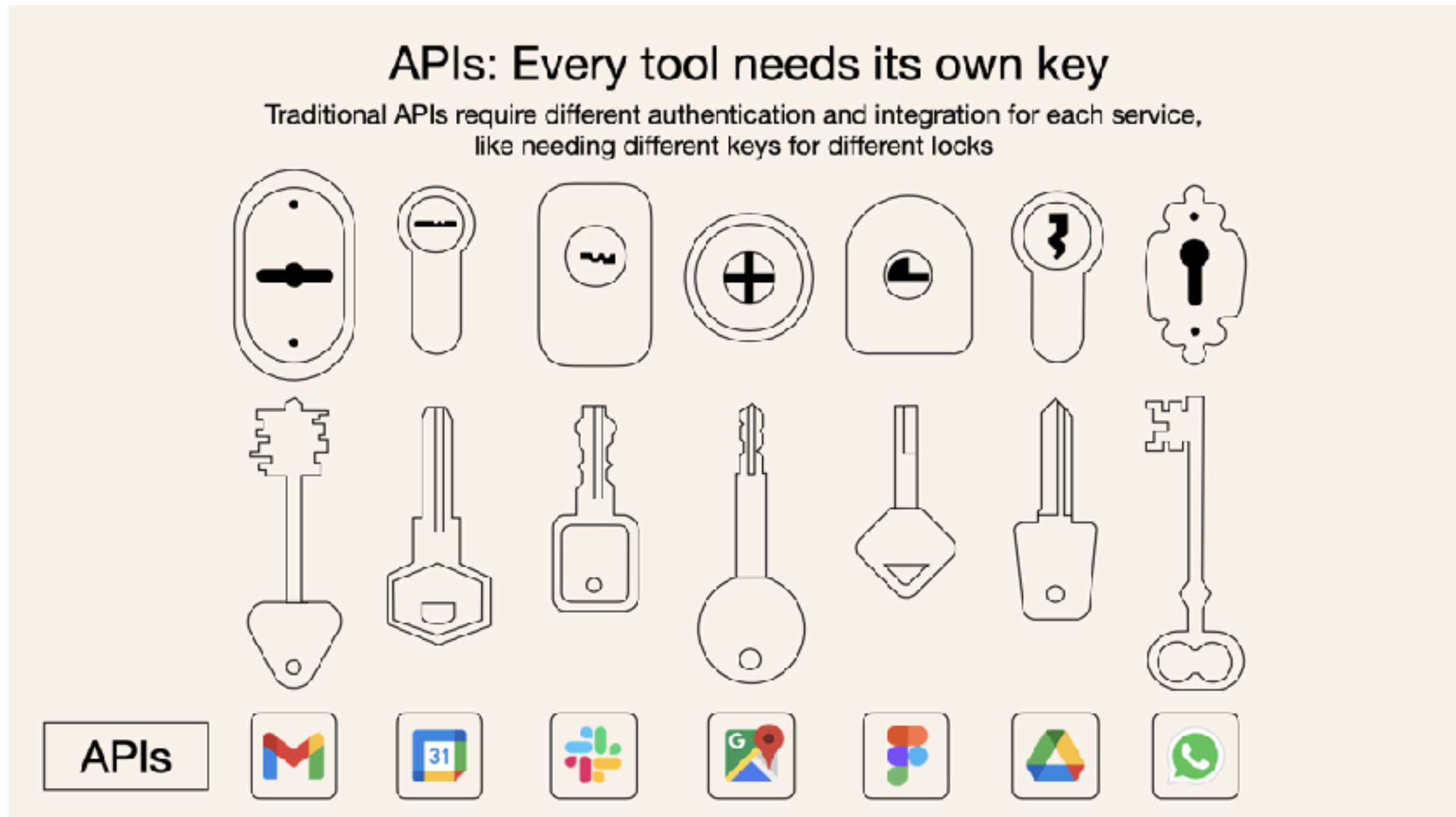
# Support function calling

| Provider name | Model name                        |
|---------------|-----------------------------------|
| OpenAI        | GPT 4                             |
| Anthropic     | Claude 3<br>(Sonnet, Haiku, Opus) |
| Google        | Gemini                            |





# Function calling !!

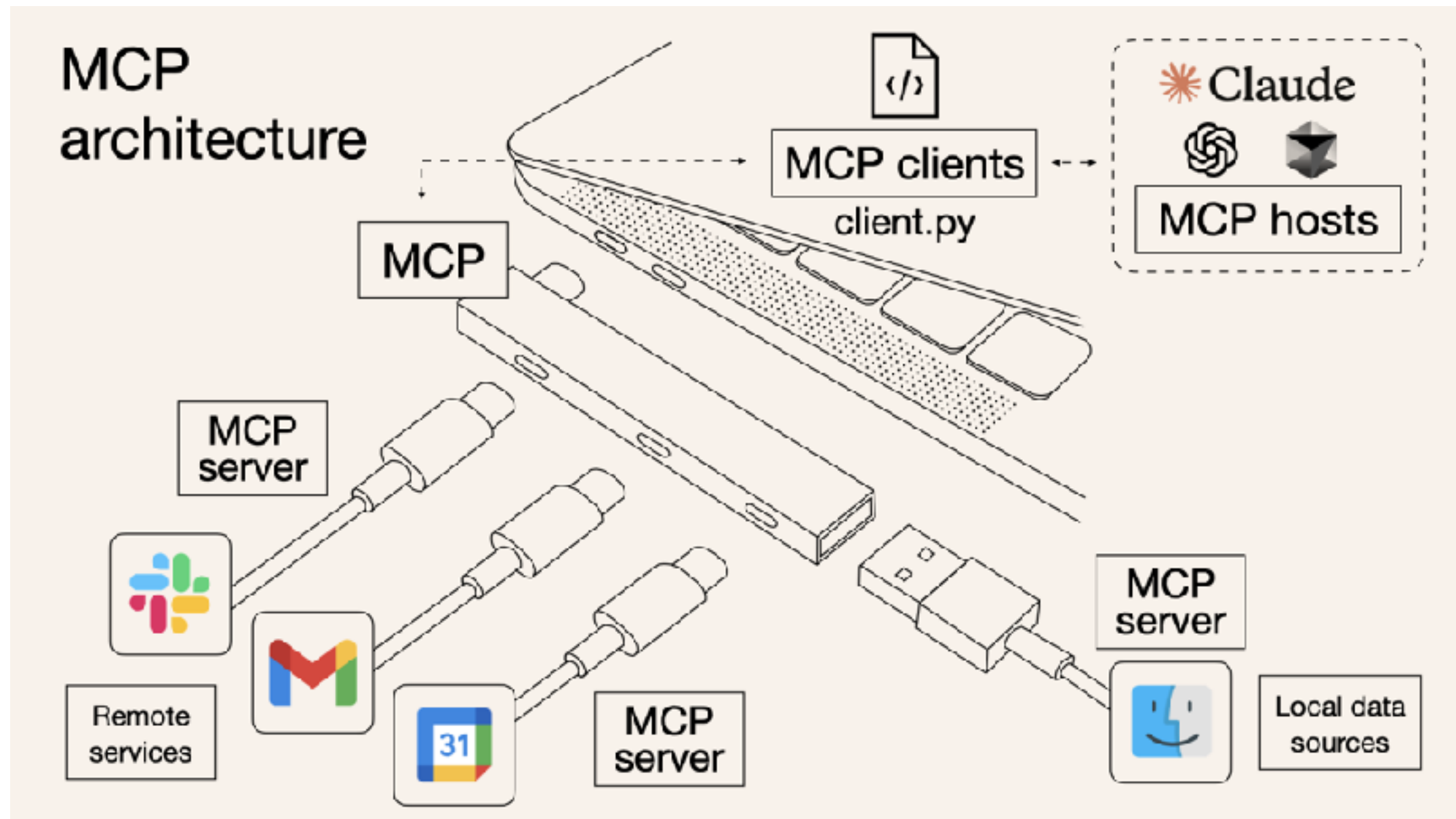


<https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/>





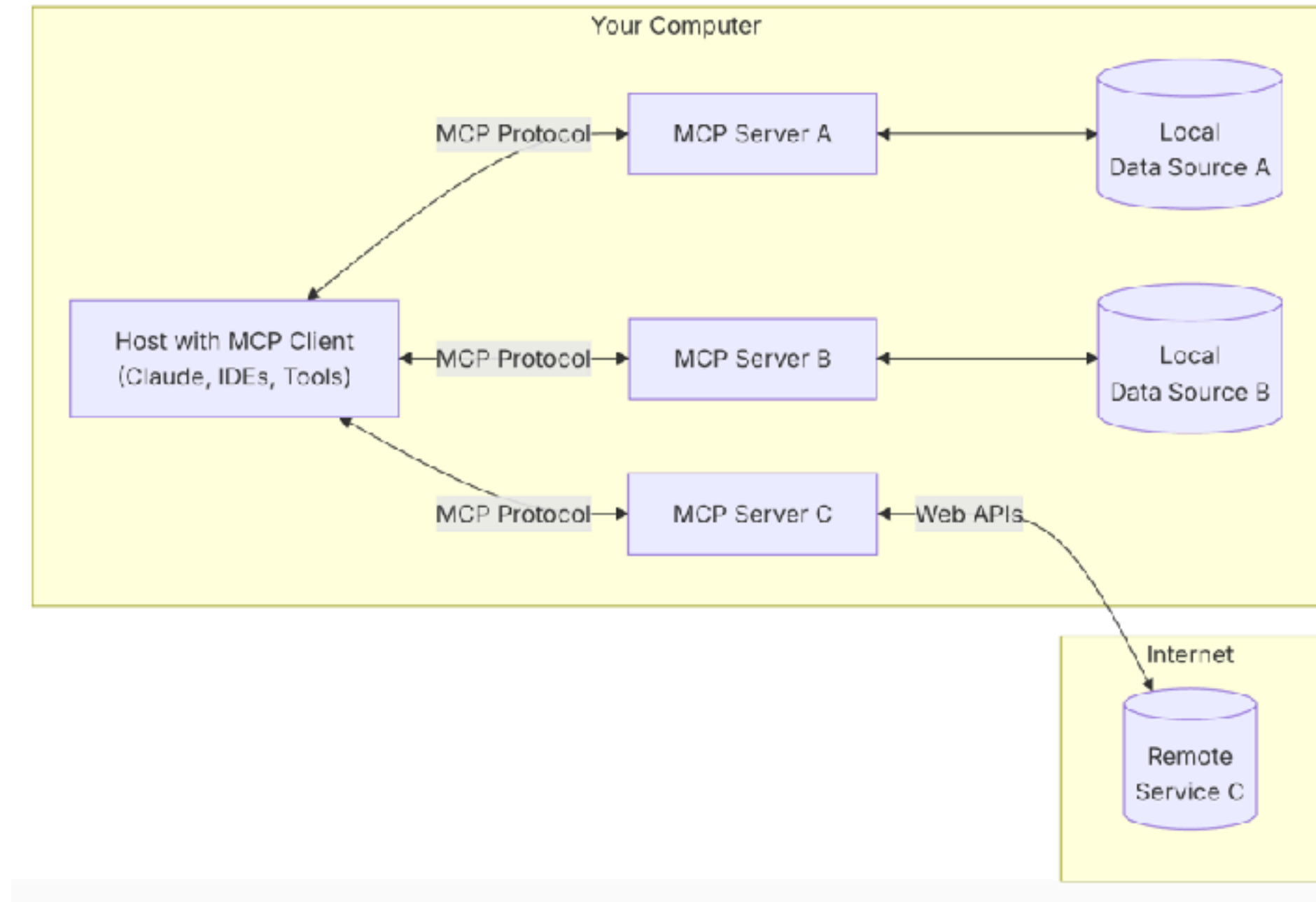
# Standardized function calling !!



<https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/>



# Model Context Protocol



<https://modelcontextprotocol.io/>



# Awesome MCP

## Find Awesome MCP Servers and Clients

MCP.so is a third-party MCP Marketplace with **17495** MCP Servers collected.

[ShipAny](#): NextJS boilerplate for building AI SaaS startups.

Today

Featured

Latest

Clients

Hosted

Official

Featured MCP Servers

View All →

**EdgeOne Pages MCP**

An MCP service designed for deploying HTML content to...

★

**AlphaVantage**

Bring enterprise-grade stock market data to agents and LLMs

★

**Time**

A Model Context Protocol server that provides time and timezone...

★

**Zhipu Web Search**

Zhipu Web Search MCP Server is a search engine specifically...

★

**MCP Advisor**

MCP Advisor & installation - Use the right MCP server for your...

★

**HowtoCook Mcp**

基于Anduin2017 / HowToCook (程序员在家做饭指南) 的mcp server, ...

★

**MiniMax MCP**

Official MiniMax Model Context Protocol (MCP) server that enabl...

★

**Serper MCP Server**

A Serper MCP Server

★

**Jina AI MCP Tools**

A Model Context Protocol (MCP) server that integrates with Jina AI...

★

**Amap Maps**

高德地图官方 MCP Server

★

**Playwright Mcp**

Playwright MCP server

★

**Baidu Map**

百度地图核心API现已全面兼容MCP协议，是国内首家兼容MCP协议的地图...

★

<https://mcp.so/>

AI for Software Development  
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172

# Retrieval-Augmented Generation (RAG)



# RAG

Enhances LLMs by retrieving external knowledge before generating response

Improve accuracy

Reduce  
hallucinations

Real-time  
knowledge updates



# Core Components

## Retriever

Fetch relevant documents from a knowledge base

## Generator

Uses the retrieved information to generate a response

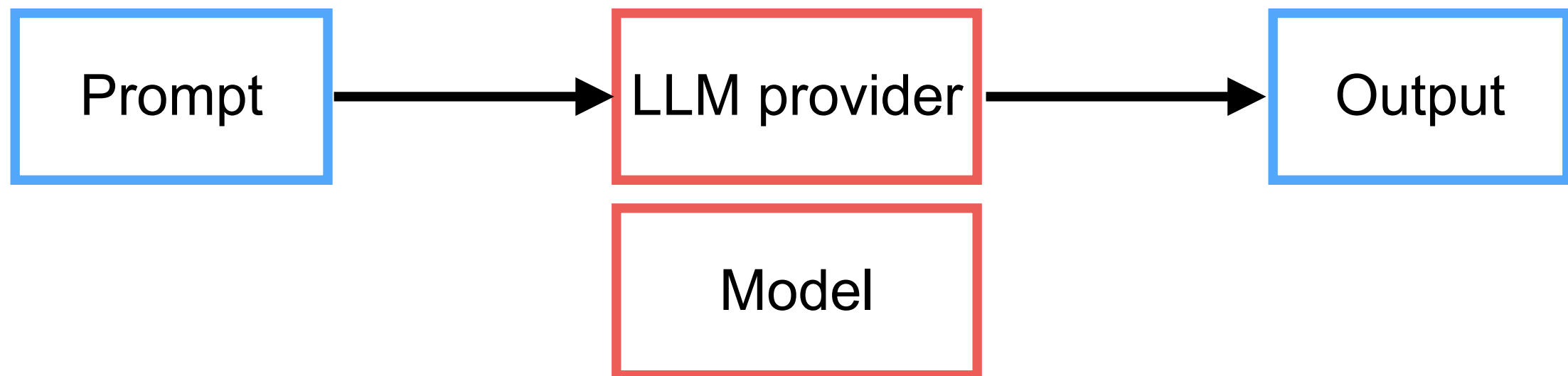
## Enhancements

Different RAG architectures modify how retrieval and generation interact to improve performance



# Limitation of LLM

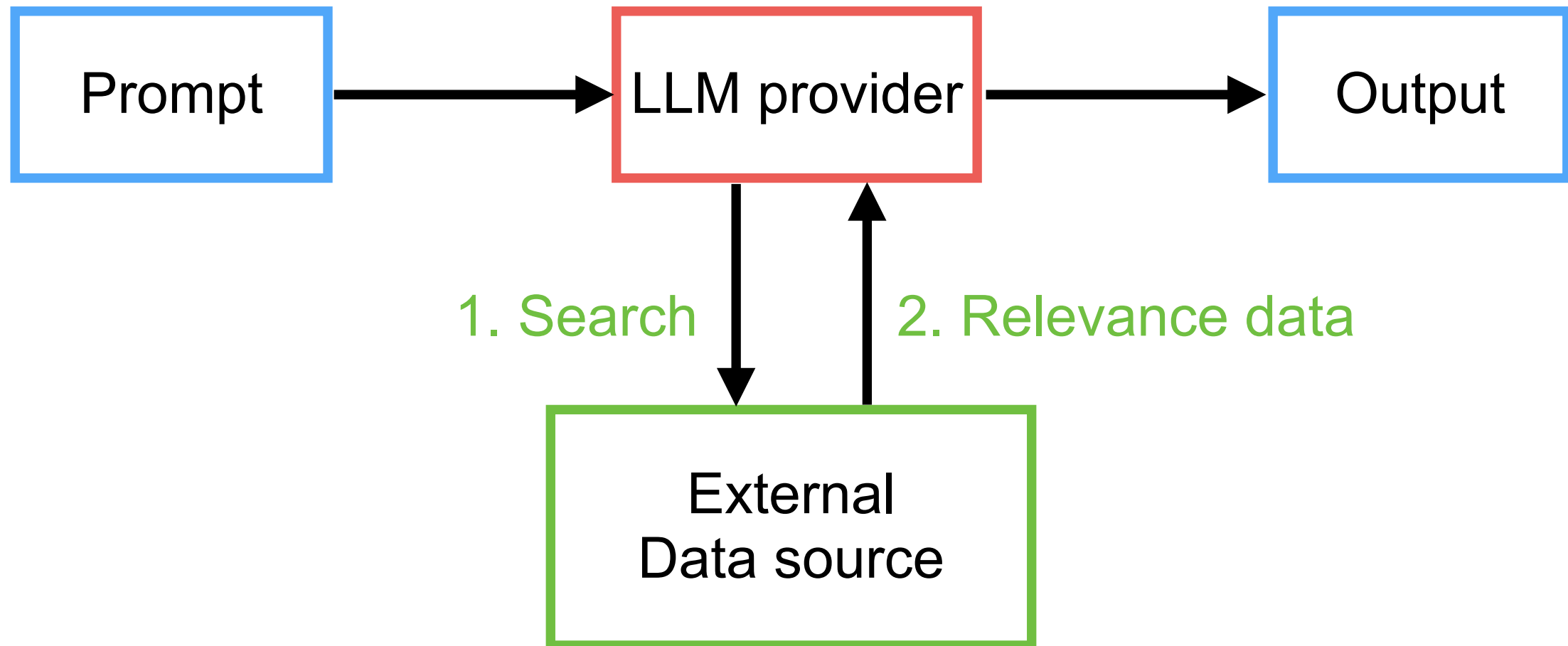
Fixed-knowledge !!





# RAG ?

Fixed-knowledge !!





# How to search/retrieve data ?

Data source

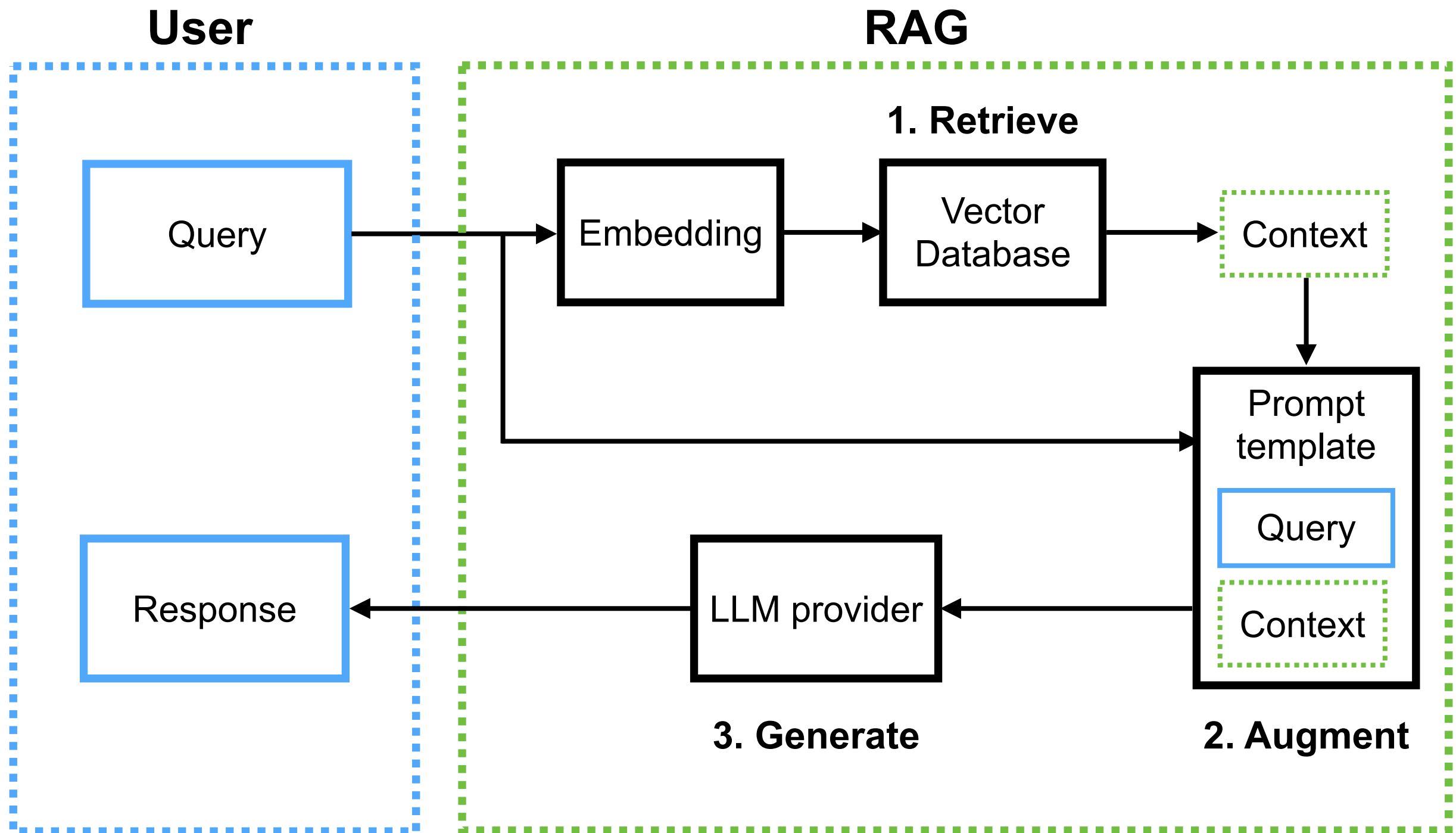
Data preparation

Search

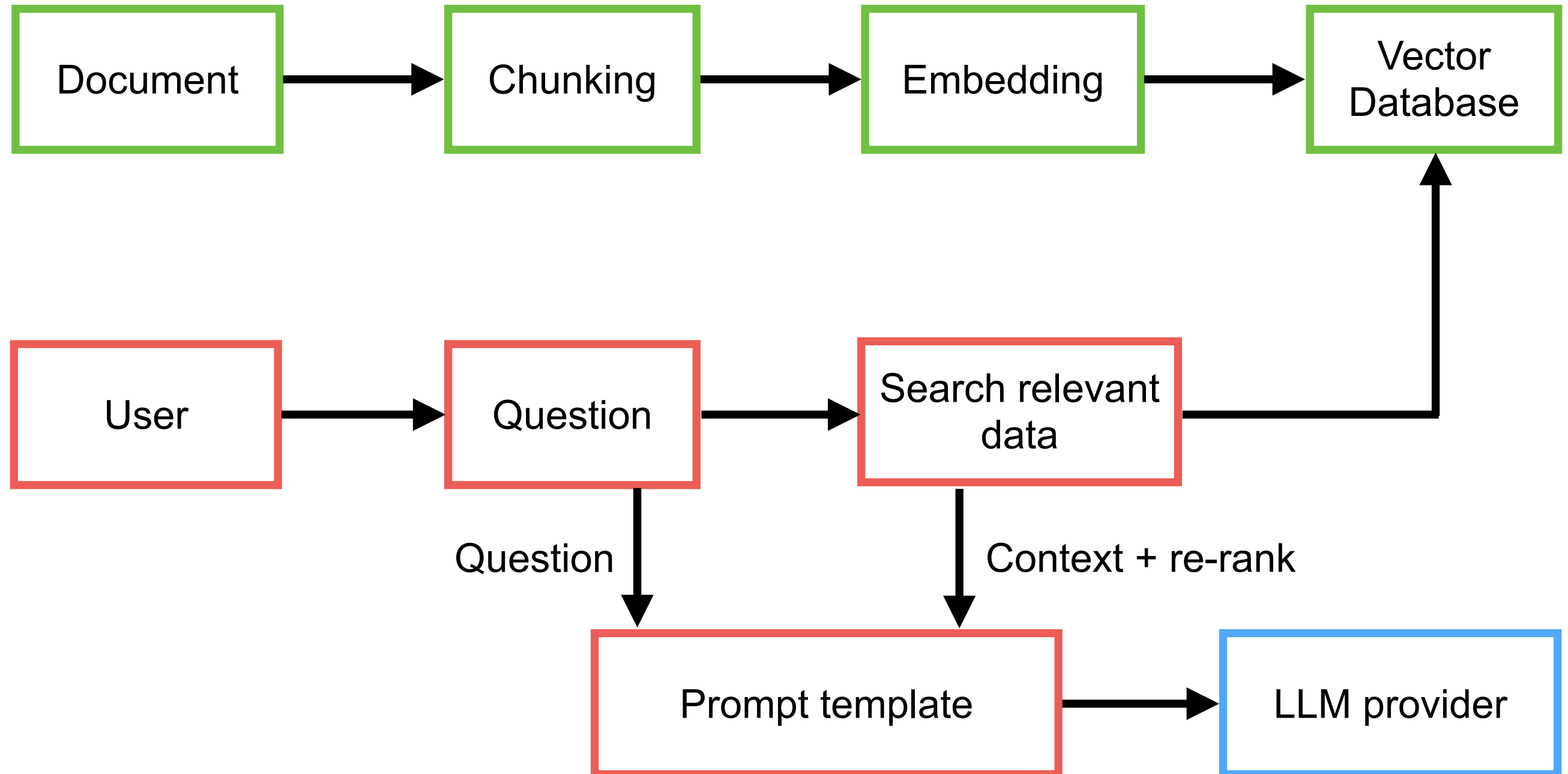
Search result



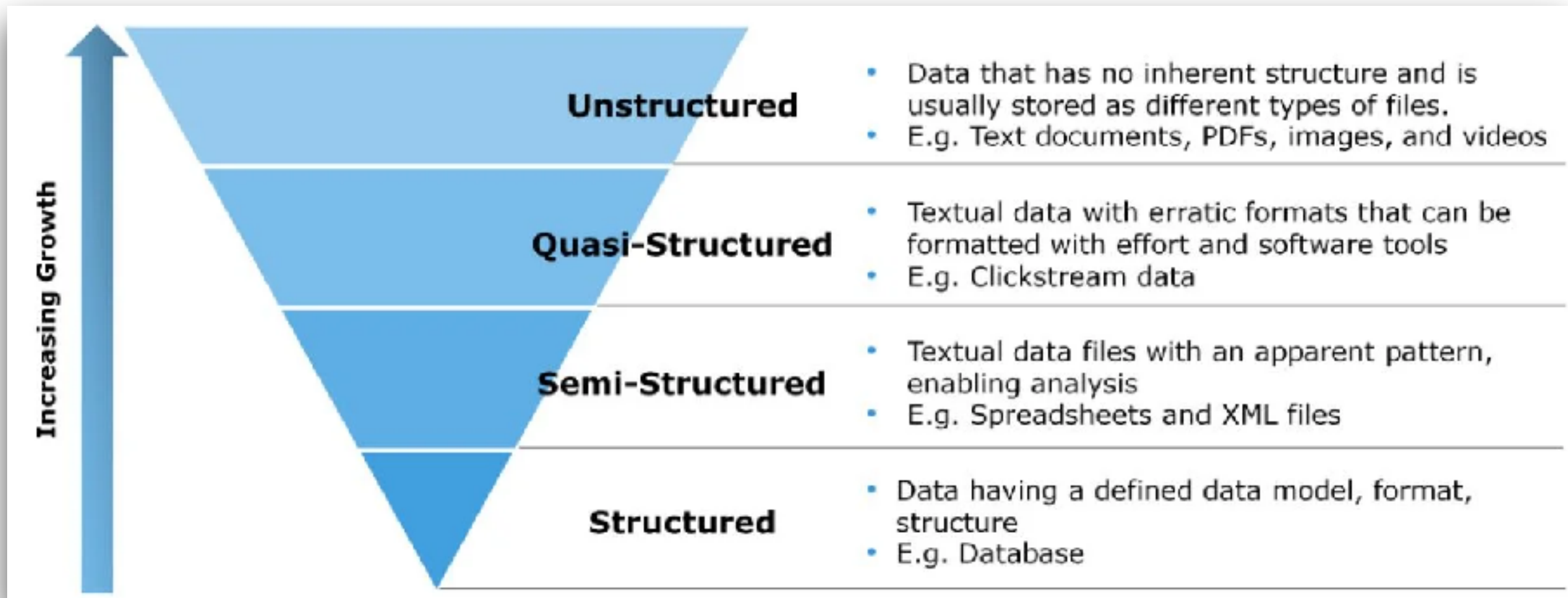
# Basic RAG Architecture



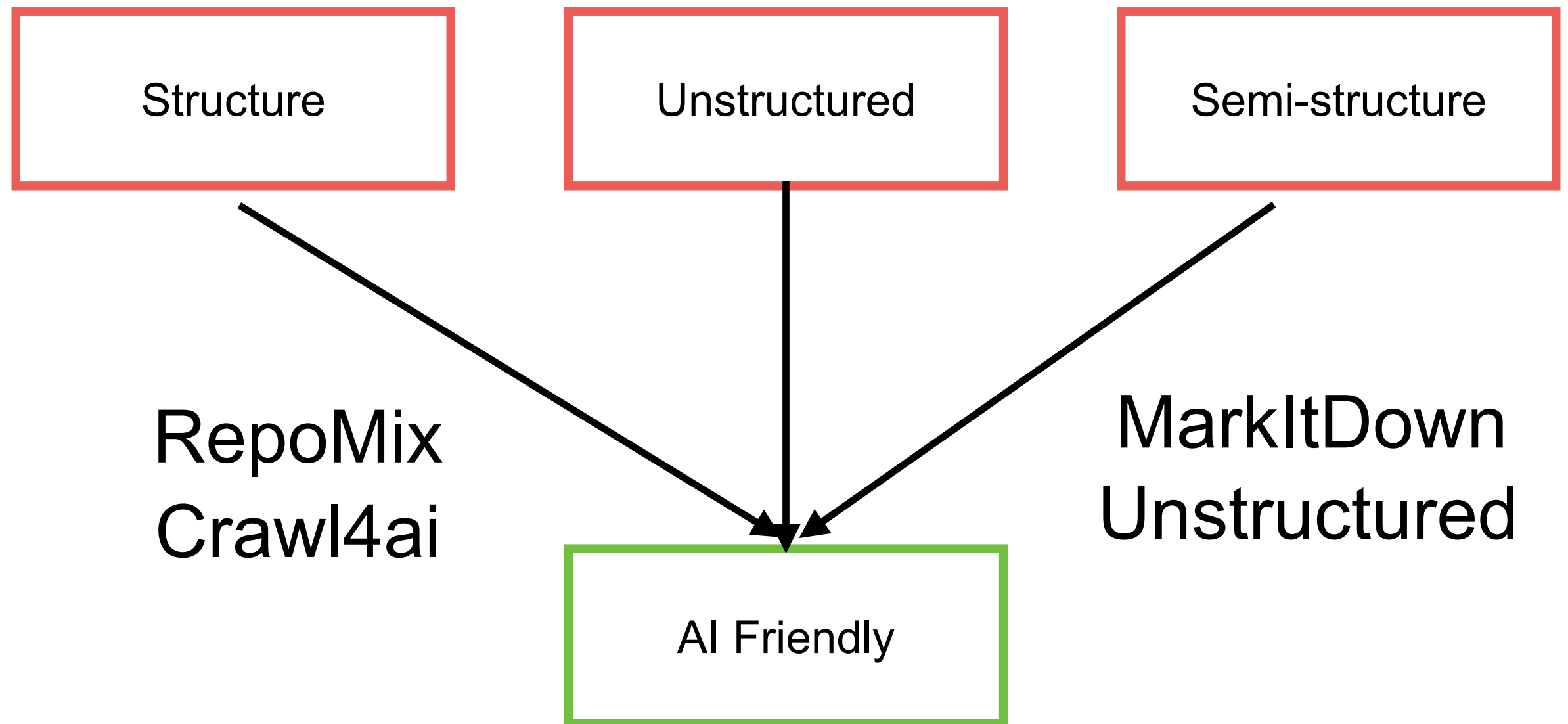
# RAG process



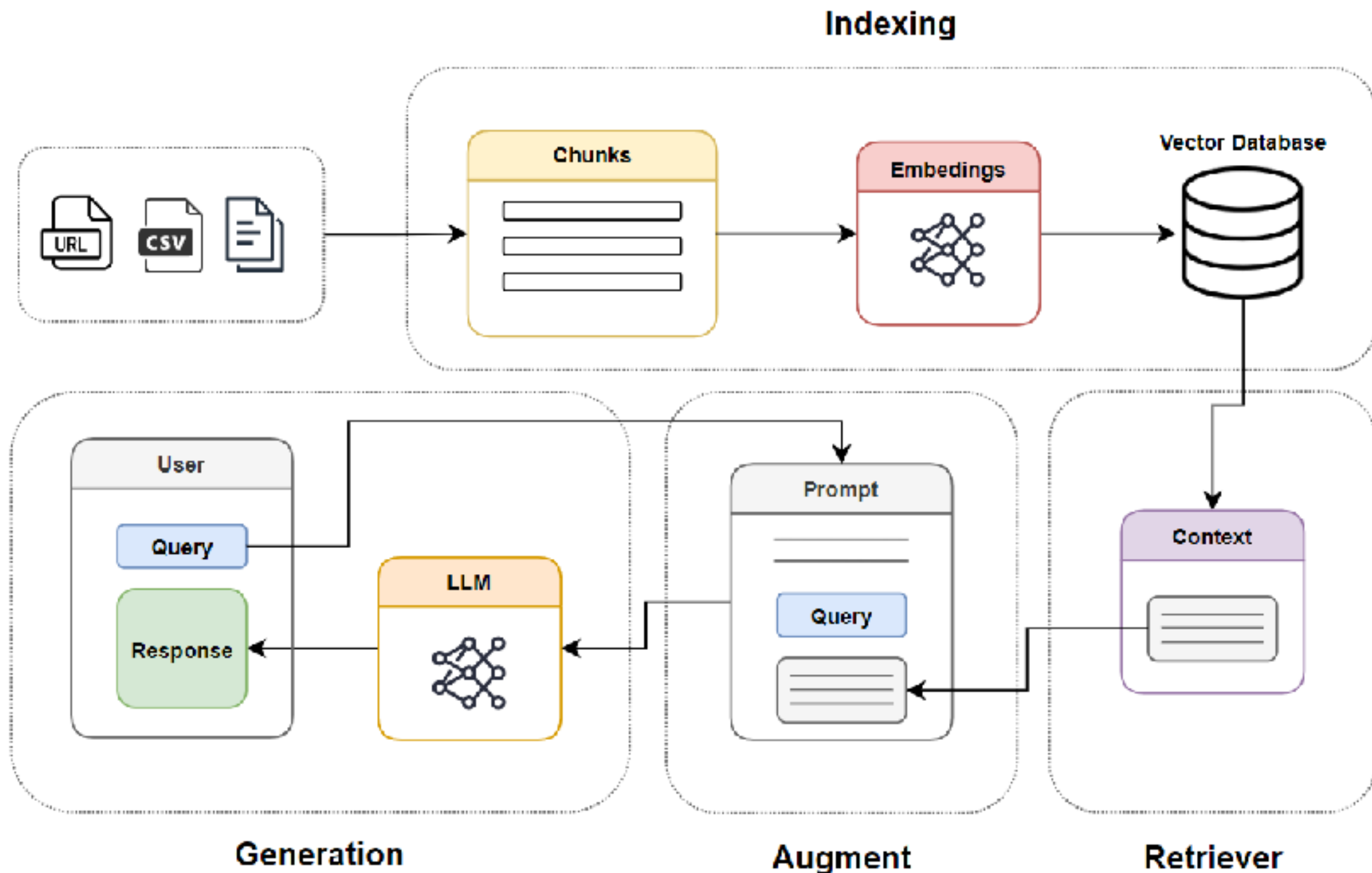
# Structures of Data ?



# Friendly Data for LLM/AI



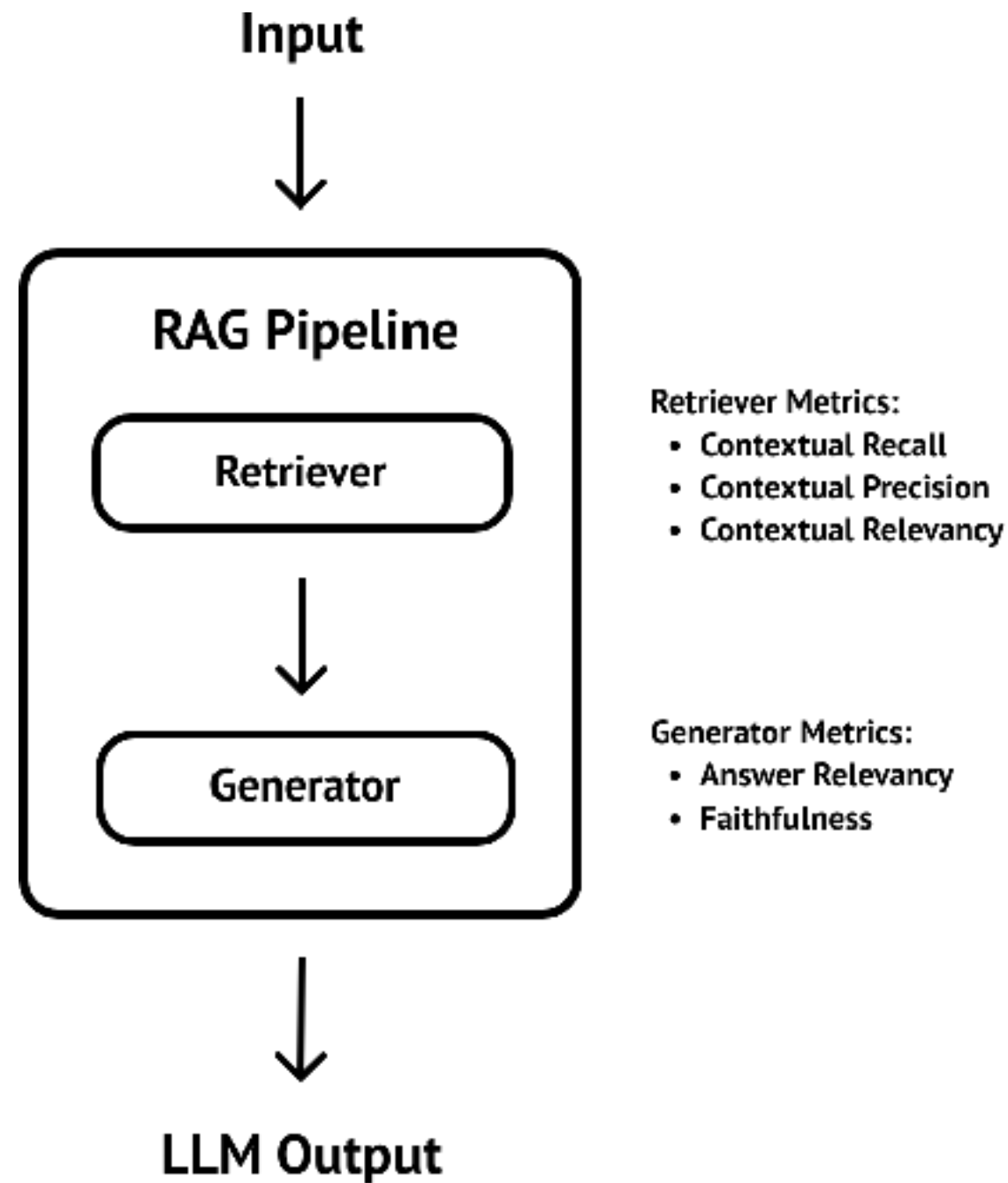
# RAG Cookbook (techniques)



<https://github.com/athina-ai/rag-cookbooks>



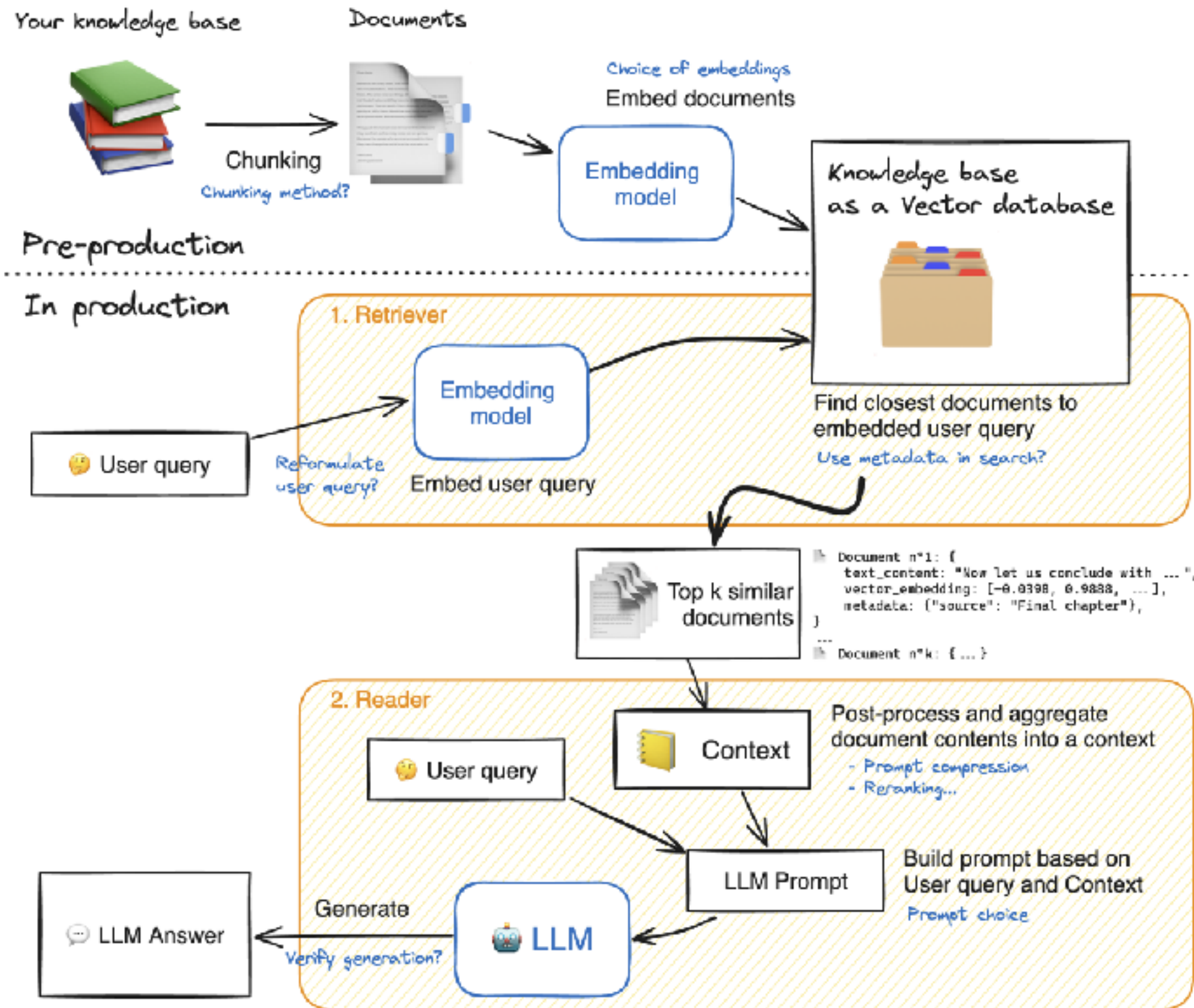
# RAG Evaluation !!



<https://www.deepeval.com/guides/guides-rag-evaluation>







[https://huggingface.co/learn/cookbook/en/rag\\_evaluation](https://huggingface.co/learn/cookbook/en/rag_evaluation)





# RAG Implementation

Document  
loading

Splitting  
or  
Chunking

Embedding

Store in  
Vector DB

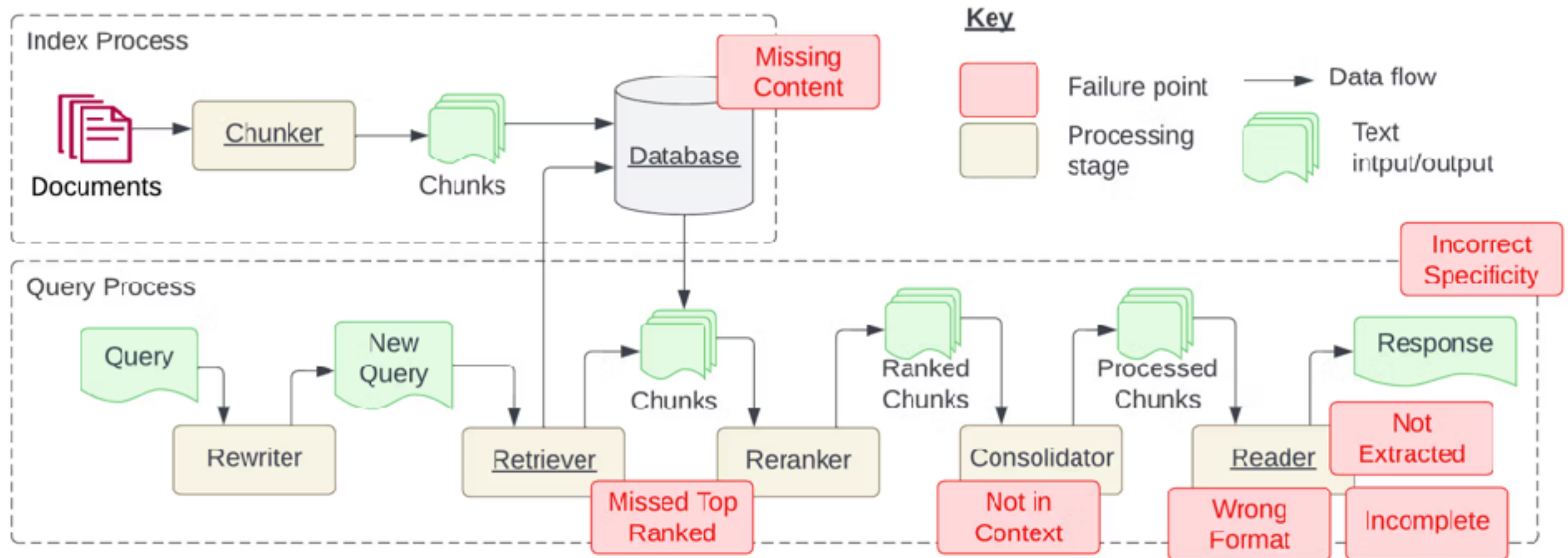
Retrieval and Re-ranking

Send Prompt to LLM

Output



# Failure Points of RAG



**Figure 1: Indexing and Query processes required for creating a Retrieval Augmented Generation (RAG) system. The indexing process is typically done at development time and queries at runtime. Failure points identified in this study are shown in red boxes. All required stages are underlined. Figure expanded from [19].**

<https://www.galileo.ai/blog/mastering-rag-how-to-architect-an-enterprise-rag-system>



# RAG is better

## But come with Cost !!

More latency

Retrieval errors

Accuracy !!

More  
Complexity

Maintenance  
overhead



# RAG Techniques ?

Semantic  
chunking

Chunk size  
selector

Context chunk  
header

Adaptive RAG

Re-ranking

Graph TAG

<https://github.com/FareedKhan-dev/all-rag-techniques>



# Pre-Retrieval optimization ?

Preprocessing and cleansing data

Chunking strategies

Add metadata

Embedding model selection



# Chunking strategies

Size of Document  
> **context window size**



Chunking

Chunk  
1

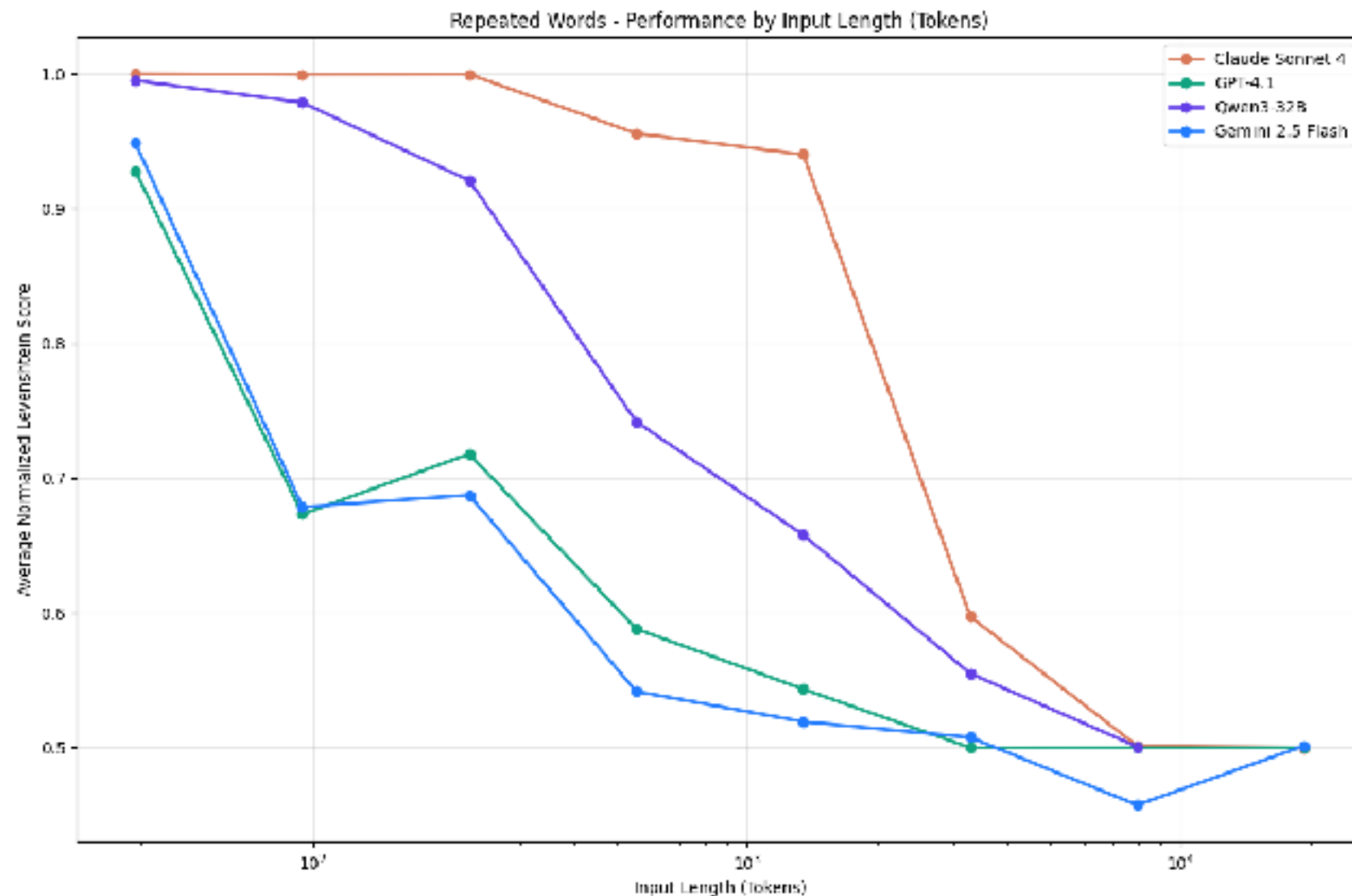
Chunk  
2

Chunk  
3



# Context Window ?

Number of tokens in the context window increases,  
the model's ability to accurately recall information from that context decreases



<https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>





# Chunking Strategies !!

Fixed size  
Recursive characters  
Document structure-based  
Semantic chunking  
Agentic chunking  
...

<https://blog.dailydoseofds.com/p/5-chunking-strategies-for-rag>





# Good Chunking

Efficiency

Relevance

Context  
preservation

Improve content  
generation

Reduce noise



# Bad Chunking

Loss context

Redundancy

Inconsistency



# Chunking Visualization

## ChunkViz v0.1

Want to learn more about AI Engineering Patterns? Join me on [Twitter](#) or [Newsletter](#).

Language Models do better when they're focused.

One strategy is to pass a relevant subset (chunk) of your full data. There are many ways to chunk text.

This is an tool to understand different chunking/splitting strategies.

[Explain like I'm 5...](#)

One of the most important things I didn't understand about the world when I was a child is the degree to which the returns for performance are superlinear.

Teachers and coaches implicitly told us the returns were linear. "You get out," I heard a thousand times, "what you put in." They meant well, but this is rarely true. If your product is only half as good as your competitor's, you don't get half as many customers. You get no customers, and you go out of business.

It's obviously true that the returns for performance are superlinear in business. Some think this is a flaw of capitalism, and that if we changed the rules it would stop being true. But superlinear returns for performance are a feature of the world, not an artifact of rules we've invented. We see the same pattern in fame, power, military victories, knowledge, and

Upload text

Splitter: Character Splitter  
Chunk Size: 25  
Chunk Overlap: 0  
Total Characters: 2658  
Number of chunks: 107  
Average chunk size: 24.8

One of the most important things I didn't understand about the world when I was a child is the degree to which the returns for performance are superlinear.

Teachers and coaches implicitly told us the returns were linear. "You get out," I heard a thousand times, "what you put in." They meant well, but this is rarely true. If your product is only half as good as your competitor's, you don't get half as many customers. You get no customers, and you go out of business.

<https://chunkviz.up.railway.app/>



# Retrieval optimization ?

Re-ranking

Hybrid search

Query  
transformation

Multi-vector  
embedding

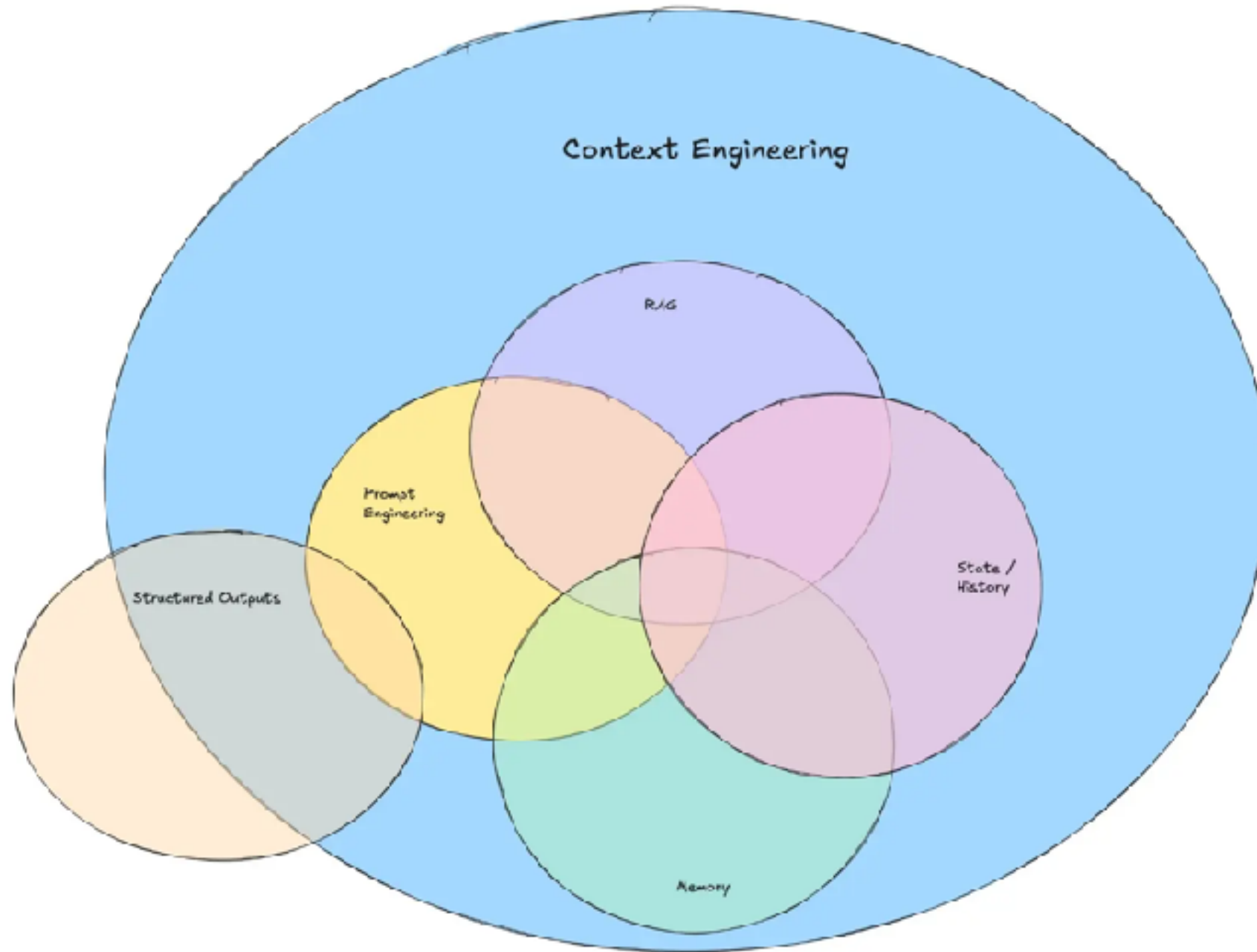
Contextual  
retrieval

Vector database  
Selection





# Context Engineering



<https://www.promptingguide.ai/guides/context-engineering-guide>



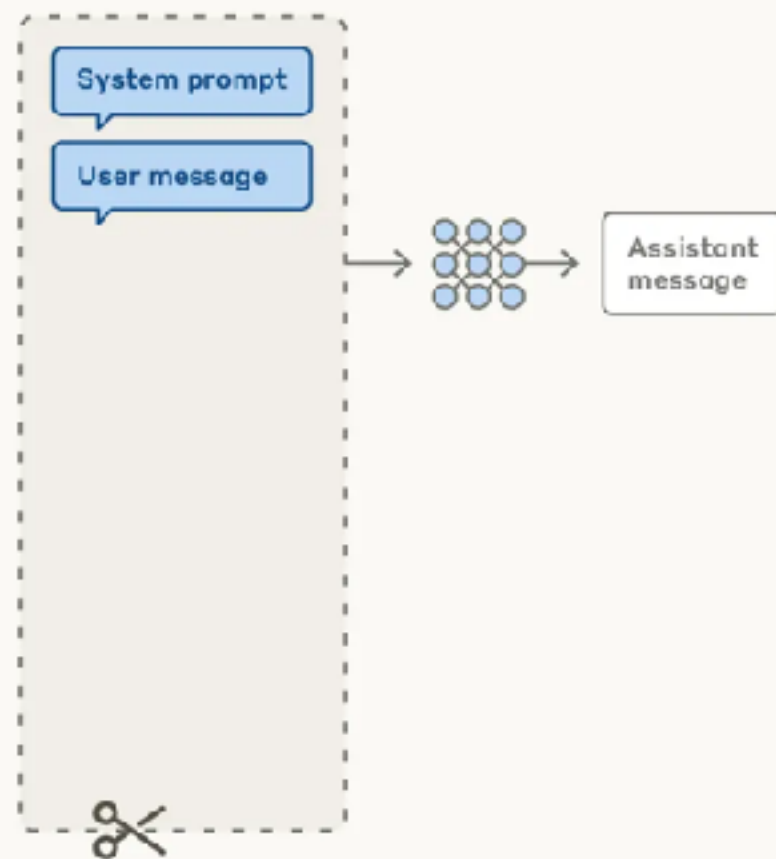


# Effective Context Engineering

## Prompt engineering vs. context engineering

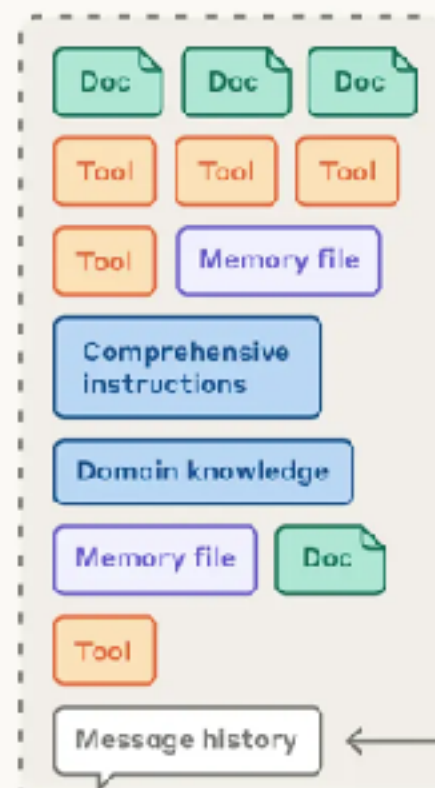
### Prompt engineering for single turn queries

#### Context window



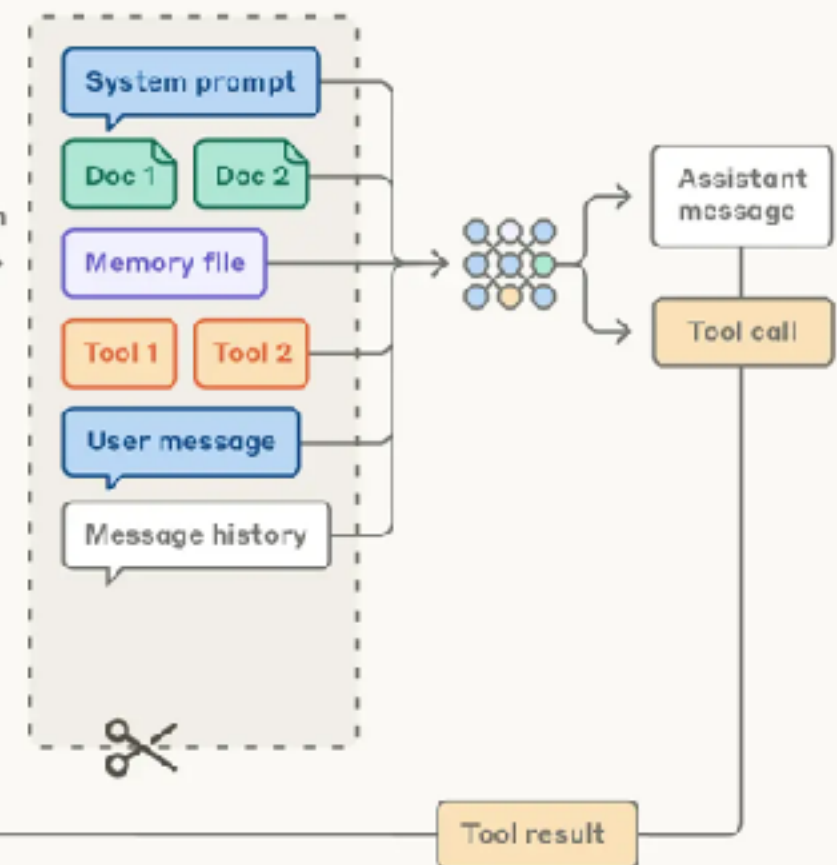
### Context engineering for agents

#### Possible context to give model



Curation

#### Context window



<https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>



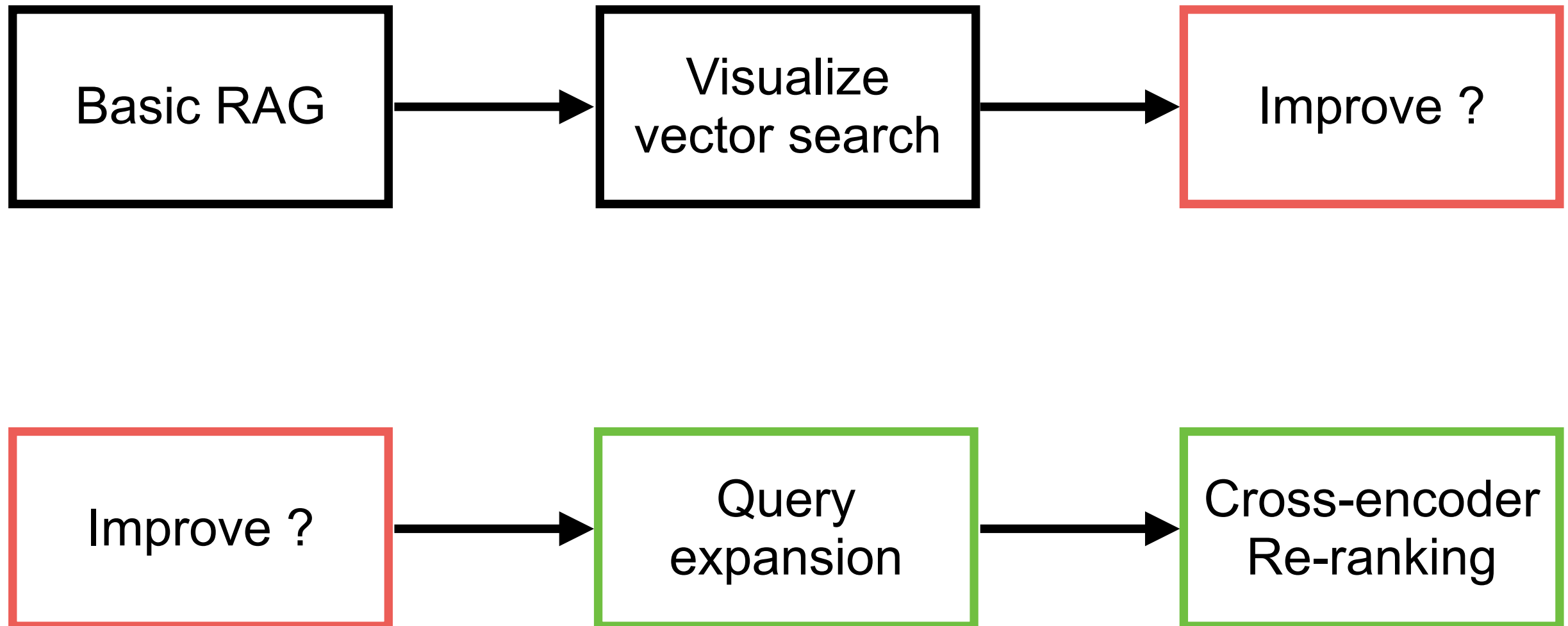
# RAG Workshop

<https://github.com/up1/workshop-basic-llm/tree/main/workshop/basic-rag>

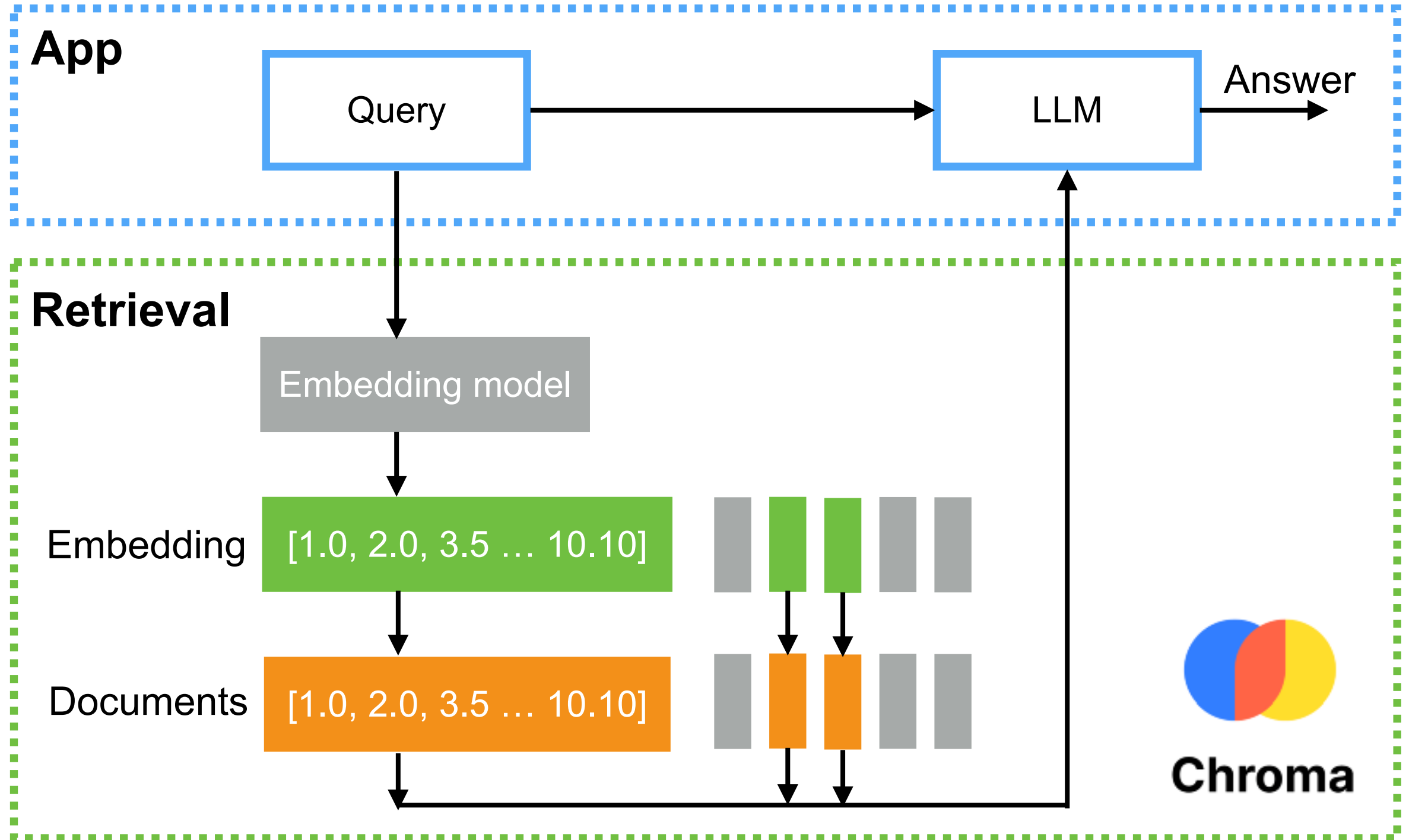




# RAG workshop



# Basic RAG



# Query Expansion

Query expansion is a widely used technique to improve the recall of search systems

Ambiguity

Vocabulary  
mismatch

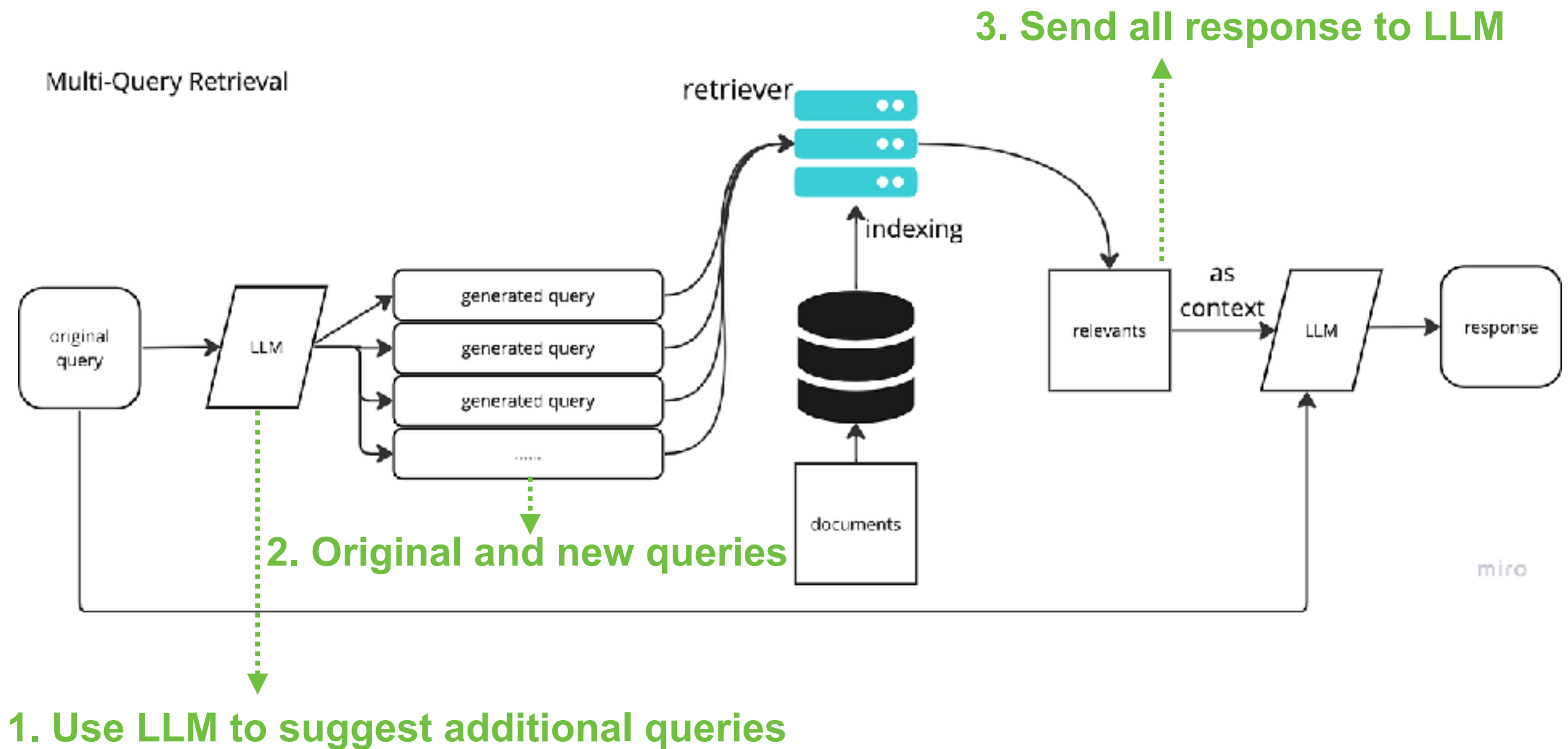
Lack of context

“Add synonyms, related context and contextual terms”

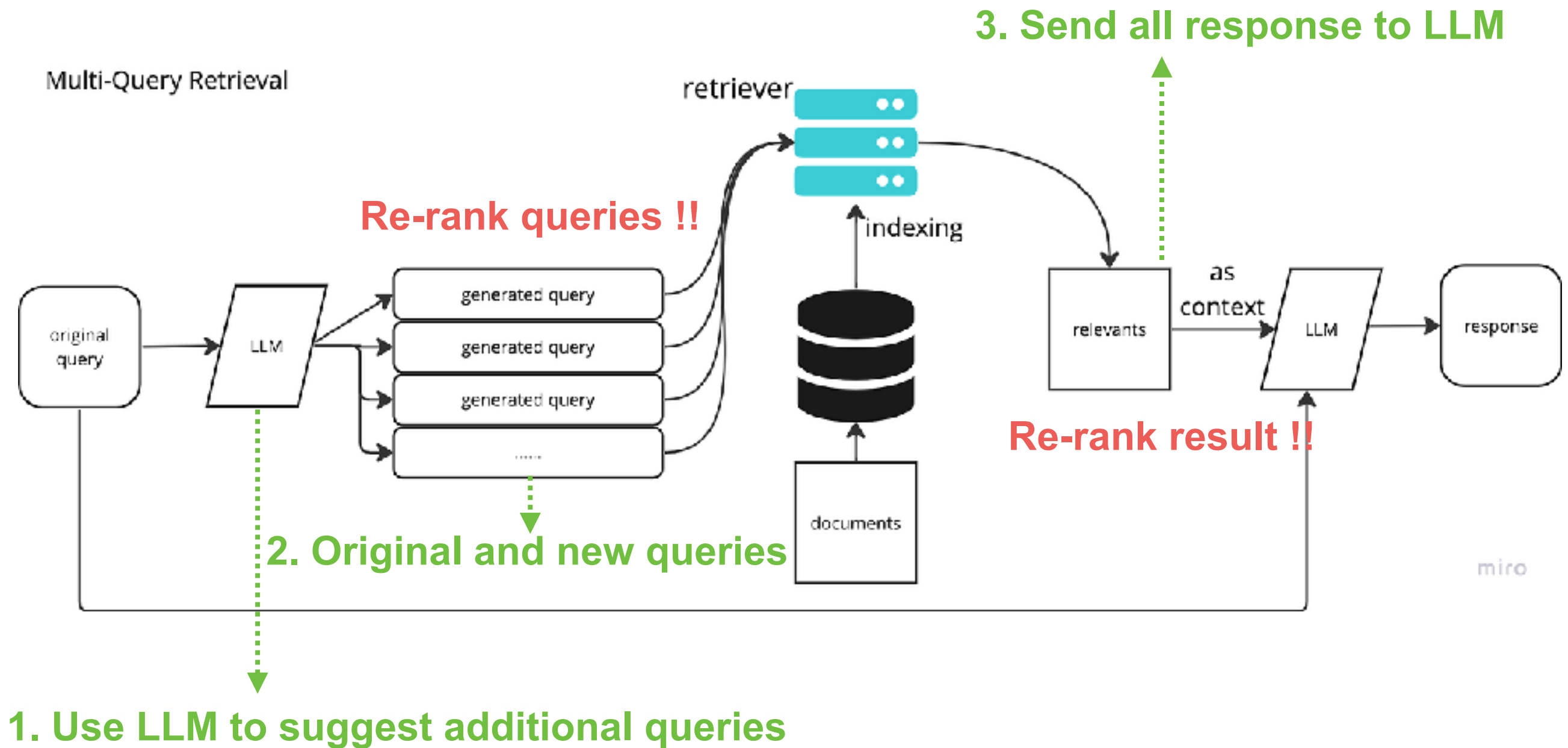
<https://arxiv.org/abs/2305.03653>



# Query Expansion

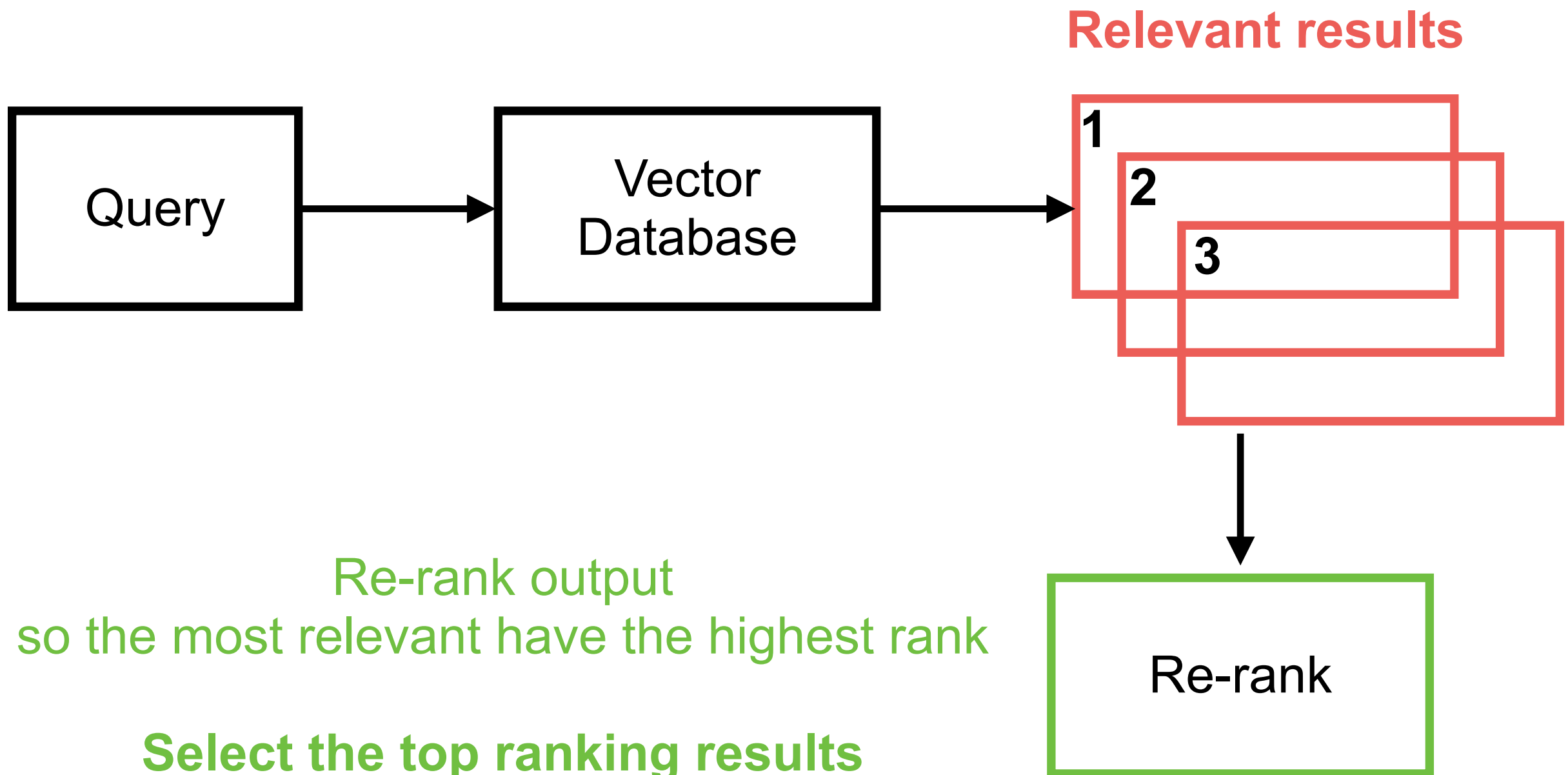


# Re-rank !!

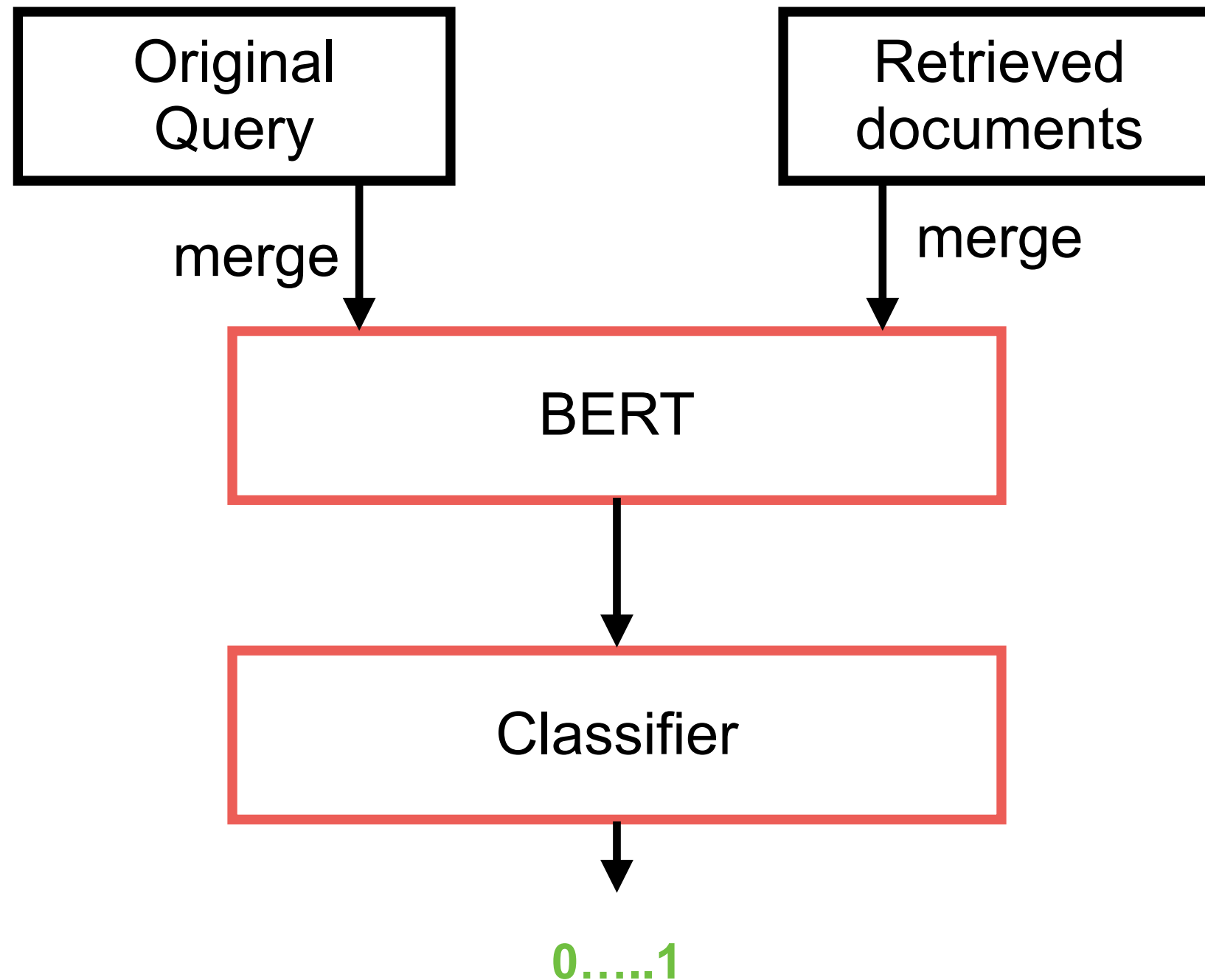


# Cross-encoder Re-ranking

How to ordering relevant results !!



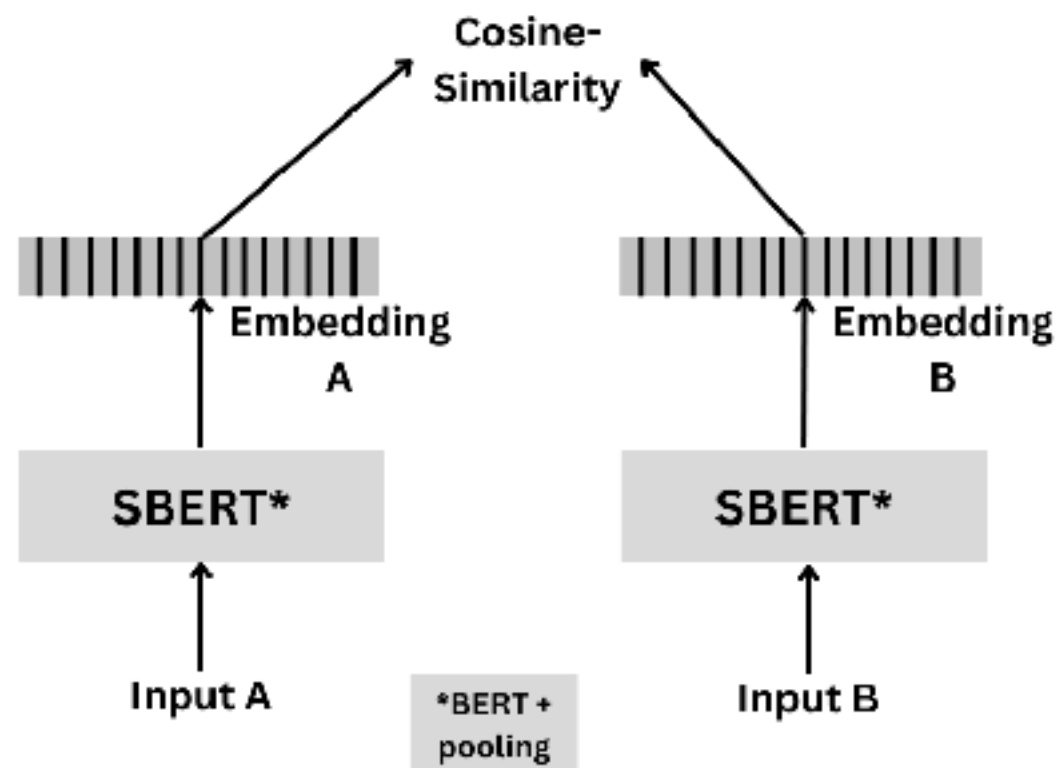
# Cross-Encoder



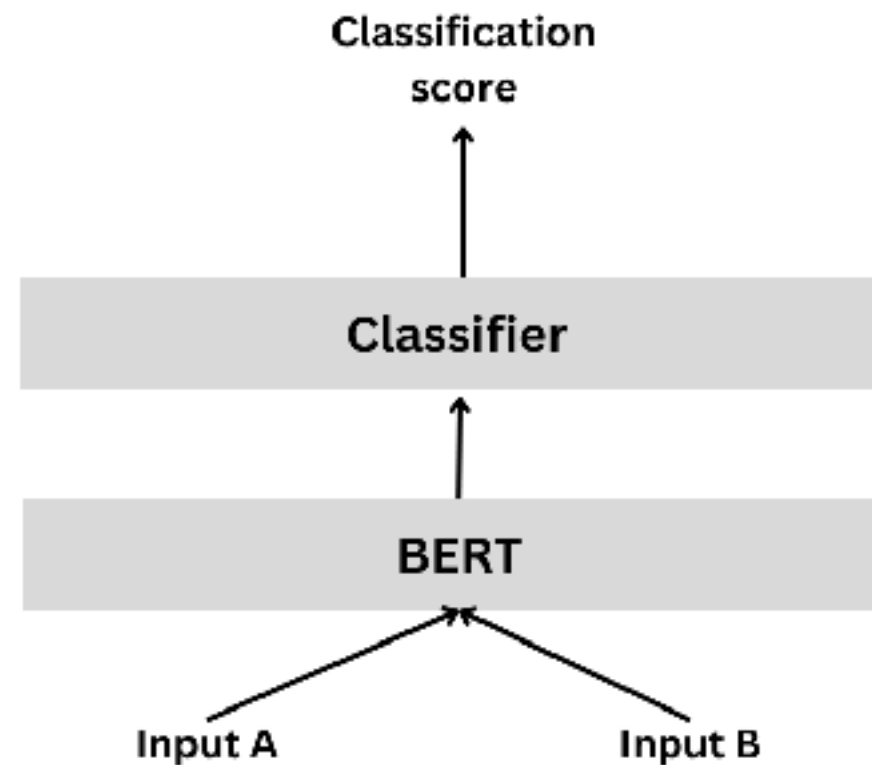


# Cross-Encoder !!

## Bi-encoder



## Cross Encoder

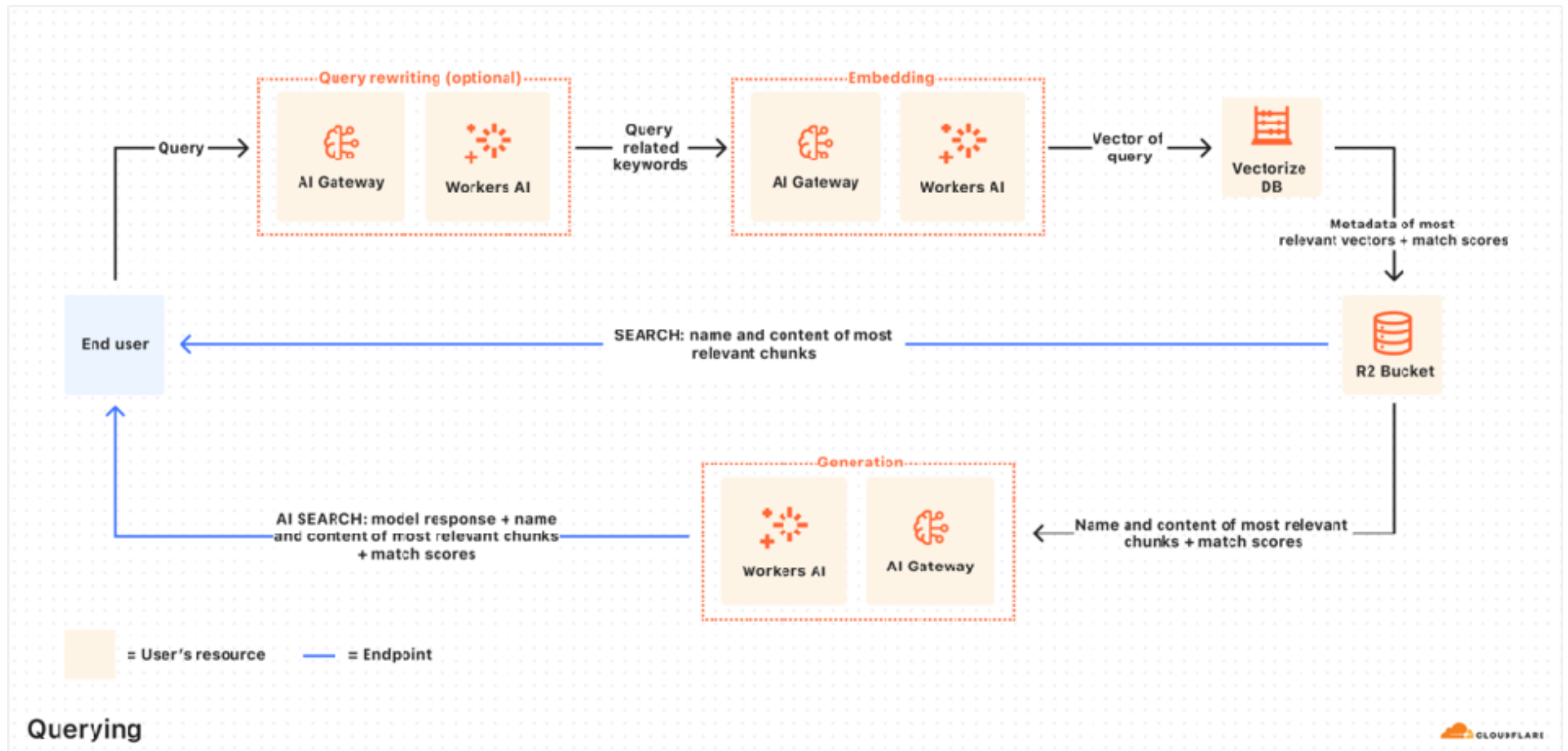


[https://sbert.net/examples/cross\\_encoder/applications/README.html](https://sbert.net/examples/cross_encoder/applications/README.html)





# Cloudflare AutoRAG



<https://developers.cloudflare.com/autorag/>



# Guardrails



<https://github.com/guardrails-ai/guardrails>



# Guardrails

Help to build reliable AI applications

Guard for  
input and output

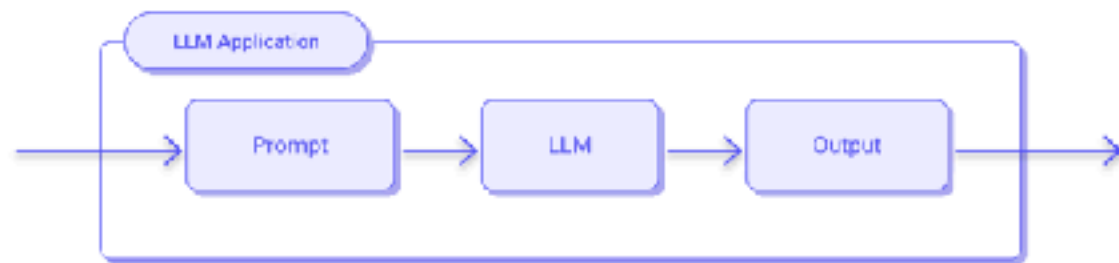
Help to generate  
Structured output

<https://github.com/guardrails-ai/guardrails>

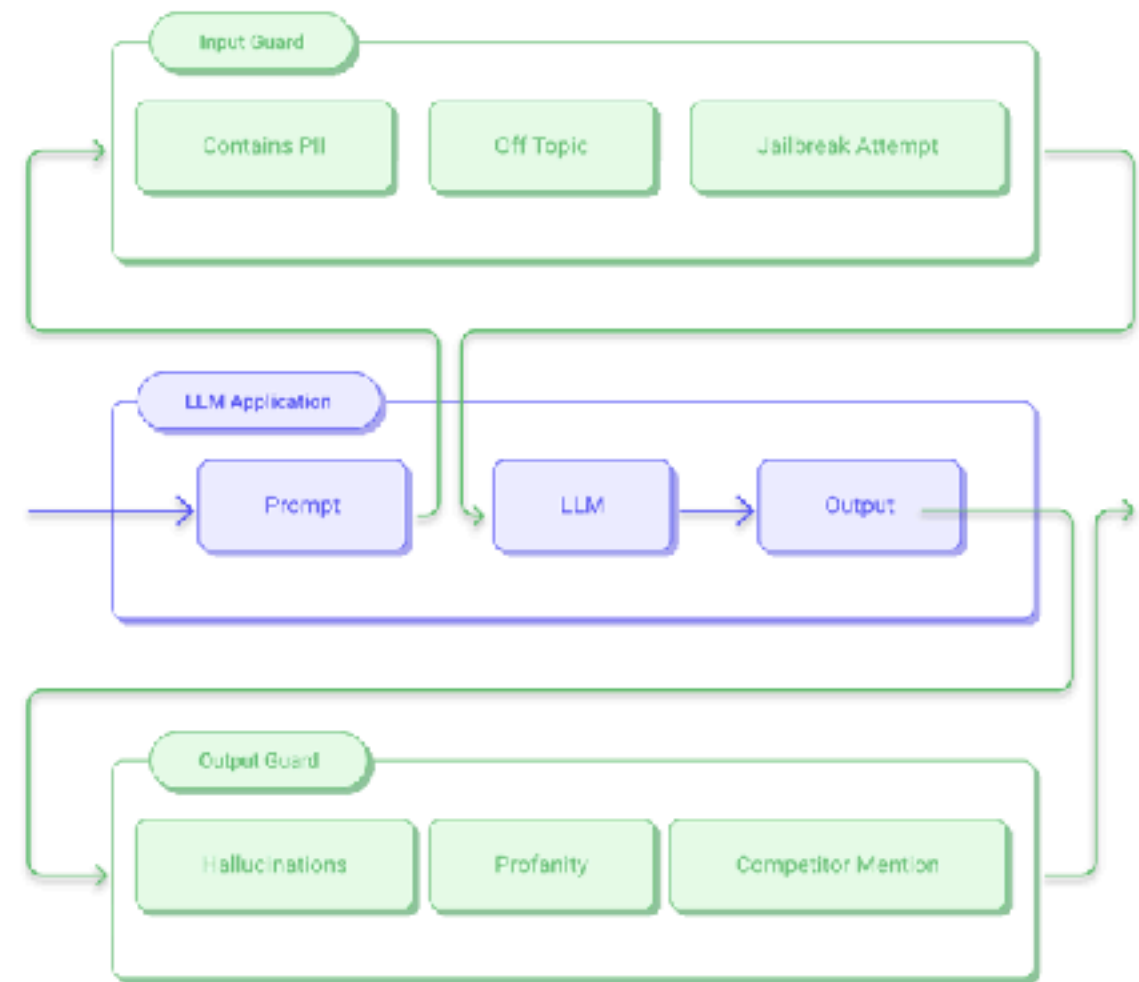


# Guardrails

*Without Guardrails*



*With Guardrails*



# Guardrails Hub

The screenshot displays the Guardrails AI Hub interface. On the left, there are several filter categories: **USE CASES** (Chatbots, Customer Support, Structured Data, RAG, Summarization, Codegen, Text2SQL), **RISK CATEGORY** (Etiquette, Brand Risk, Factuality, Formatting, Invalid Code, Jailbreaking, Code Exploits, Data Leakage), **INFRASTRUCTURE REQUIREMENTS** (ML, LLM, NA, Rule), **CONTENT TYPE** (String, Object, List, Integer, Float, SQL, Code, CSV, Python), **CERTIFICATION** (Guardrails Certified), and **LANGUAGE** (EN). The main section is titled **Validators** and includes a search bar, a 'Generate Code' button, and a list of validators. Each validator card shows its name, description, last update time, and associated risk categories. The validators listed are: Arize Dataset Embeddings, Ben List, Bespoke MiniCheck, Bias Check, Competitor Check, Correct Language, Cucumber Expression Match, and Detect PII.

**Guardrails AI** Hub Blog Docs Sign in to get started

## Validators

Validators are basic Guardrails components that are used to validate an aspect of an LLM workflow. Validators can be used a to prevent end-users from seeing the results of faulty or unsafe LLM responses.

Search

Showing 66 of 66 validators

Generate Code

**Arize Dataset Embeddings** Select

Validates that user-generated input does not match the dataset of jailbreak...  
Last updated 8 months ago

STRING BRAND RISK ML

**Bespoke MiniCheck** Select

Validates that the LLM-generated text is supported by the provided context using...  
Last updated 7 months ago

STRING BRAND RISK FACTUALITY ML

**Competitor Check** Select

Flags mentions of competitors. Fixes responses by filtering out competitor names.  
Last updated 5 months ago

STRING BRAND RISK ML

**Cucumber Expression Match** Select

Validates that the input string matches a specified cucumber expression.  
Last updated 6 months ago

STRING BRAND RISK ML

**Ben List** Select

Validates that the output does not contain banned words, using fuzzy search.  
Last updated 8 months ago

STRING BRAND RISK ML

**Bias Check** Select

Validates that the text is free from biases related to age, gender, sex, ethnicity,...  
Last updated 1 week ago

STRING BRAND RISK ML

**Correct Language** Select

Validate that an LLM-generated text is in the expected language. If the text is not ...  
Last updated 11 months ago

STRING ETIQUETTE ML

**Detect PII** Select

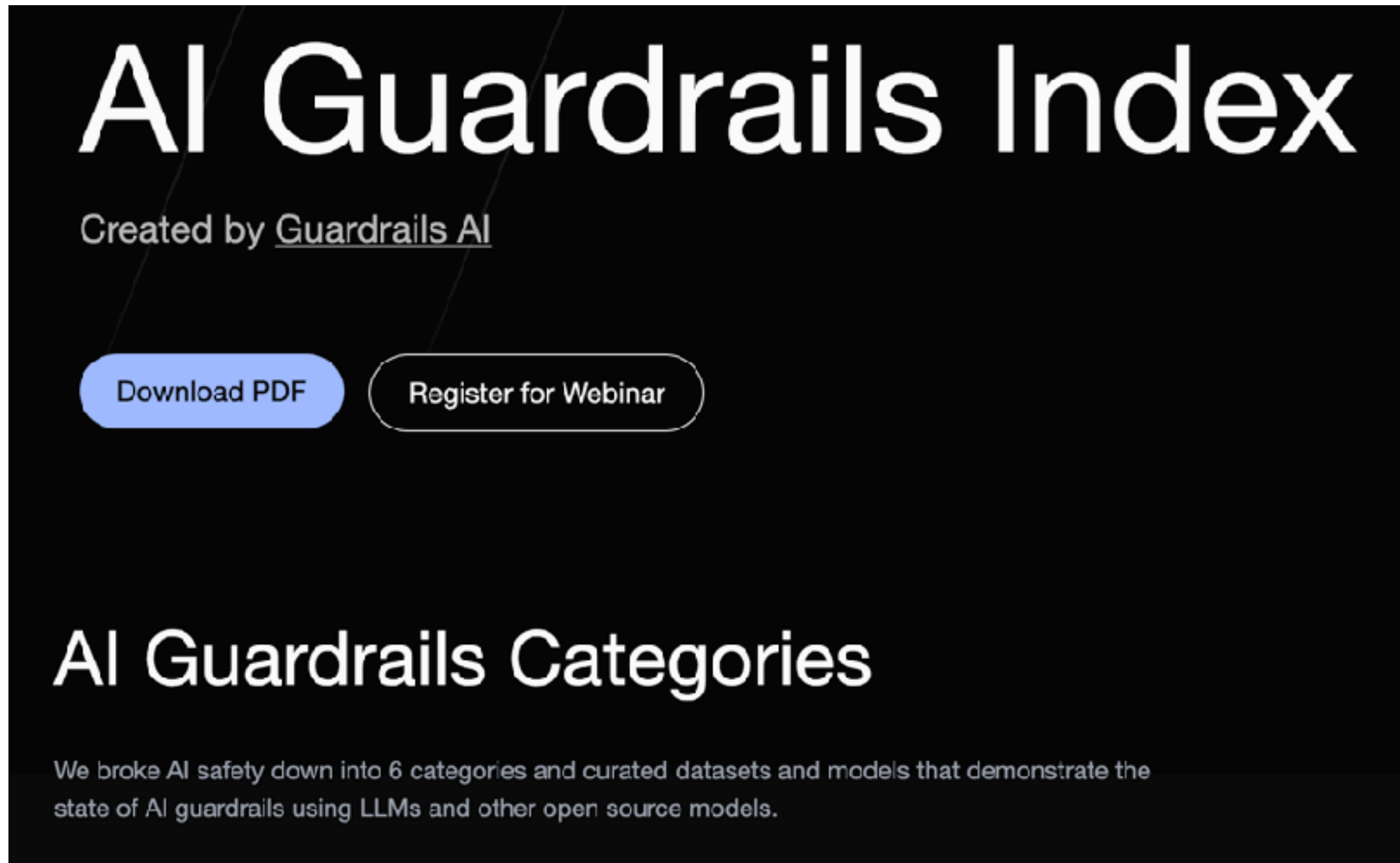
Detects personally identifiable information (PII) in text, using Microsoft Presidio.  
Last updated 6 months ago

STRING DATA LEAKAGE ML

<https://hub.guardrailsai.com/>



# Guardrails Index

A screenshot of the AI Guardrails Index website. The page has a dark background with white text. At the top, the title "AI Guardrails Index" is displayed in a large, bold font. Below it, the text "Created by [Guardrails AI](#)" is shown. There are two buttons: a blue "Download PDF" button and a white "Register for Webinar" button. Further down, the section "AI Guardrails Categories" is titled, followed by a paragraph explaining that the index breaks down AI safety into 6 categories and curates datasets and models demonstrating the state of AI guardrails using LLMs and other open source models.

## AI Guardrails Index

Created by [Guardrails AI](#)

[Download PDF](#) [Register for Webinar](#)

### AI Guardrails Categories

We broke AI safety down into 6 categories and curated datasets and models that demonstrate the state of AI guardrails using LLMs and other open source models.

<https://index.guardrailsai.com/>



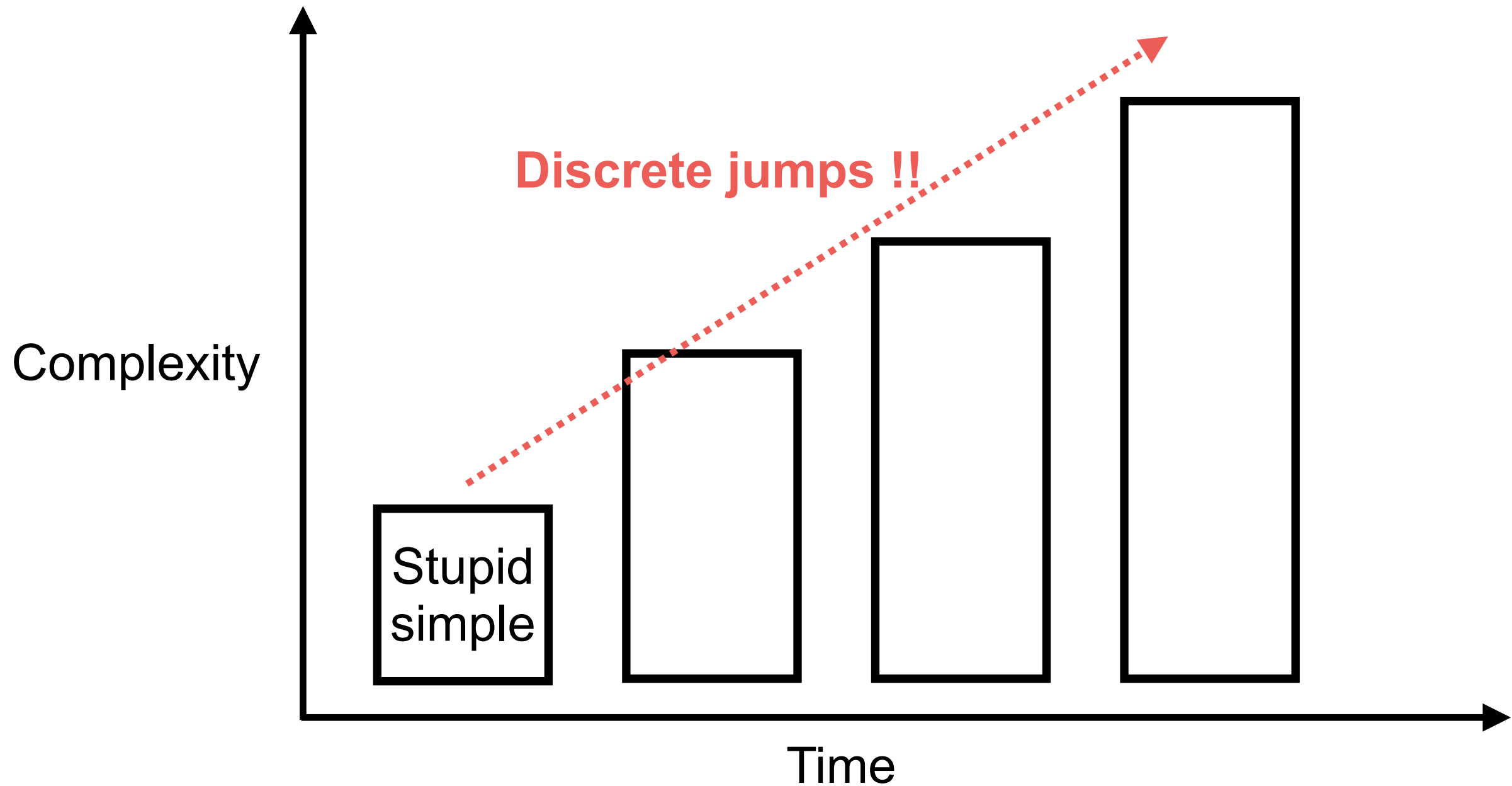
# PDCA process





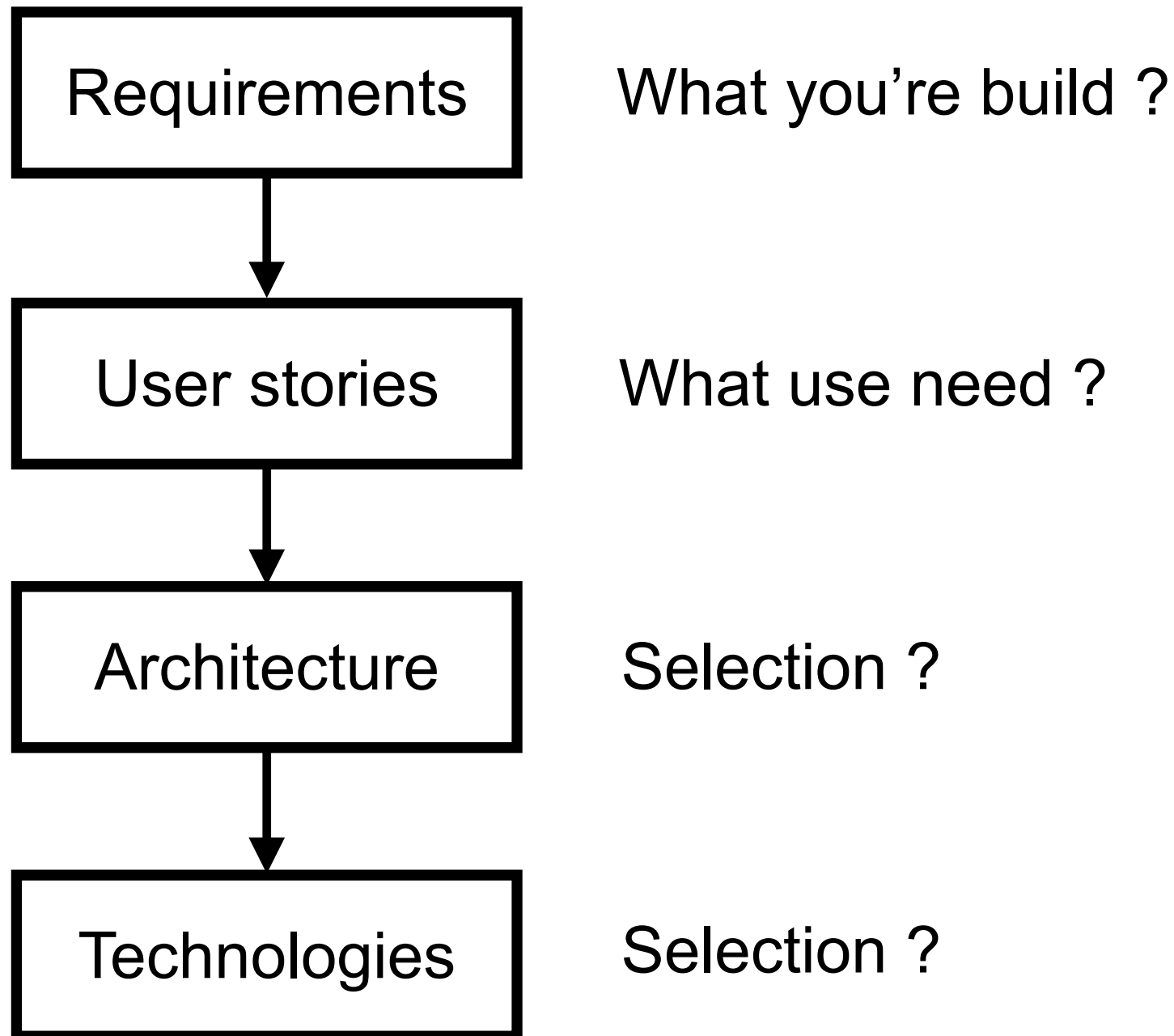
# Staged Complexity

Start small and get a feature working first

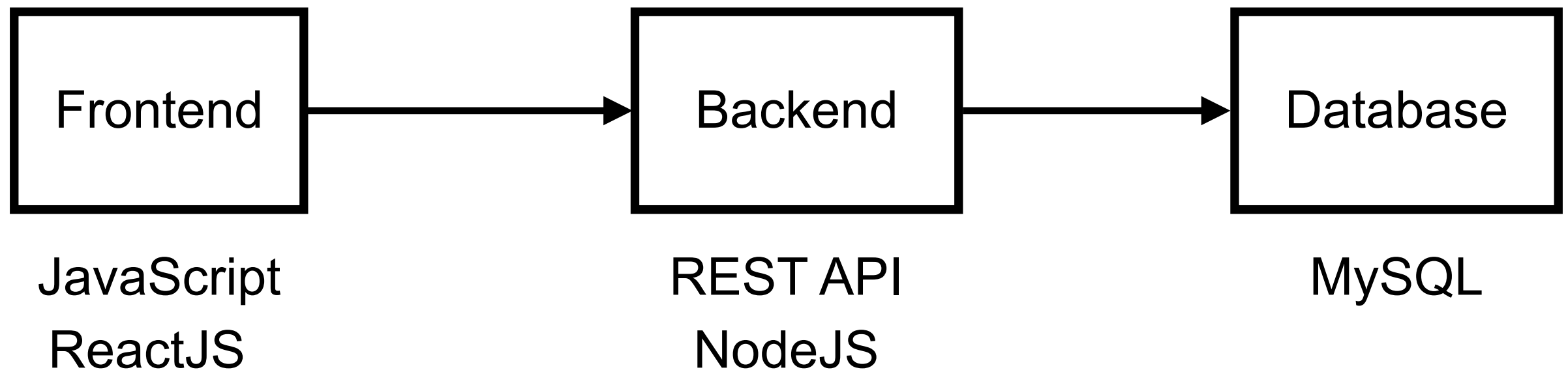




# Planning workflow



# Architecture of project

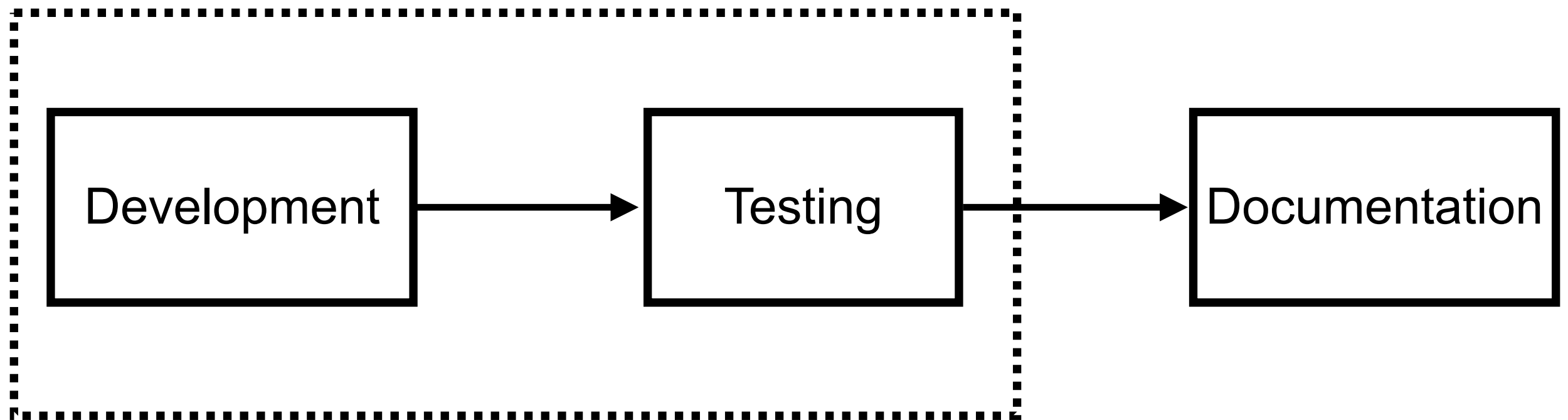


Choose your tech stack !!



# Processes

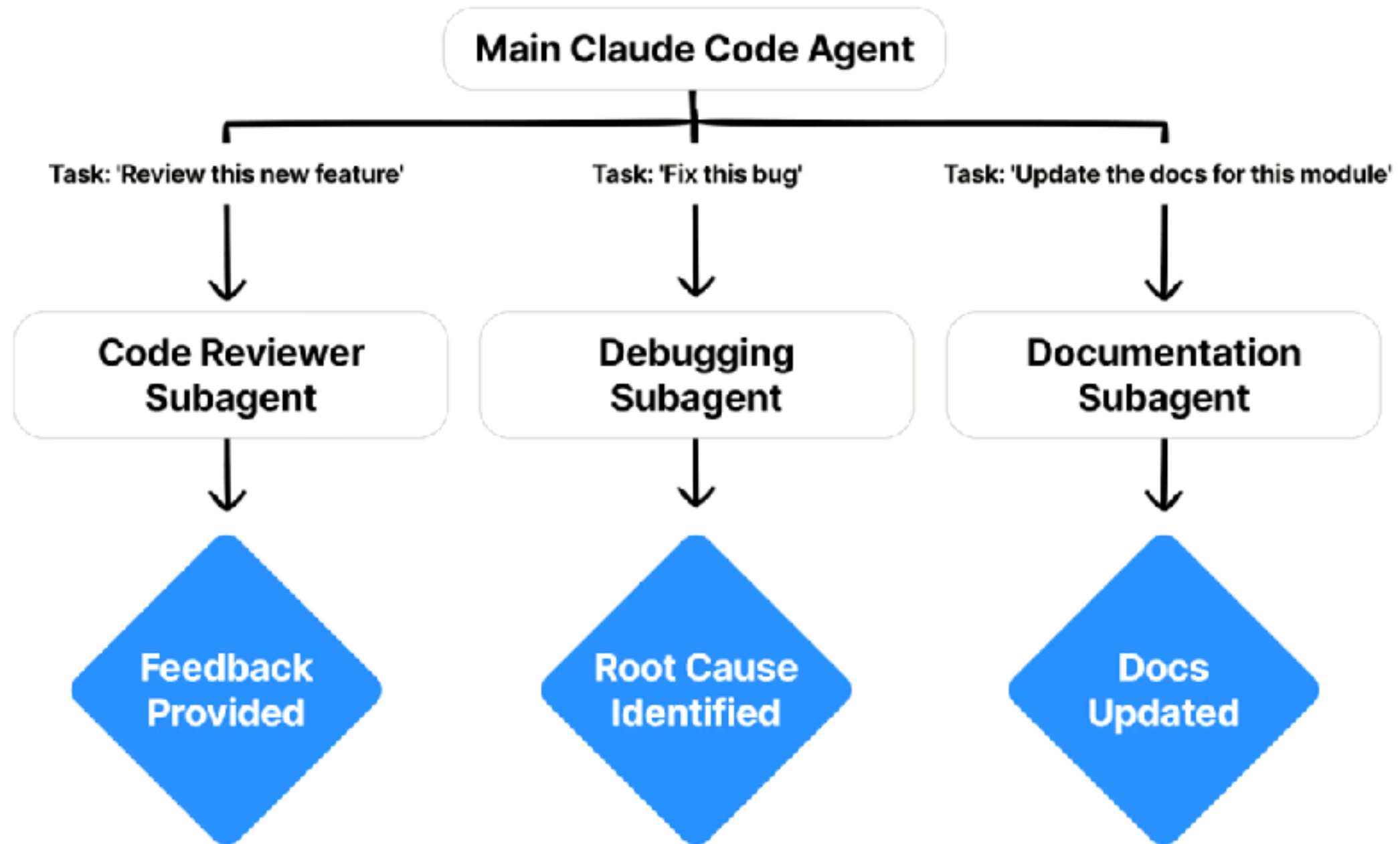
## Iterative and incremental process



# AI Subagent



# AI Subagent



<https://code.claude.com/docs/en/sub-agents>



# AI Subagent



<https://github.com/VoltAgent/awesome-claude-code-subagents>



# Start your journey

